

UNIVERSITÀ DI PISA
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Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Sources and code: <https://github.com/matteodisante/soaring-anomalous-transport>

Abstract

Soaring flight couples persistent horizontal motion to the search for atmospheric lift and the management of altitude. This thesis investigates which stochastic descriptions can account for that motion using paraglider and hang-glider records from the French free-flight federation. It combines empirical observables, probabilistic flight-phase segmentation and a programme of mathematical modelling and Monte Carlo validation.

This volume contains the introduction, the data chapter and an analysis of horizontal transport. The latter compares circuit types, launch altitudes, geographic regions and equipment groups, and tests displacement scaling with a common flight population, quantiles, moment spectra and kurtosis. Appendices 2.A and 2.B provide the spectral conventions and geodetic transformations used in Chapter 2. The final *Experimentals* section presents an exploratory joint factorial analysis extending the conditional comparisons of Chapter 3.

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Chapter 1

Introduction

1.1 A flight is a transport process with an energy budget

A soaring pilot travels by repeatedly exchanging height for distance and recovering height in rising air. The atmosphere supplies the energy, but the pilot must locate that supply and decide how to use it. A long displacement is therefore the outcome of several coupled processes: motion through the air, encounters with lift, decisions made with incomplete information, and constraints imposed by wind and terrain. Because the recorded trajectory contains all these contributions, inferring a stochastic description requires more than fitting its geometry: the description must also preserve the physical mechanisms that allow the pilot to remain airborne.

Two elementary relations explain why neither an unconstrained random walk nor a purely geometrical description is sufficient. Writing $\mathbf{r}$ for horizontal position and z for altitude, a local kinematic description is

$$\dot{\mathbf{r}}(t) = \mathbf{v}_{\text{air}}(t) + \mathbf{w}_h(\mathbf{r}(t), z(t), t), \quad (1.1)$$

$$\dot{z}(t) \simeq w_z(\mathbf{r}(t), z(t), t) - s(|\mathbf{v}_{\text{air}}(t)|, \phi(t)). \quad (1.2)$$

Here $\mathbf{w}_h$ and w_z are the horizontal and vertical air velocities, $s > 0$ is the sink rate relative to the air, and ϕ is bank angle. The second relation is a quasi-steady approximation: it makes explicit the competition between lift and aerodynamic sink without attempting to model every manoeuvre. Altitude is therefore a finite resource, and a transport model must account for its use and replenishment if its trajectories are to represent sustained flight.

This physical picture suggests three descriptions of behaviour: directed *transition*, exploratory *search*, and *climb* in lift. They are hypotheses about the organisation of motion, rather than labels directly supplied by a recorder. A pilot can find lift during a transition, climb without completing a circle, or search while losing height. Determining which distinctions the data support is part of the scientific problem.

Soaring control has been studied through the problem of locating and exploiting thermals, including adaptive strategies based on local information [1]. The present work asks a complementary question: what statistical laws emerge over entire flights, and which mechanisms can account for those laws? A direct empirical precedent is the analysis of soaring transport and flight phases by Vilpellet et al. [2]. This thesis revisits that connection using the FFVL archive, explicit data-quality controls, and a separation between measured properties and assumptions used to interpret them.

1.2 The questions and the evidence needed to answer them

1.2.1 Which stochastic models survive the observations?

A first observable is the squared displacement over an interval τ ,

$$M_2(\tau) = \left\langle |\mathbf{r}(t + \tau) - \mathbf{r}(t)|^2 \right\rangle. \quad (1.3)$$

Ordinary diffusion without drift has $M_2 \propto \tau$; directed motion can produce a ballistic law $M_2 \propto \tau^2$. An intermediate slope is informative, but it does not identify a mechanism, since correlated Gaussian motion, persistent motion observed during a crossover, and mixtures of flight behaviours can produce similar second moments. Random-walk frameworks offer competing explanations of anomalous transport [3]. The purpose here is to discriminate between such explanations using more than one curve.

The empirical programme therefore calls for measurements of the MSD, displacement quantiles and directional anisotropy, and how each depends on route, duration and equipment. Quantile ratios test whether the distribution changes shape as the lag increases. Open and closed tasks help expose the effect of the flight's intended geometry. These measurements, together with the spectrum of moments, the shape of the increment distribution and velocity correlations, must be compared with explicit model predictions: higher moments compare the growth of ordinary and large displacements, while velocity correlations resolve persistence in time. A model is challenged where its predictions disagree with these observations under a comparable measurement protocol.

This logic also sets the limit of an exclusion. A discrepancy with a homogeneous Gaussian model for pooled flights need not exclude Gaussian fluctuations within a flight: different conditions can create a non-Gaussian mixture. Similarly, evidence against a particular constant-speed renewal Lévy walk is not a theorem against every process with finite-speed legs [4]. Each comparison must specify the hypotheses being tested and leave unsupported exclusions open.

1.2.2 How do the flight phases generate global transport?

Although an unsegmented flight supplies constraints on the whole process, it does not identify which behaviour generates a long displacement, a change of heading or a period of persistence. A probabilistic segmentation into transition, search and climb can address this missing link. It requires an explicit mathematical model, readable trajectory examples, and a distinction between internal consistency and accuracy against independent annotations.

The next step will be a chapter devoted to observables conditioned on phase: episode durations, distance and height changes, speed and turning distributions, displacement scaling, and dependencies between successive episodes. A useful model must reproduce both these conditional statistics and the global ones, a stronger requirement than generating a visually plausible sequence of coloured flight segments. Windows that cross phase boundaries must also be retained when reconstructing global observables.

1.2.3 Does information from other pilots change the motion?

A nearby pilot can reveal lift before a follower reaches it. This creates a possible information channel that is absent from a model of independent searchers. A later chapter will compare solitary and group flight using both global and phase-conditioned observables. The operational definition of a group will concern simultaneous proximity and plausible visibility, not membership of a competition. A competition can contain isolated pilots, and pilots outside a competition can travel together.

The comparison will ask where differences occur: in time spent searching, climb selection, transition direction, episode persistence, or the efficiency of converting altitude into distance.

Site, weather, wing and pilot characteristics must be controlled because they can produce the same differences without social interaction. Incomplete recording of nearby pilots will limit what can be called solitary flight. These are conditions for an interpretable comparison, not effects assumed in advance.

1.3 From empirical constraints to a physical model

The thesis combines empirical analysis, mathematical modelling and simulation, with a distinct role for each. Data analysis establishes the constraints that a candidate process must satisfy; mathematics then makes that process precise and derives its predictions, which Monte Carlo simulation tests under finite flight durations, heterogeneous environments and the actual observation protocol.

Model construction will follow the global analysis and the analysis conditioned on phase. Possible ingredients include finite-speed transitions, lift encounters, distributions of episode duration, and dependencies between successive directions or episodes. Renewal independence is one hypothesis to test; correlated random walks provide alternatives when that independence fails [5, 6]. The choice among these ingredients must be justified by the observations and an explicit physical mechanism, rather than fixed by the outline of the thesis. Parameters fitted on one set of flights will be checked on separate flights or days, using several observables jointly.

A reference model without wind and terrain may help isolate mechanisms, but obtaining it requires an inference about the environment: mean ground velocity alone does not determine wind, and rotating a trajectory into its principal axes does not remove orography. Artificially making a covariance matrix isotropic changes the measured object; it does not recreate a flight in still air. The analysis consequently preserves the motion in geographical coordinates and studies environmental strata before asking which properties could survive in a common reference model.

A further part of the programme concerns the spatial organisation of thermals and route choice. It will first estimate the two-dimensional distribution of *visited* lift regions, accounting for where pilots have searched. Recovering the distribution of available thermals requires an observation model: unvisited terrain is not evidence of absent lift. A mathematical description of thermal availability and strength could then support separate optimisation problems, such as minimum-time travel between two locations, a closed route, or maximum distance within a specified flight budget. Wind and, subsequently, possible geometrical constraints from terrain will be introduced in stages. A spatial density alone cannot determine an optimal path without assumptions about thermal strength, spatial dependence, altitude and the pilot's information.

1.4 Structure and current scope

Figure 1.1 shows how the measurements support the later modelling. The present document contains the introduction, the data chapter and the analysis of horizontal transport and displacement scaling, together with the two appendices cited in Chapter 2. Flight-phase segmentation, conditional observables, social flight, stochastic modelling and route optimisation form the subsequent research programme.

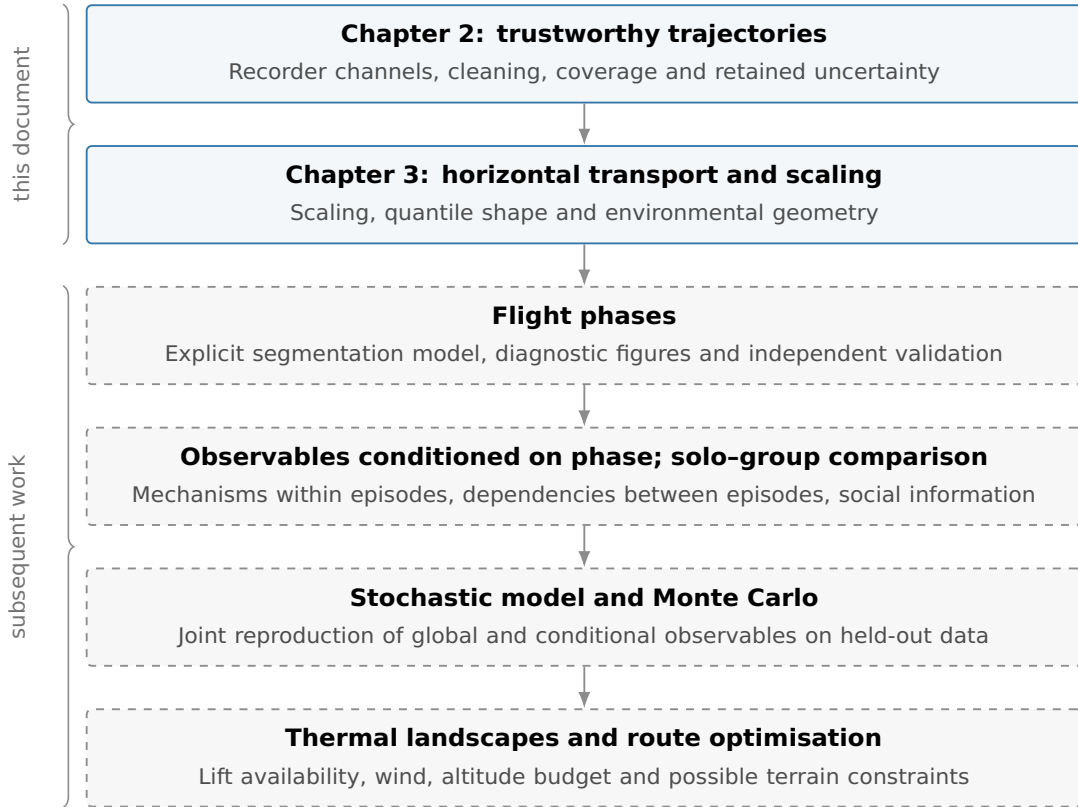


Figure 1.1. Logical structure of the thesis. The solid boxes are developed in this document; dashed boxes describe subsequent work. Model comparison must return to the measured observables, not stop at a fitted trajectory or a single exponent.

Chapter 2 follows the records from acquisition to retained trajectories, including the treatment of missing or inconsistent channels. The two appendices state the spectral conventions and the geodetic transformation used in that chapter. Chapter 3 measures horizontal transport and distributional scaling with fixed flight populations, and resolves circuit and launch-altitude differences.

1.5 Chapter summary

The thesis asks which stochastic description can account for soaring trajectories while respecting the physical roles of lift, travel and environmental constraints. The route from observations to a model begins with reliable trajectories and global transport statistics, then uses validated flight phases to identify the mechanisms mixed by those statistics. Comparisons between solitary and interacting flights, followed by simulation and thermal-landscape modelling, extend this programme. Their conclusions remain future work: the present document establishes the cleaned trajectories and the global transport statistics that those comparisons must explain.

Chapter 2

The dataset

The FFVL archive provides flight metadata and recorded trajectories for paragliders and hang gliders. We describe its coverage, then the cleaning, coordinate conversion and resampling that produce the trajectories used in the transport analysis.

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The *Coupe Fédérale de Distance* (CFD) is the cross-country competition of the French free-flight federation (FFVL). Its two archives separate paragliders (`parapente.ffvl.fr`) from hang gliders (`delta.ffvl.fr`). Each entry contains flight metadata and, usually, an IGC trajectory.

Seasons are named by their starting year: 1999 means 1999–2000. The paraglider archive begins in that season; the hang-glider archive begins in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Acquisition and verification

Two command-line tools, one per discipline, share the same acquisition code. Downloads are resumable and rate-limited; each response must contain a readable IGC file with an A header and at least one B fix record (Sec. 2.2). The archived XMLs preserve the source lists, and the output location is configurable.

An independent check of the files on disk verified all 186,052 paraglider and 6,716 hang-glider tracklogs against the same content requirements. Section 2.5 compares these downloads with the full list of declared flights.

2.1.4 Further data sources

The present analysis uses the FFVL archive. WeGlide could add sailplanes, and XContest could broaden paraglider and hang-glider coverage. Both would feed the same catalog format.

Research-access requests were pending at both portals on 20 July 2026. At that date, WeGlide’s default quota of roughly sixty API requests per day precluded bulk acquisition. These additional sources are not included here.

2.2 The IGC tracklog format

An *IGC file* is a plain-text flight record with fixed-position fields, defined by the FAI/IGC specification [7]. The first character identifies the record type:

Record	Content
A	Recorder identifier.
H	Headers: date, pilot, glider, datum, and related metadata.
I	Declaration of optional extension fields.
B	UTC time, WGS84 latitude and longitude, GNSS validity flag, pressure altitude and GNSS altitude (both in metres).

The B-record flag describes the GNSS solution: **A** means a three-dimensional fix; **V** means a two-dimensional fix or no GNSS data. We invalidate GNSS altitude at **V** fixes and retain their horizontal coordinates provisionally, pending the checks in Sec. 2.7.2. A pressure value alone does not establish a working barometer.

The parser rejects malformed records: wrong length, illegal times, coordinates or hemisphere letters. Physically implausible but legally encoded fixes pass to cleaning. The resulting sequence of fixes has a device-dependent cadence, typically a few seconds, which need not be uniform.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	Content / caveat
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
aile	wing model name (raw XML field; French for “wing”)
aile_class	wing class, one system per discipline (Sec. 2.4)
flight type	the <i>scoring</i> category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, ... (<i>caveat</i> : a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

The derived **catalog** (`catalog.csv`) has one row per flight: the metadata above, a `local_path` and a `downloaded` flag. The **season index** (`seasons_index.csv`) counts declared flights, track links and downloads. Both can be rebuilt from the archived XMLs.

2.4 Glider categories

Aircraft design changes speed, glide performance and turning geometry, motivating separate transport analyses [2]. Paragliders have a flexible canopy, hang gliders a framed flexible or rigid wing, and sailplanes a rigid airframe. The FFVL source identifies the first two families; sailplanes are absent from this dataset.

Within each family, `ail_e_class` defines the equipment groups in Table 2.2. Paraglider EN and legacy LTF codes, such as “C ou 2”, are harmonized as EN A–D; CCC and tandem/non-certified entries remain distinct. Blank and placeholder entries stay unclassified.

EN labels describe safety-test responses and recovery demands, rather than measured speed or glide ratio [8]. The descriptions here use the consulted 2012 final draft; the 2013 standard and later amendments were not compared. FAI hang-glider classes distinguish design and control systems [9].

For the conditional transport analysis, two operational equipment proxies are defined:

beginners : EN A/B/C, experts : EN D/CCC.

These names distinguish the recorded wing classes, not measured pilot experience: an experienced pilot may fly an A, B or C wing, and a C label does not identify a novice. Comparisons between the groups therefore combine equipment differences with any differences in the pilots who select them. Tandem, non-certified and unclassified entries enter neither group. No equally spaced performance scale is assumed. The harmonization does not reassess certificates, and a tandem label alone does not establish EN certification.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. EN labels summarize test-based recovery characteristics; FAI classes distinguish control and design. They are used as equipment groups, without assigning pilot ability. Sources: the FAI Sporting Code, Common Section 7 [9] (§1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the consulted 2012 final draft [8].

Class	Notes
Paragliders — harmonized EN/LTF catalogue groups	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI <i>Standard</i> sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI <i>Sport</i> sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9

Season	Flights	With GPS	Downloaded	Coverage (%)
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the “With GPS” column of Tables 2.3 and 2.4.

- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights. GNSS altitude is used for every admitted flight, with the channel recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries carry the placeholder date 0000-00-00, written as a sentinel at acquisition. Such entries cannot support a calendar-conditioned comparison until a date is recovered independently; the relative-time preprocessing does not itself enforce this metadata exclusion. The count is small: 8 of 203,343 paraglider entries carry the sentinel (0.0039%), 0 of 9,259 hang-glider ones, and the largest within-season fraction is only 0.17% of its entries (2001-2002).

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude. The pipeline then proceeds in a fixed order:

- (i) choose the altitude channel (Sec. 2.7.1);
- (ii) clean individual fixes (Sec. 2.7.2);
- (iii) trim the ground phases (Sec. 2.7.3);
- (iv) filter out unfit whole flights (Sec. 2.7.4);
- (v) convert to a local Cartesian frame (Sec. 2.7.5);
- (vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);
- (vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output consists of fix, segment and attempted-flight tables, accompanied by a table of suspect intervals retained for later sensitivity analysis. What each file holds and how the disk is organised is set out in Sec. 2.8.

Steps (i)–(iv) use raw geographic coordinates: great-circle distances give horizontal speeds, and altitude is read as logged. Step (v) places the cleaned, trimmed trajectory in a local East–North–Up frame, anchored at its first surviving fix. Resampling and differentiation then act on Cartesian components.

Table 2.8 collects the working thresholds. They are specified independently of the transport exponents; Table 2.7 records their effect on retention.

Status of the numerical snapshot. The census, transport measurements and phase analysis use the archive regenerated with pipeline version 2.3.0. It applies the local paired-barometer requirement, the 10 m s^{-1} vertical-speed threshold and the propagation of reconstructed altitude through derivative-estimation windows. Both inter-fix gaps and intervals between finite altitudes obey the same bridge-or-split bound; longer altitude gaps split the complete trajectory. The checks establish agreement with these numerical rules, not the absence of every recorder error. The witness audit below concerns the isolated version-2.0.1 correction. The targeted comparison reran the legacy and corrected rules on 2,388 candidates: 189 flights changed. Frozen-run counts changed in 188 cases; interior-ground counts changed in 2 cases (the categories can overlap). The corrected witness retained 22,468 additional horizontal fixes; 0 flight admission decisions changed. These counts describe the targeted set, not detector error rates. They do not replace regeneration of the archive and its observables.

2.7.1 Choice of altitude channel

The analysis uses *logged GNSS altitude* throughout: $z = h_{\text{GNSS}}$. This keeps one channel across flights but leaves the recorder’s vertical datum unresolved:

Recorder specification	GNSS height convention
IGC-approved	Ellipsoidal height [7].
CIVL hang-/paragliding	Geoid-referenced height in the consulted document, which contains draft annotations [10].

Compliance has not been established across this historical archive, and the parser preserves the numerical field. Absolute-height comparisons therefore need a datum audit.

A constant height offset cancels from increments; spatially varying geoid corrections and changing sensor biases do not. Pressure altitude also depends on atmospheric pressure and temperature. Using GNSS consistently does not make it uniformly more accurate.

Comparing the barometric and GNSS channels. From 3,000 sampled raw flights per discipline, we retain those with at least 95% pressure-altitude presence and median cadence within 0.25 s of the pooled mode (median intervals rounded to seconds).

Each pair of spectra uses the same longest block of finite, nonzero readings with valid GNSS flags. Duplicate/backward timestamps and gaps above 1.5 modal intervals break the block; ties in fix count select the earliest block. Linear resampling at the modal interval requires at least 256 raw and 256 grid fixes.

Figure 2.1 shows a time-varying channel difference: a constant offset does not align the 30 min traces. The high-frequency median spectra nearly coincide, with broad variation between flights. These comparisons exclude missing-channel runs and do not represent every sampled recorder. Appendix 2.A defines the spectral estimator.

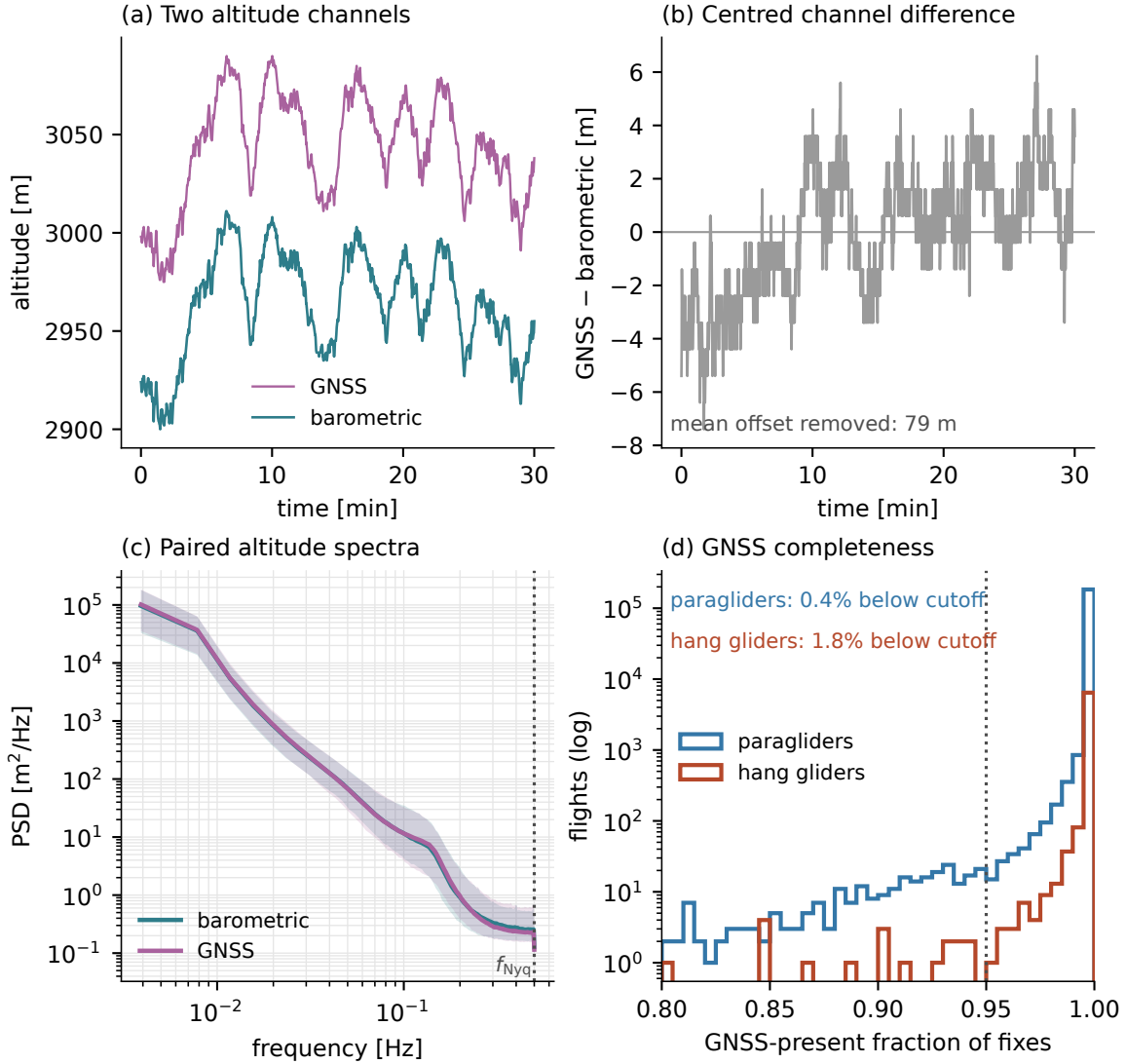


Figure 2.1. Barometric and GNSS altitude. (a) Paired raw channels over 30 min; (b) their demeaned difference. (c) Per-frequency median Welch PSD with the 10–90% range across paired flights, pooled over both disciplines (Appendix 2.A). The band describes flight variability; the dotted line marks $f_{\text{Nyq}} = 1/(2\Delta t)$. (d) GNSS-presence fractions by discipline, shown over 0.80–1.00 with logarithmic counts. The dotted line is the 95% admission threshold.

Variation between flights. Similar medians can conceal different upper tails. We therefore measure each flight’s median PSD in the 0.35 Hz to 0.50 Hz band and compare it with the median of these values across barometric records. Of the 2,008 paired flights contributing to the spectrum, 6.5% carry a GNSS floor above three times the barometric channel’s typical one, and 2.1% above ten times it. Multipath is one possible GNSS error source [11]; this comparison does not attribute the measured excess to a unique physical mechanism.

Sensitivity to the observation interval. A same-flight comparison separates the selection of valid intervals from subsequent trajectory smoothing. Using the same decoder and Welch estimator, we compare a whole-record rule, which can retain missing-value codes and interpolate across recording gaps, with the paired valid-block rule above. On the same 2,008 flights, the 90th percentile of the per-flight GNSS spectral floor decreases from 3.758 to 0.590 m² Hz^{−1}; the corresponding barometric values change from 0.545 to 0.508 m² Hz^{−1}. Here the floor is

each flight’s median PSD over 0.35 Hz to 0.50 Hz; its percentile is distinct from a per-frequency percentile in Fig. 2.1.

Neither comparison uses the pipeline’s Savitzky–Golay output: both use logged altitudes, linear resampling and Welch detrending. The smaller GNSS upper tail therefore reflects the changed observation interval in this comparison. It does not demonstrate lower physical sensor noise. The experiment changes the combined validity, missing-value and gap-selection rule, without identifying the contribution of each criterion separately. These spectra remain conditional on recorders and intervals with adequate paired-channel support.

Availability. A flight needs finite, nonzero GNSS altitude at 95 % of its fixes and an altitude range of at least 30 m. These criteria reject incomplete or nearly constant channels, including genuine flights with too little vertical excursion. No pressure altitude is substituted.

The presence criterion excludes 0.4 % of paraglider and 1.8 % of hang-glider flights. Both channels fall below their presence thresholds in 0.14 % and 1.13 %, respectively. Within admitted flights, missing GNSS altitude initially leaves the horizontal fix intact. At resampling, short altitude holes are reconstructed and flagged; long holes split the full trajectory (Sec. 2.7.6). Missing readings affect 6.1 % of paraglider and 10 % of hang-glider flights, with a median of 5 missing altitudes among affected paraglider flights.

Local barometric support. The pressure channel has a separate role: it can corroborate a suspected frozen GNSS position and an interior ground interval (Secs. 2.7.2 and 2.7.3). Its flight-level eligibility requires at least 95 % nonzero finite readings and a range of at least 30 m. The corresponding archive fractions, 72.1 % and 84.7 %, describe eligibility, not coverage of every candidate interval.

At fix i , define $b_i = 1$ when a finite, nonzero pressure altitude accompanies the horizontal position in the same decoded record, and $b_i = 0$ otherwise. For a candidate interval I , the barometric test is available only if

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i = 1. \quad (2.1)$$

Zero pressure values are treated as missing before duplicate timestamps are merged: the pair $\{0, 1000\}$ therefore supplies one 1,000 m reading. No temporal interpolation supplies a witness. Equation (2.1) requires pressure and position in every record of I , even when GNSS altitude is invalid.

An eligible barometer with incomplete local coverage causes the barometric test to abstain. Flights without an eligible barometer remain in the analysis, subject to the weaker recorder-declaration rule below. Restricting a sensitivity analysis to $B_I = 1$ would measure the effect of this difference in evidence.

2.7.2 Fix-level cleaning

Fix-level cleaning distinguishes three cases (Fig. 2.2):

- An *off-the-trend defect*, such as a horizontal spike, may receive a Hampel flag. The separate impossibility gate decides deletion.
- An *on-the-trend defect*, such as a duplicate timestamp or frozen position, requires timestamp or recorder checks.

- A *vertical defect* invalidates altitude while provisionally retaining the horizontal fix.

These local tests leave slowly varying and plausible-looking errors unresolved. Deletions, boundaries and reconstruction flags record what the pipeline changed.

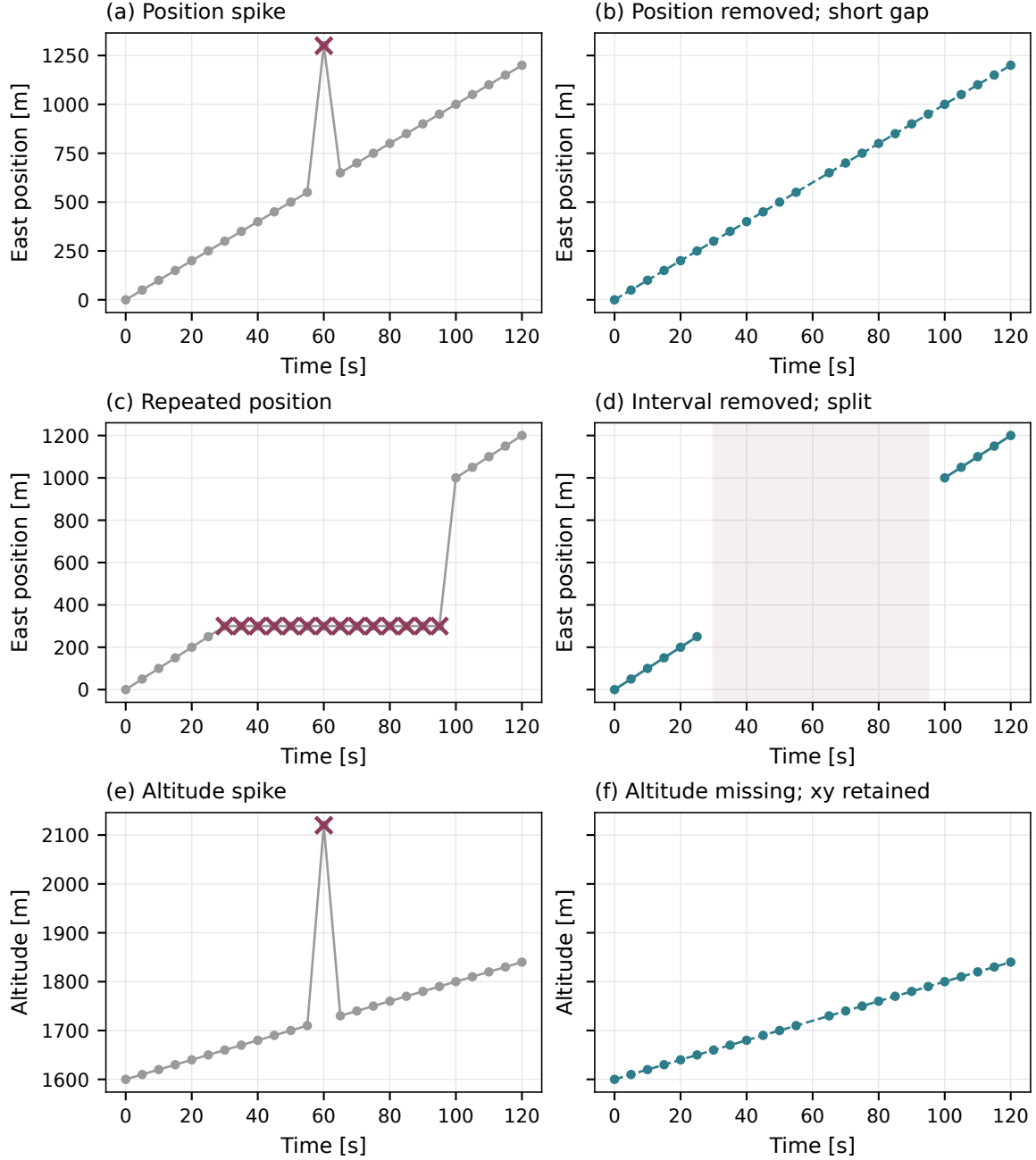


Figure 2.2. Synthetic cleaning examples. Crosses mark suspect readings and dots retained information. A horizontal spike is deleted after the speed/rejoin test; a frozen-position run is removed and split when duration and witness conditions hold; an altitude spike leaves horizontal fixes intact. Dashed paths indicate possible reconstruction at resampling.

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i) , before the Cartesian conversion (Sec. 2.7.5), so distances between fixes are great-circle

(haversine) distances [12] on a sphere of mean radius $R_\oplus = 6,371$ km:

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \quad (2.2)$$

The same distances give step velocities, total path length and extent:

$$\begin{aligned} v_{xy,i} &= \frac{\Delta s_i}{t_{i+1} - t_i}, & v_{z,i} &= \frac{h_{i+1} - h_i}{t_{i+1} - t_i}, \\ L &= \sum_i \Delta s_i, & D &= \max_i \Delta s(\text{fix}_0, \text{fix}_i). \end{aligned} \quad (2.3)$$

Near 45° latitude, the spherical distance error is at most about 0.3%: 3 cm on a 10 m step, below metre-scale GNSS noise. Appendix 2.B derives the ellipsoidal correction and its latitude dependence.

The Hampel identifier. Let the window of fix k be the fixes within $\pm w$ of t_k , and $\tilde{\varphi}_k, \tilde{\lambda}_k$ the component-wise window medians. The residual is the great-circle distance from that median point,

$$r_k = \Delta s((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)), \quad (2.4)$$

with local scale $\text{MAD}_k = \text{med}\{r_j : |t_j - t_k| \leq w\}$, and the fix is flagged when

$$r_k > \max(k \sigma_k, \varepsilon_{\min}), \quad \sigma_k = 1.4826 \text{MAD}_k, \quad (2.5)$$

σ_k a robust local scale [13]. The factor 1.4826 is the usual one-dimensional Gaussian MAD normalization; for these two-dimensional radial residuals it is a scale convention, not an exact standard deviation or a calibrated false-alarm probability.

The working values are $k = 5$, $w = 20$ s and $\varepsilon_{\min} = 15$ m, with at least 5 fixes per window.

Flags contribute to the anomaly count; they never gate deletion. A thermalling circle can depart from its local median while remaining reachable at the absolute speed bound.

Reach, rejoin, or cut. Write a for the last fix the scan accepted, the *anchor*, and measure every candidate against the chord from it,

$$V(a, i) = \frac{\Delta s((\varphi_a, \lambda_a), (\varphi_i, \lambda_i))}{t_i - t_a}. \quad (2.6)$$

An unreachable fix opens a pending block. Its anchor a stays fixed while elapsed time grows. The scan then applies three actions:

Condition	Action
Reachable fix, no pending block	Keep it and advance the anchor.
First rejoin within w	Delete the pending predecessors; keep the rejoin as the new anchor.
No rejoin within w	Keep the block, split at its first fix and re-anchor on the current fix.

The cap starts at the pending block's first fix. It reuses the Hampel half-window without a shared calibration. Gaps left by deletion are bridged or split at resampling (Sec. 2.7.6).

Figure 2.3 compares a spike, a moderate offset and a large offset. The moderate offset becomes reachable as time grows and can cost genuine fixes; the large offset remains unreachable and is split. This gate cannot identify which branch contains the true positions.

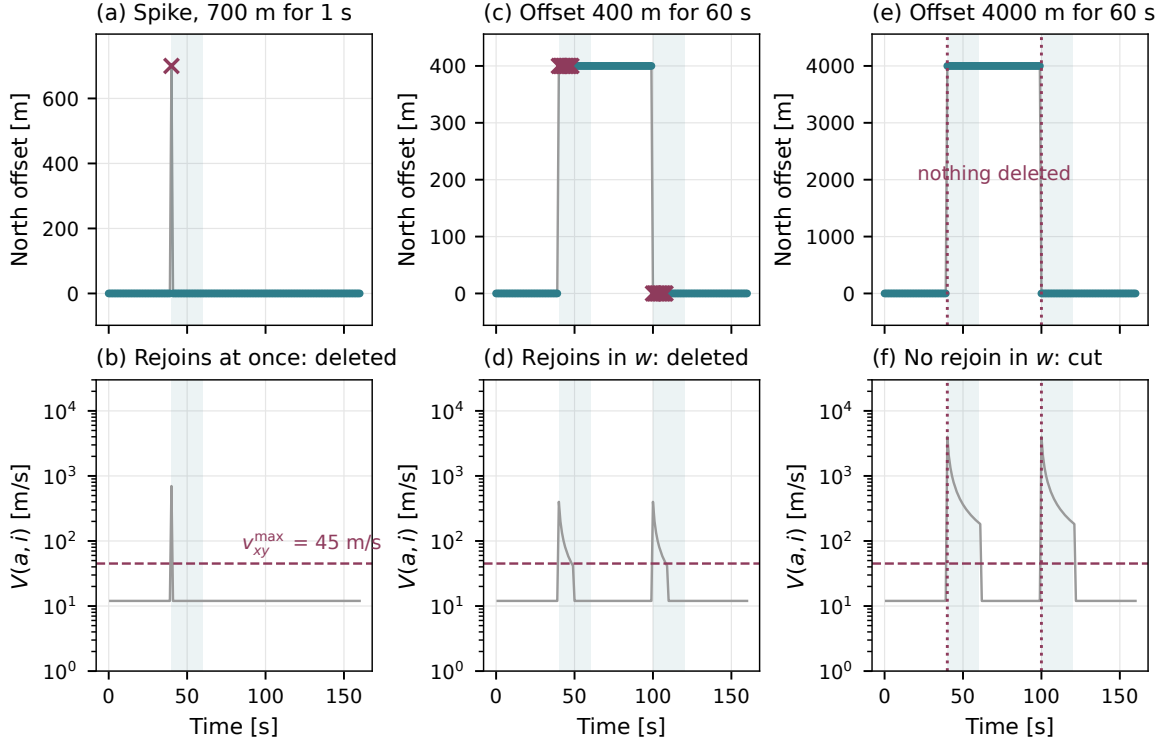


Figure 2.3. Implemented gate on synthetic tracks cruising east at 12 m s^{-1} , sampled every second. Northward offsets are 700 m for one fix (a,b), 400 m for 60 s (c,d), and 4,000 m for 60 s (e,f). Crosses mark deletions, dotted lines boundaries, and shading the block cap w . Lower panels show chord speed from the current anchor and the bound $v_{xy}^{\max}$. The moderate offset removes nine fixes at each end, including genuine ones; the large offset is kept as a separate segment.

Vertical-speed checks. Write the observed altitude as $h_i = z_i + \epsilon_i$. Even a smooth true trajectory produces noisy finite differences,

$$v_{z,i}^{\text{obs}} = \frac{h_{i+1} - h_i}{t_{i+1} - t_i} = \bar{v}_{z,i}^{\text{true}} + \frac{\epsilon_{i+1} - \epsilon_i}{t_{i+1} - t_i}. \quad (2.7)$$

For independent errors of variance σ_z^2 , the differenced noise has variance $2\sigma_z^2/\Delta t^2$. Short cadences therefore amplify altitude error. This calculation illustrates the effect; real GNSS errors may be correlated.

The sustained-excess statistic is

$$\tilde{v}_j = \text{med}\{|v_{z,i}^{\text{obs}}| : |m_i - m_j| \leq w_v, v_{z,i}^{\text{obs}} \text{ finite}\}, \quad m_i = \frac{1}{2}(t_i + t_{i+1}), \quad (2.8)$$

with $w_v = 5 \text{ s}$. The statistic measures local speed magnitude. Each of the $N - 1$ steps is assigned to its midpoint m_i . Figure 2.4 illustrates window selection and the median: for $2r + 1$ finite values, at least $r + 1$ must exceed the bound for the median to do so. For an even count, the two central values are averaged.

With fewer than 5 finite steps, the implementation does not use the median. Writing n_j for that count, the quantity used by the cleaning is

$$u_j = \begin{cases} \tilde{v}_j, & n_j \geq 5, \\ |v_{z,j}^{\text{obs}}|, & n_j < 5. \end{cases}$$

The second branch is the sparse-window fallback. It can occur at a slow recording cadence, near missing altitudes, or at the ends of a record. The sustained-excess rule marks the altitude

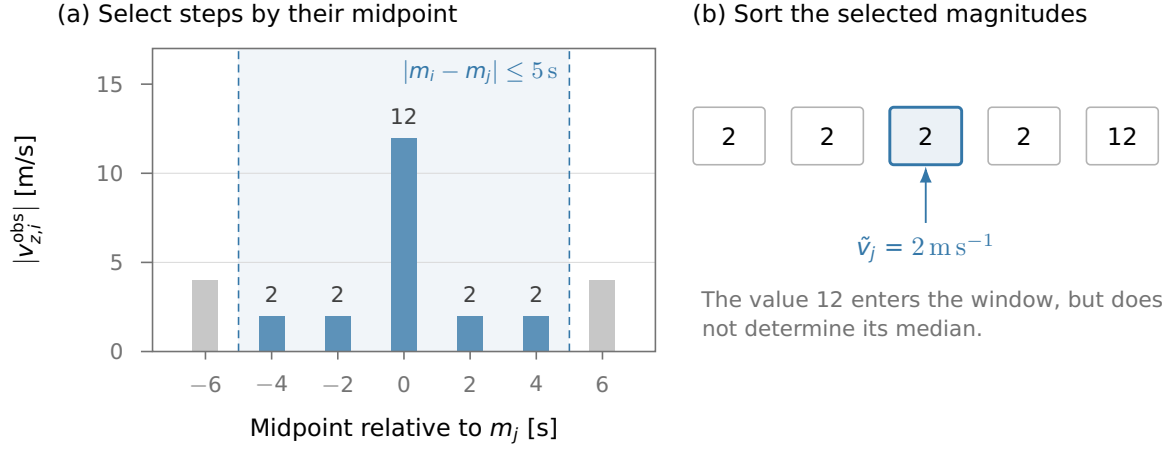


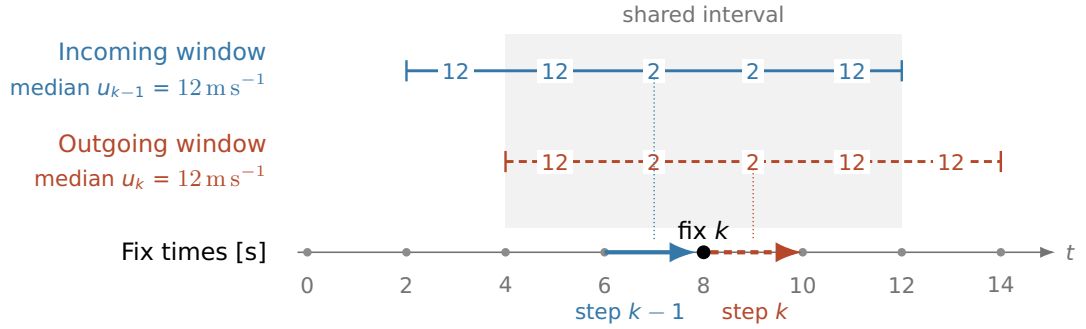
Figure 2.4. How Eq. (2.8) is evaluated. This schematic uses fixes every 2 s and a half-window of 5 s: five step midpoints lie inside it. Grey bars are outside the window and are excluded, regardless of their height. Missing or non-finite speeds are also excluded. Taking absolute values makes an ascent and a descent of equal magnitude contribute equally. These are illustrative values, not flight data.

of an interior fix k as missing when

$$u_{k-1} > v_z^{\max} \quad \text{and} \quad u_k > v_z^{\max}, \quad v_z^{\max} = 10 \text{ m s}^{-1}. \quad (2.9)$$

Both the incoming and outgoing window statistics must exceed the bound strictly. With populated windows, the rule can invalidate altitude even when the two steps touching fix k are below the bound (Fig. 2.5). The overlapping windows locate a neighbourhood of excessive speeds; they do not identify its erroneous altitude independently.

(a) Two windows meet at one decision



(b) Apply the AND condition

u_{k-1} [m/s]	u_k [m/s]	Both $> 10 \text{ m s}^{-1}$?	Result of this rule
2	2	No	Keep altitude
2	12	No	Keep altitude
12	2	No	Keep altitude
12	12	Yes	Mark altitude missing

Figure 2.5. Two overlapping windows determine the altitude decision at fix k . Fixes are 2 s apart and the half-window is 5 s; numbers along the bars are speed magnitudes in m/s. Both medians equal 12 m s^{-1} , although the steps touching k have magnitude 2 m s^{-1} . Equation (2.9) therefore invalidates its altitude at the 10 m s^{-1} bound. The table illustrates the AND condition. Timestamp and horizontal position remain intact.

An isolated spike on a low-speed background leaves a well-supported median low (Fig. 2.6).

A separate out-and-back rule invalidates it when both adjacent steps exceed $v_z^{\max}$ with opposite signs. Sustained excess instead raises the median, whether caused by recorder error or exceptionally strong physical motion.

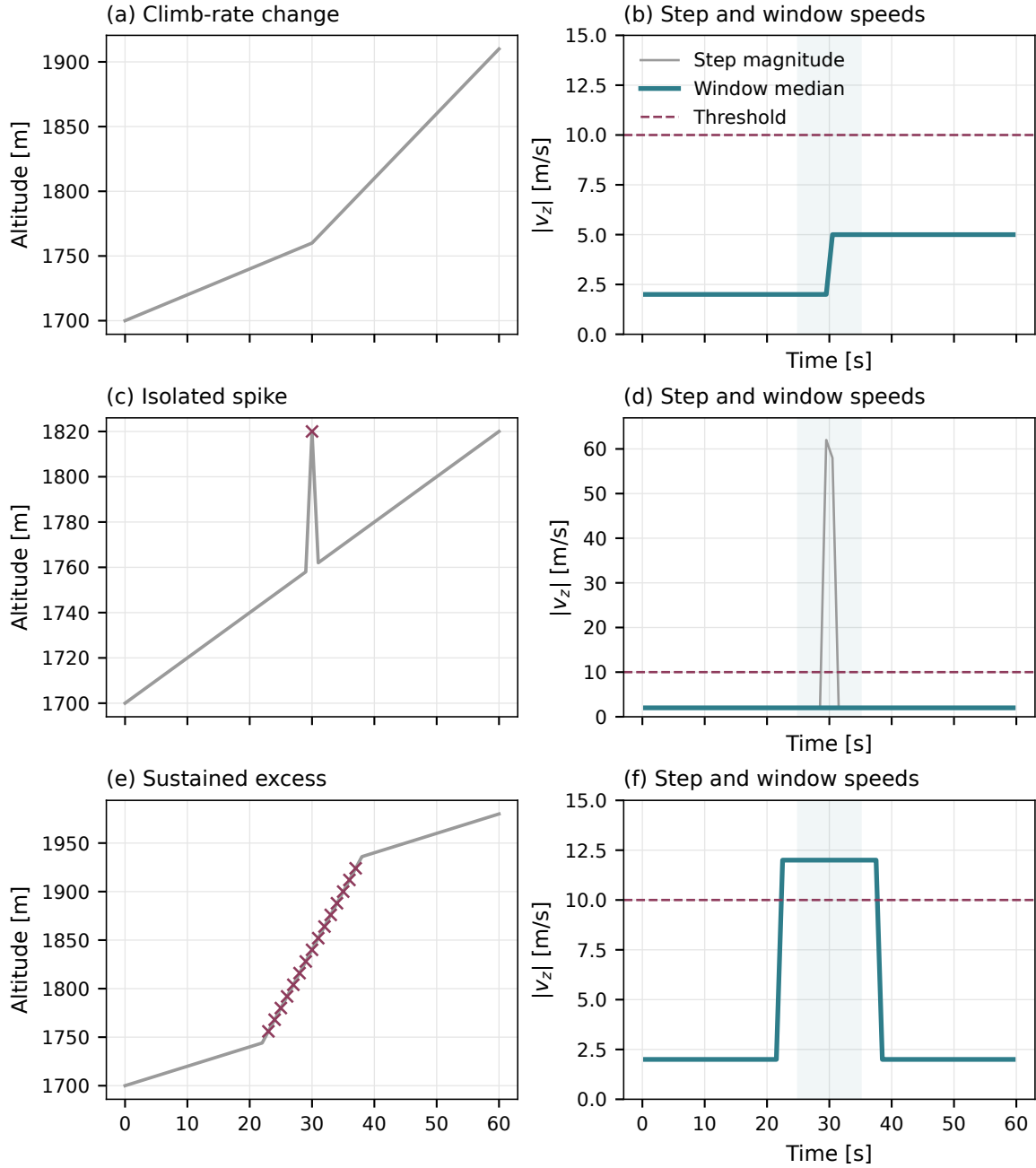


Figure 2.6. The median test at a one-second cadence, using the configured window and speed threshold. Left: synthetic observed altitude. Right: absolute per-step speed (grey), window median (solid), and the bound (dashed). A change between plausible climb rates remains below the bound; an isolated spike barely affects the median and is handled by the separate spike rule; a sustained excessive interval raises the median. Crosses mark altitudes invalidated by the actual implementation.

Unreturned vertical steps. An isolated jump without a prompt return can escape both checks. Define

$$e_i = 1\{|v_{z,i}^{\text{obs}}| > v_z^{\text{max}}\}, \quad J_i = |h_{i+1} - h_i|, \quad (2.10)$$

$$d_i^{\text{return}} = \min_{\substack{j=i+1, \dots, i+5 \\ h_j \text{ finite}}} |h_j - h_i|. \quad (2.11)$$

The minimum uses the next five fix positions, fewer at the record end, and equals $+\infty$ if none has finite altitude. Set $e_i = 0$ outside the observed steps. The counter `n_alt_level_shift` records candidates satisfying

$$e_i = 1, \quad e_{i-1} = e_{i+1} = 0, \quad d_i^{\text{return}} > 0.3J_i. \quad (2.12)$$

This short return check does not establish a permanent offset. It runs before trimming and segment selection, and can overlap other altitude rules.

Among retained-flight metadata, 47,440 paraglider and 1,350 hang-glider records contain at least one candidate. Reprocessing eight previously selected examples with the current pipeline reproduces their counters. In each discipline these include two high-count records and two seeded positive-count records. All eight still have candidate times near usable phase-feature centres. This is temporal association, not proof that an uncorrected sensor defect enters a feature. These counts describe candidates before later cuts, rather than surviving confirmed defects. Automatic repair is not applied: a level shift and a real large increment can have similar signatures. Their effect on phase features still requires a sensitivity analysis.

Figure 2.7b motivates the working bound 10 m s^{-1} : around 9 m s^{-1} to 10 m s^{-1} , the density begins to decay more slowly and the paraglider curve crosses the hang-glider curve. The transition is gradual. The bound is an empirical cleaning choice, so extreme genuine descents can be invalidated too. In the full diagnostic sample, 0.069 % of window medians exceed it. This exceedance fraction differs from the removal rate, which also depends on neighbouring windows and the spike rule.

Deletion and mandatory boundaries. Altitude rules retain timestamp and horizontal coordinates; position and timestamp rules may remove the whole fix. Reconstruction is deferred to resampling. After all cleaning checks, every surviving step above the horizontal speed bound becomes a mandatory segment boundary, counted as `n_boundaried`. This also catches an impossible return from a block that the forward scan kept.

The forward scan does not re-evaluate its initial anchor, so an erroneous first fix can survive this local procedure and displace the origin of the later Cartesian frame. A separate flight-level test therefore rejects a track containing a fix beyond the distance reachable from its first retained position (Sec. 2.7.4). The reach test acts on spherical distances before projection and smoothing, with a 50 m tolerance. The written Cartesian trajectories undergo separate diagnostic checks (Sec. 2.9); those operations do not preserve the reach inequality as an exact invariant.

Frozen-position runs and the evidence for GNSS lock loss. A logger may repeat its last position after losing GNSS lock while timestamps continue. An unremoved run then resembles a period of spatial confinement. The detector uses three joint conditions,

$$D_I < \delta_{xy}, \quad t_{\text{max } I} - t_{\text{min } I} \geq \tau_{\text{freeze}}, \quad W_I = 1, \quad (2.13)$$

where D_I is the great-circle diagonal of the latitude–longitude bounding box of candidate run I , evaluated after longitude unwrapping. This is the implemented extent proxy, not an exact all-pairs diameter. The working values are $\delta_{xy} = 15 \text{ m}$ and $\tau_{\text{freeze}} = 60 \text{ s}$.

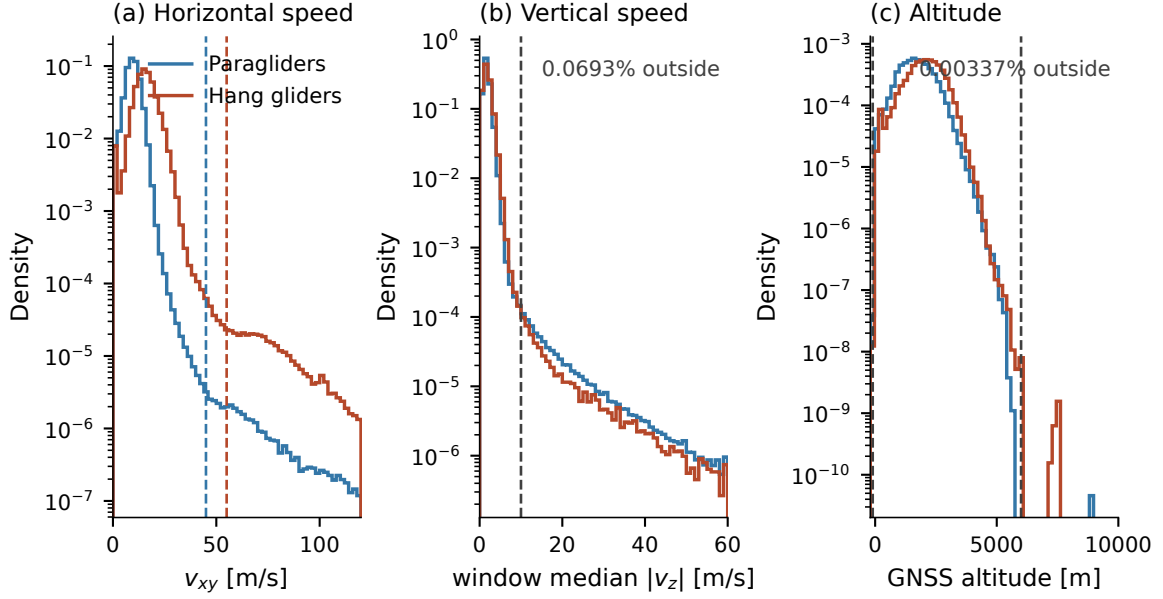


Figure 2.7. Raw diagnostics for the absolute bounds: (a) horizontal step speed, (b) windowed GNSS med $|v_z|$, and (c) GNSS altitude. Dashed lines mark the working thresholds. Densities pool a large paraglider subsample and the complete hang-glider population, using the full finite-sample denominator even where the displayed range cuts off a tail. Logarithmic ordinates expose rare values; abscissae are linear. Annotations count bound exceedances, while actual deletion and invalidation also depend on the rejoin and neighbouring-window rules.

Let E_I denote exact equality of both coordinate fields throughout the run, d_I the fraction of its fixes with a V flag or missing GNSS altitude, and Δb_I the median pressure altitude of the final quarter minus that of the first quarter. With B_I defined in Eq. (2.1), the witness is

$$W_I = E_I \vee \begin{cases} |\Delta b_I| < \delta_z, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1, B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0, \end{cases} \quad \delta_z = 5 \text{ m}. \quad (2.14)$$

Removed frozen-lock runs always split the trajectory. Exact coordinate repetition is a separate witness; otherwise incomplete local pressure coverage forces abstention.

The signatures have limits. Quantized coordinates can repeat for a stationary receiver, and sparse fixes do not establish continuous stillness between them. Small net pressure change can hide an excursion; a barometer can also continue climbing during a horizontal GNSS freeze. Shared logger failures can affect both channels.

For scale, a coordinated turn in still air has

$$R = \frac{v_{\text{air}}^2}{g \tan \phi}, \quad T_{\text{turn}} = \frac{2\pi R}{v_{\text{air}}}. \quad (2.15)$$

At $v_{\text{air}} = 10 \text{ m s}^{-1}$ and bank angle $\phi = 35^\circ$, $2R \simeq 29 \text{ m}$ exceeds δ_{xy} , while a sustained 1 m s^{-1} climb changes pressure altitude far more than δ_z . These are reference scales, not universal minimum diameters or guarantees for every wind, bank angle and recorder cadence.

A genuine small manoeuvre can satisfy all signatures and be cut; a jittering freeze without a witness can survive. Runs shorter than τ_{freeze} are retained. Independent inspection or labelled injections are needed to estimate these error rates.

Defect	Detected by	Action
Off the trend — <i>horizontal position defects</i>		
Position spike, out-and-back jump	Unreachable block with a rejoin within w (Eq. 2.6). The Hampel flag [14] only records local disagreement.	delete block; apply gap rule at resampling
Position discontinuity (re-acquisition offset)	No reachable rejoin within the block cap w	keep fixes; split at the discontinuity
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — <i>record structurally wrong, horizontal position agrees with its neighbours</i>		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	Spatial collapse, minimum duration and witness (Eqs. 2.13–2.14)	remove run; split trajectory
Out of range — <i>horizontal fix intact; altitude unavailable or suspect</i>		
Missing / out-of-band altitude, sustained $ v_z $ excess, or $ v_z $ spike	Absent reading; outside $[-100, 6000]$ m; both adjacent window statistics above $v_z^{\max}$ (Eq. 2.9); or an opposite-sign spike pair above that bound	invalidate altitude; keep horizontal fix
Policy: altitude-only invalidation retains the timestamp and horizontal coordinates. Position and time rules may remove complete fixes. These are operational decisions, not known sensor truth. At resampling, short altitude holes are reconstructed and long ones exclude the full interval and split the trajectory (Sec. 2.7.6).		

Table 2.5. Fix-level decisions. Thresholds appear in Table 2.8; Fig. 2.7 shows the absolute-bound diagnostics. Horizontal flags, deletions and altitude invalidations remain separate operations.

Integrity and validation. The integrity gate rejects flights whose cleaning burden exceeds 10 %. Counts use only the airborne window selected by trimming (Sec. 2.7.3), so discarded ground recordings do not affect this decision. Let N_{del} count deleted fixes, N_{keep} surviving fixes and N_{zbad} survivors whose altitude was invalidated. The burden is

$$f_{\text{clean}} = \frac{N_{\text{del}} + N_{\text{zbad}}}{N_{\text{keep}} + N_{\text{del}}}. \quad (2.16)$$

Altitude invalidation includes out-of-band values, sustained excess, spikes and V declarations. Duplicate-second merges and later reconstruction are excluded. This gate removes 550 paraglider and 53 hang-glider flights. The larger loss occurs later when no segment survives resampling: 20,798 and 193 flights, respectively.

Removal fractions measure the amount changed; threshold sweeps measure sensitivity. Classification error rates require independent labels. Synthetic regressions and the real examples in Fig. 2.8 check specific failure modes, while a representative labelled archive audit remains to be done.

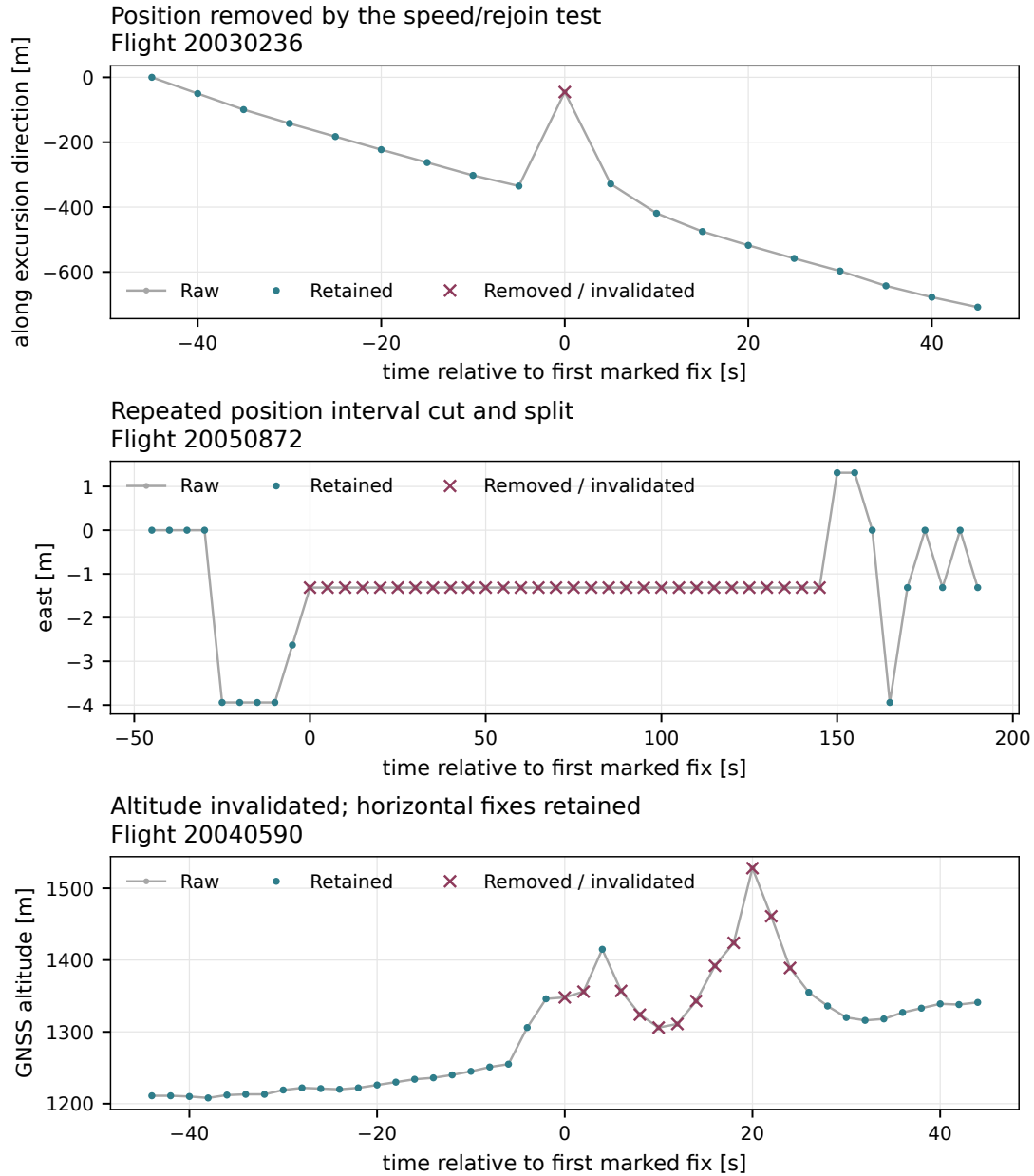


Figure 2.8. Short examples extracted from the interior of raw flight records and reprocessed with the current cleaning implementation. Top: the horizontal coordinate projected along the detected excursion direction, making the removed spike visible. Middle: a collapsed-position interval; crossed marks are removed and the surviving sides split. Bottom: GNSS altitude invalidation, with all horizontal fixes retained. Flight identifiers and plotted raw values are saved with the figure. The examples document detector decisions; no independent sensor truth is available for these intervals.

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis, using the horizontal speed v_{xy} alone. With $v_0 = 5 \text{ m s}^{-1}$ and $T_0 = 30 \text{ s}$ (Table 2.8), the retained airborne segment is $[t_{\text{on}}, t_{\text{off}}]$ with

$$t_{\text{on}} = \min\{t : v_{xy} > v_0 \text{ on } [t, t + T_0]\}, \quad t_{\text{off}} = \max\{t : v_{xy} > v_0 \text{ on } [t - T_0, t]\}, \quad (2.17)$$

the same sustained-speed rule read forward and time-reversed. Here v_{xy} is the speed inferred on each observed step, not a continuously measured velocity. Interior slow periods do not define these outer boundaries, although slow flight at either end can be clipped. A flight with no sustained stretch at all is discarded.

The persistence time T_0 reduces sensitivity to brief speed excursions. It does not bound take-off or landing error by T_0 : a sustained fast ground movement can be retained, while a long airborne interval below v_0 at either end can be removed. The trimmed fraction is recorded per flight, and such cases require direct inspection.

The median time removed before onset and after landing is 33 s and 71 s for paragliders, and 19 s and 62 s for hang gliders (Fig. 2.9). Counts cover 185,131 and 6,580 flights reaching a defined window. The stored `trimmed_fraction` combines the two ends. Longer post-landing intervals are consistent with continued logging after touchdown.

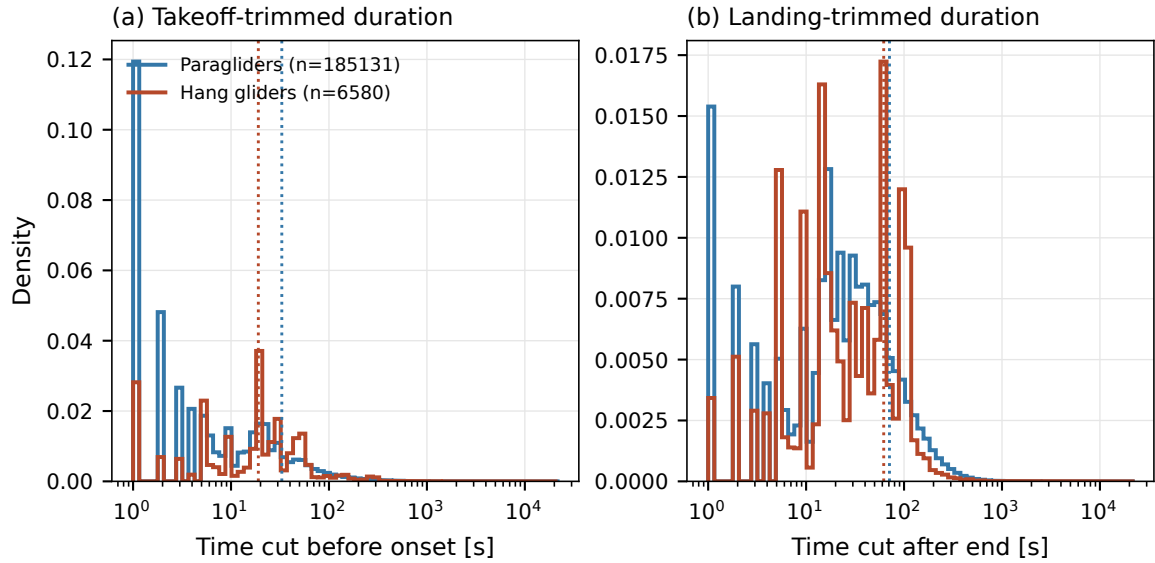


Figure 2.9. Time removed (a) before estimated onset and (b) after estimated landing, with one observation per flight per panel. Dotted lines mark discipline medians. The sustained-speed interval T_0 does not bound the time removed; the tails extend to hours.

Trimming also moves the first altitude of the track. The launch altitude z_0 used to classify flights by orography in Section 3.2.2 is the GNSS altitude of the first fix of the trimmed track, not of the raw log. Let z_{raw} be the first finite GNSS altitude of the raw log and z_{trim} that of the trimmed track, so that $z_0 = z_{\text{trim}}$. Typically the two are close: the median of $|z_{\text{trim}} - z_{\text{raw}}|$ is 7.0 m for paragliders and 3.0 m for hang gliders, and 57.7 % and 77.4 % of flights differ by at most 10 m (Fig. 2.10). The tail is long: the difference exceeds 100 m in 9.9 % of paraglider and 2.5 % of hang-glider flights, and 500 m in 1.6 % and 0.7 %. These differences are almost always positive (97 % and 90 %), and the time cut before onset is long for them (medians 442 s and 295 s, against 33 s and 19 s overall). The likely cause is a logger switched on at the bottom of

the site and carried up to the launch by road, lift or on foot: the raw log then starts at the valley, and trimming removes that approach and starts at the launch. We inferred this from these statistics and a small sample of flights, not from inspecting each one. This is why z_0 is taken from the trimmed track. Using z_{raw} instead of z_{trim} would place 6.1 % of paraglider and 1.9 % of hang-glider flights in a different altitude class of Section 3.2.2.

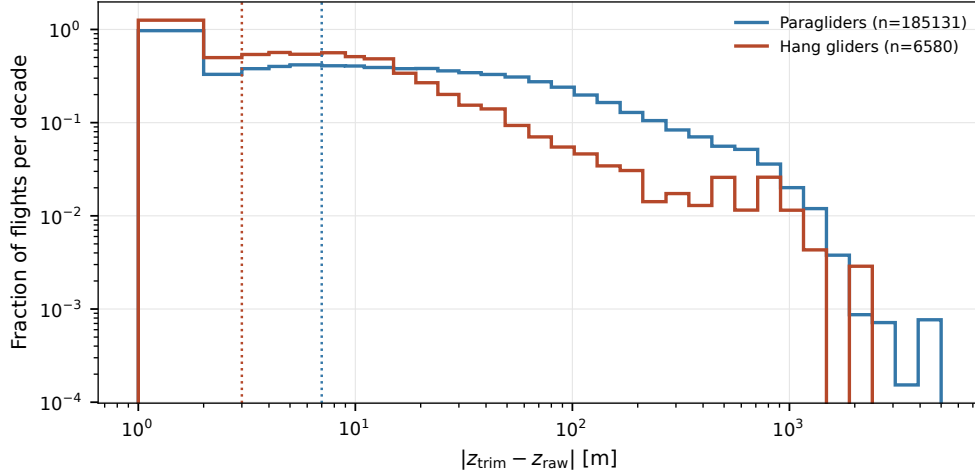


Figure 2.10. Absolute difference between the GNSS altitude of the first fix of the trimmed track and that of the first fix of the raw log, one observation per flight. Bins are logarithmic in whole metres, and the vertical axis is the fraction of flights per decade. Differences of 0 m and 1 m share the first bin, and values above 5,000 m are drawn in the last one. Dotted lines mark discipline medians.

Interior ground intervals. A landing and relaunch in one log can imitate a long waiting episode. The detector requires sustained low horizontal speed on an interval I containing at least three fixes, all with paired barometry. Fit $h_i^{\text{baro}} = \hat{\alpha} + \hat{\beta}t_i + r_i$ by least squares. The joint conditions are

$$\begin{aligned} v_{xy} < v_0 \text{ on } I, \quad |I| &\geq T_{\text{ground}}, \quad B_I = 1, \\ Q_{.95}(r) - Q_{.05}(r) &\leq \delta_z^{\text{gr}}, \quad |\hat{\beta}| \leq \delta_z / \tau_{\text{freeze}}, \end{aligned} \quad (2.18)$$

where $|I|$ denotes duration, $T_{\text{ground}} = 10$ min and $\delta_z^{\text{gr}} = 5$ m. The residual span tolerates isolated extremes; the slope bound prevents detrending from erasing a steady climb. Without complete eligible barometry the detector abstains, since slow ground-relative flight can otherwise resemble a landing.

Excised intervals split the trajectory. Subsequent flight-level cuts apply to the parent flight, summing duration and path length over its remaining blocks; they are not reapplied separately to each segment.

Interior excisions occur in 231 paraglider flights (0.12 %) and 3 hang-glider flights (0.05 %). Most affected flights have one excision; the maxima are 5 and 2 (Fig. 2.11).

Shorter intervals satisfying the same speed and barometric conditions are *flagged* when their duration reaches 60 s, the configured minimum passed to this stage. They remain in the trajectories. The stored flags support a future sensitivity analysis with and without these observations; that comparison has not yet been carried out.

2.7.4 Flight-level filtering

The cuts below define the analysed population. They act on the trimmed parent flight, with duration T_{rec} and path L accumulated only within contiguous blocks. Forced boundaries and

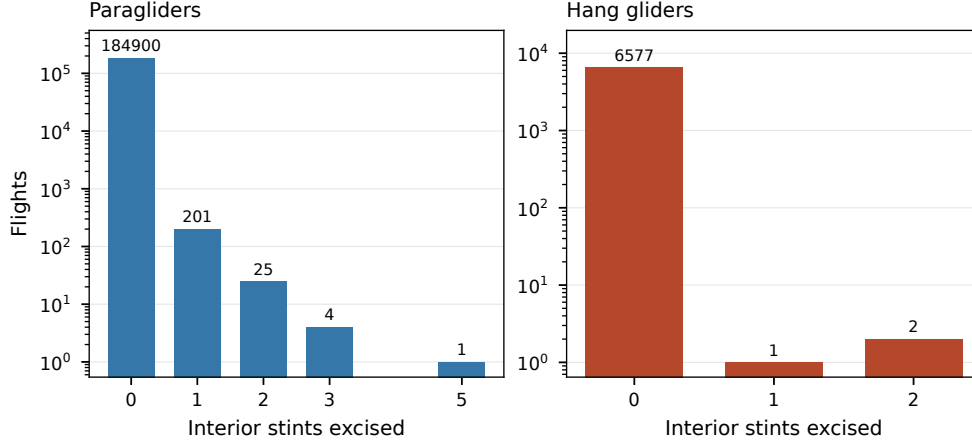


Figure 2.11. Interior intervals excised per flight among all flights reaching this stage. Logarithmic counts expose the rare nonzero cases. Empty bars denote counts absent from the archive.

gaps longer than $g_{\max}$ contribute to neither sum; elapsed span T_{span} includes them.

Quantity	Condition	Purpose
Recorded duration	$T_{\text{rec}} \geq 40 \text{ min}$	Exclude short records.
Elapsed span	$T_{\text{span}} \leq 16 \text{ h}$	Screen long recording sessions.
Path length	$L \geq 20 \text{ km}$	Require horizontal travel.
GNSS altitude range	$\max z - \min z \geq 600 \text{ m}$	Require vertical excursion.
Mean speed	$L/T_{\text{rec}} \leq v_{xy}^{\max}$	Check consistency with step bounds.

These are sample restrictions, not a soaring classifier: a genuine flight with little vertical excursion can fail. Mean speed adds no independent evidence when all included steps already satisfy the bound. Figure 2.12 shows sample-size sensitivity; a plateau does not establish unbiased selection.

Raw records near the cuts. The raw census contains 9 paraglider records spanning 16 h to 166 h, and no overlong hang-glider records. Among raw records above the duration floor, the path cut excludes

	Below path floor	Above duration floor
Paragliders	236	184,541
Hang gliders	7	6,638

These raw counts are distinct from the sequential post-trimming removal counts.

Reach from the first fix. Before projection and smoothing, every fix must satisfy

$$\Delta s(\text{fix}_0, \text{fix}_i) \leq v_{xy}^{\max} t_i + \varepsilon_{\text{reach}}, \quad \varepsilon_{\text{reach}} = 50 \text{ m}, \quad \forall i, \quad (2.19)$$

with t_i measured from the first trimmed fix. Testing each elapsed time prevents a later long duration from excusing an early impossible displacement. It also catches a corrupt initial anchor whose first step is no longer summed into L .

This rejects 217 paraglider and 5 hang-glider flights in the stored snapshot. The written Cartesian trajectories receive separate checks with operational tolerances (Sec. 2.9).

Unclassified equipment remains in pooled analyses but is excluded from class-resolved comparisons (Sec. 2.4). Sampling regularity is tested after resampling (Sec. 2.7.6).

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates: resampling and smoothing operate on the coordinates *componentwise* (Sec. 2.7), and latitudes and longitudes cannot play that role, being angles on a curved surface whose componentwise differences are not displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6,378,137$ m, flattening $f = 1/298.257223563$, $e^2 = f(2 - f)$) [15, 16],

$$\begin{aligned} N(\varphi) &= \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, & X &= (N + h) \cos \varphi \cos \lambda, \\ & & Y &= (N + h) \cos \varphi \sin \lambda, \\ & & Z &= (N(1 - e^2) + h) \sin \varphi, \end{aligned} \quad (2.20)$$

$N(\varphi)$ the prime-vertical radius of curvature and h the ellipsoidal height along the normal, derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. WGS84 reports geodetic latitude φ , measured along the ellipsoid normal. Geocentric latitude ψ follows the radius from the Earth's centre (Fig. 2.13). Their difference is approximately

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad \max |\varphi - \psi| \simeq 0.19^\circ.$$

Confusing them can shift a position by about 20 km (Appendix 2.B). Equation (2.20) already takes the geodetic latitude recorded in IGC; no latitude conversion is needed.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin — close to but not the take-off point, since the ground phase has been removed (Sec. 2.7.3) — the ECEF displacement $\Delta \mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix} \Delta \mathbf{X}, \quad (2.21)$$

The rotation is orthogonal with unit determinant: it preserves the three-dimensional ECEF chord and angles (Appendix 2.B). Keeping only (E, N) then projects the chord onto the tangent plane.

For a point at surface distance d from the origin on a sphere,

$$r_{EN} = R_\oplus \sin(d/R_\oplus), \quad d - r_{EN} = \frac{d^3}{6R_\oplus^2} + O(d^5/R_\oplus^4). \quad (2.22)$$

The horizontal shortening is about 0.03 m at 20 km and 0.51 m at 50 km; it reaches about 33 m at 200 km. The often-used $d^3/(24R_\oplus^2)$ expression instead compares an arc with its full

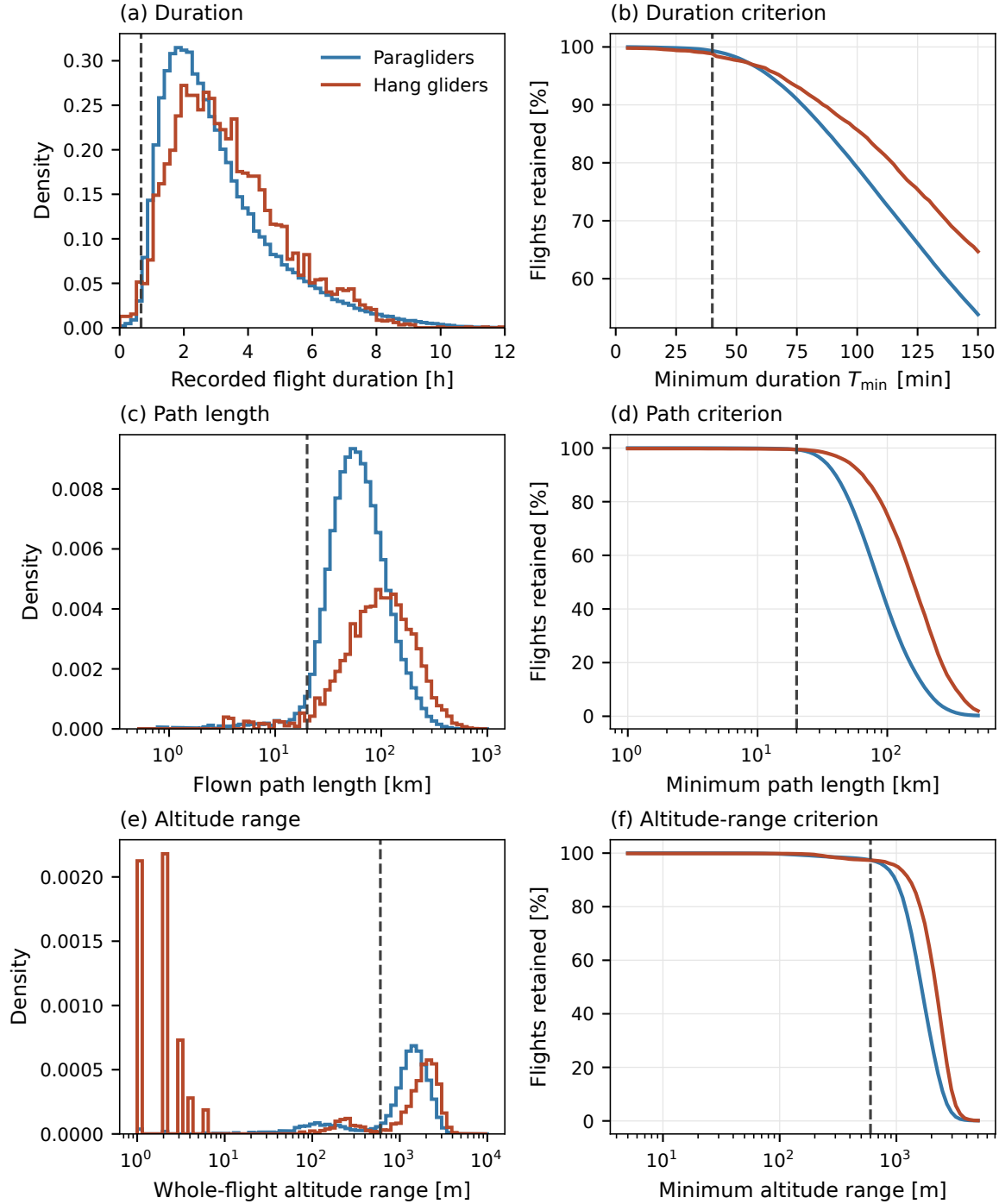


Figure 2.12. Raw flight-level diagnostics by discipline. Left column: (a) recorded duration, (c) flown path length and (e) GNSS altitude range. Right column: retention under the corresponding single criterion, evaluated separately on these raw summaries. Dashed lines mark the working thresholds ($T \geq 40$ min, $L \geq 20$ km and altitude range ≥ 600 m). The altitude panels require the configured minimum GNSS coverage. Histogram densities retain their full finite-sample denominators when tails extend outside the displayed range. These marginal raw diagnostics are not the retention fractions of the sequential cleaned-flight cuts.

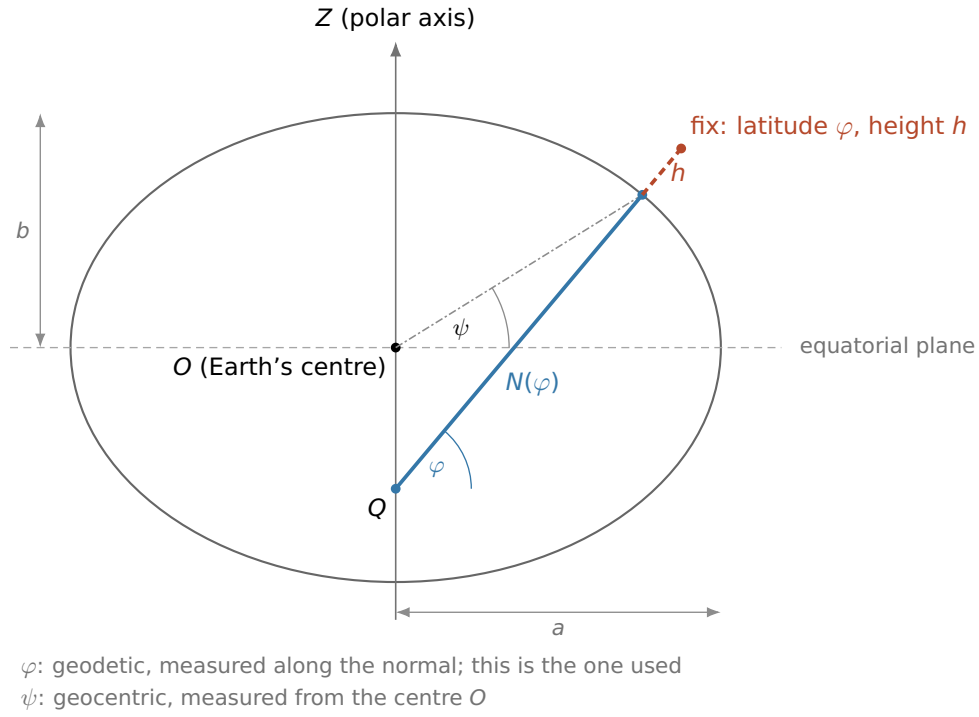
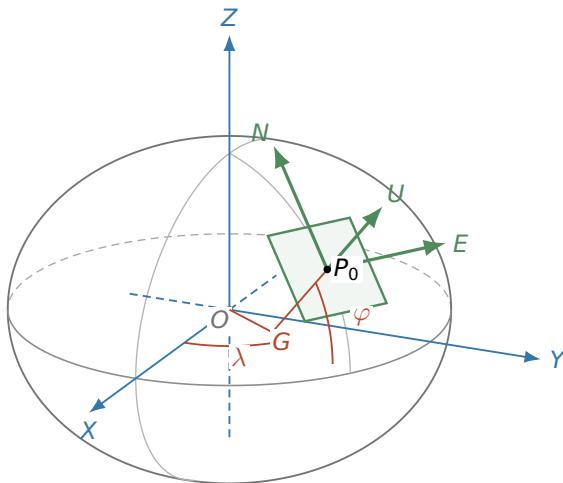


Figure 2.13. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b ; the geodetic latitude φ , the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O , so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.14.

(a) The global frame and the two angles



(b) The tangent frame, close up

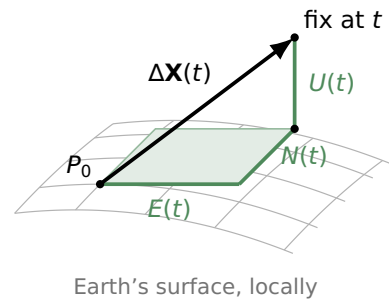


Figure 2.14. The local ENU frame. (a) The origin P_0 is the first fix of the trimmed track. Longitude λ and geodetic latitude φ orient its green unit vectors within the blue ECEF frame. Up follows the ellipsoid normal through G , with flattening exaggerated. (b) A later fix's displacement is resolved into E , N and U : the box edges project its space diagonal.

three-dimensional chord. Neither estimate is a uniform error bound for every pair of fixes in a fixed frame.

The height input is the logged GNSS altitude, whose datum has not been harmonized. Its use as ellipsoidal height introduces a horizontal error of order

$$|\delta \mathbf{r}_{EN}| \sim |\delta h| \frac{d}{R_{\oplus}}.$$

For example, a 100 m height error at 50 km contributes about 0.8 m. This illustrates the scale; it is not an archive-wide bound.

The vertical coordinate is $z(t) = h_{\text{GNSS}}(t)$, without subtracting the origin altitude. The rotated component U also contains Earth curvature and is not used for vertical dynamics. Velocities depend on altitude increments; absolute heights retain the datum limitation of Sec. 2.7.1.

2.7.6 Uniform sampling within a flight

The smoothing and derivative estimates in the next section require equally spaced observations. Each flight keeps its native interval, $\Delta t = \text{median}_i(t_{i+1} - t_i)$: a flight recorded every second stays at one second, and a flight recorded every five seconds stays at five seconds. Figure 2.15 shows these intervals in the archive.

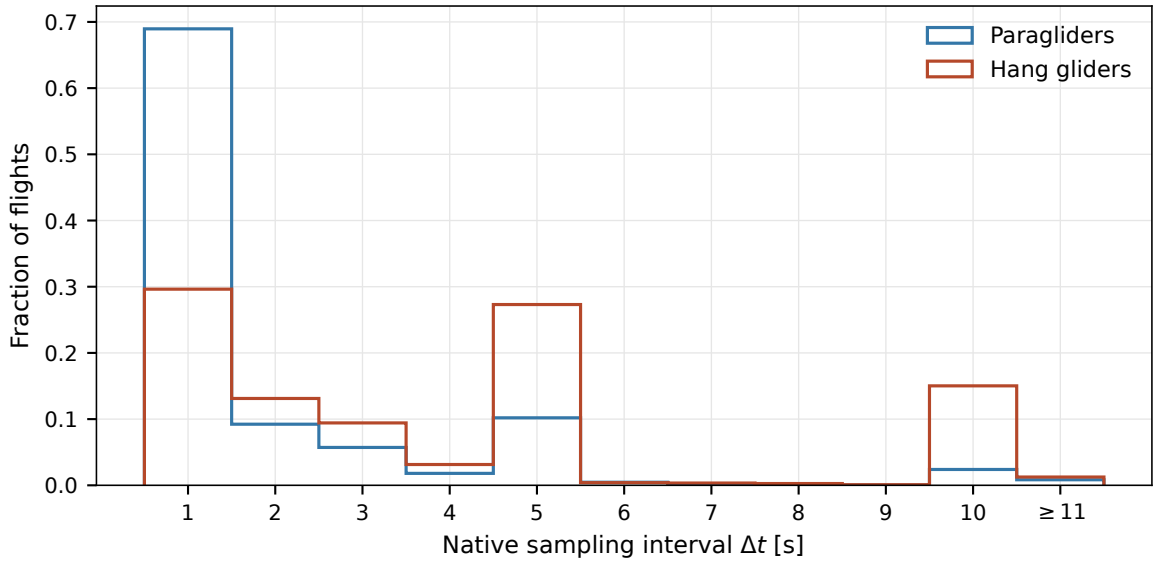


Figure 2.15. Native sampling interval per flight, measured as the median inter-fix time in the raw archive. Bars give the fraction of flights in one-second bins; the last bin also includes intervals beyond 11 s. Paragliders concentrate at 1 s, while hang gliders are spread across several cadences. Resampling preserves each flight’s native interval.

Fill a short gap; split at a long one. For a gap $g = t_{i+1} - t_i$ between surviving fixes, interpolation is allowed when $g \leq g_{\text{max}}$, with

$$g_{\text{max}} = \min(10 \Delta t, \max(20 \text{ s}, 2 \Delta t)). \quad (2.23)$$

At fast cadences the relative limit binds; at slower cadences the absolute cap binds, with a $2 \Delta t$ floor allowing one missed fix. Thus the configured 20 s value is not an absolute ceiling for

every logger: at $\Delta t = 15$ s, the bound is 30 s. For example,

$$\Delta t = 1 \text{ s} \Rightarrow g_{\max} = 10 \text{ s}, \quad \Delta t = 5 \text{ s} \Rightarrow g_{\max} = 20 \text{ s}.$$

A longer gap starts a new segment. The same bound applies between consecutive finite altitude readings, even with horizontal fixes in between. Cleaning boundaries always force a split. The cap limits reconstruction but cannot ensure that manoeuvres inside short gaps are resolved.

Local examples: $\Delta t = 5$ s, $g_{\max} = 20$ s

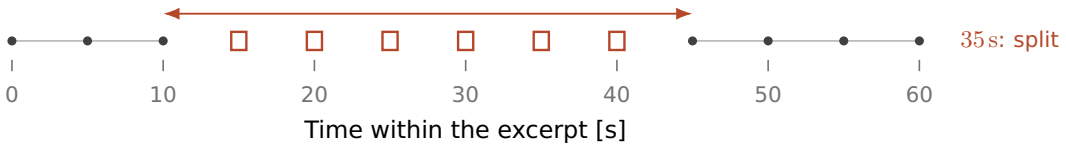
(a) Short gap between fixes



(b) Long gap between fixes



(c) Fixes present, altitude missing



• retained fix ○ time grid filled □ excluded fix, missing altitude

Figure 2.16. Three local examples of resampling. Circles mark new grid times in a short gap; a long gap leaves separate segments. In (c), squares mark existing horizontal fixes without valid altitude. The separation between valid altitude endpoints exceeds $g_{\max}$, so those fixes are excluded and the full trajectory is split.

Reconstructing the coordinates. Within segment s , the grid is $t_k^{(s)} = t_{\text{start}}^{(s)} + k\Delta t$, ending at or before its last fix. An already uniform, complete series is unchanged. Otherwise, E and N are evaluated with a shape-preserving cubic interpolant (PCHIP) [17], and z with linear interpolation. Linear filling avoids overshoot in altitude but can smooth away a climb–glide transition. Missing-altitude prefixes and suffixes are excluded before the grid is built; altitude is never extended beyond finite support.

For altitude missing at existing fixes, let g_z separate the surrounding finite altitude readings. The rule is

$$\begin{cases} g_z \leq g_{\max} : & \text{fill altitude linearly,} \\ g_z > g_{\max} : & \text{exclude intervening fixes and split the full trajectory.} \end{cases}$$

It applies to both missing raw readings and values invalidated by cleaning (Fig. 2.16c). Unsupported intervals remain recorded as rejected segment candidates. Sacrificing their horizontal coverage avoids constructing long unsupported three-dimensional paths.

Three flags record reconstruction for subsequent phase analysis:

Flag	Marked grid times
<code>interpolated</code>	No surviving fix within $\Delta t/2$.
<code>z_reconstructed</code>	The above, plus times whose nearest fix has no altitude.
<code>z_derivative_reconstructed</code>	Derivative windows touched by altitude reconstruction.

The flags identify grid times or derivative windows touched by reconstruction; numerical completeness alone does not establish accuracy. The altitude-gap bound concerns the separation of finite source readings, not the duration of a consecutive run of flags on the new grid. With irregular source timing, nearest-fix flagging can merge adjacent reconstructed portions into a longer flagged run even when every finite-altitude bracket satisfies the bound.

Which segments are retained. A segment must span at least 90 s, have at most 10 % of its grid times flagged as *interpolated*, and yield finite values in all three coordinates. The temporal-coverage fraction and the altitude-gap bound are separate criteria; the latter is applied before interpolation. A rejected segment leaves the other segments available; the parent's flight-level cuts are not reapplied.

Segments retain the flight's spatial origin and elapsed time. Time-lagged displacement pairs and analysis windows are formed within segments. Splitting therefore changes the available coverage and can truncate flight phases; it must be accounted for when comparing observables. Figure 2.17 describes gaps in the raw archive; observables that depend on the recording interval, such as fix-to-fix turning angles, require a common temporal scale or a comparison within cadence groups.

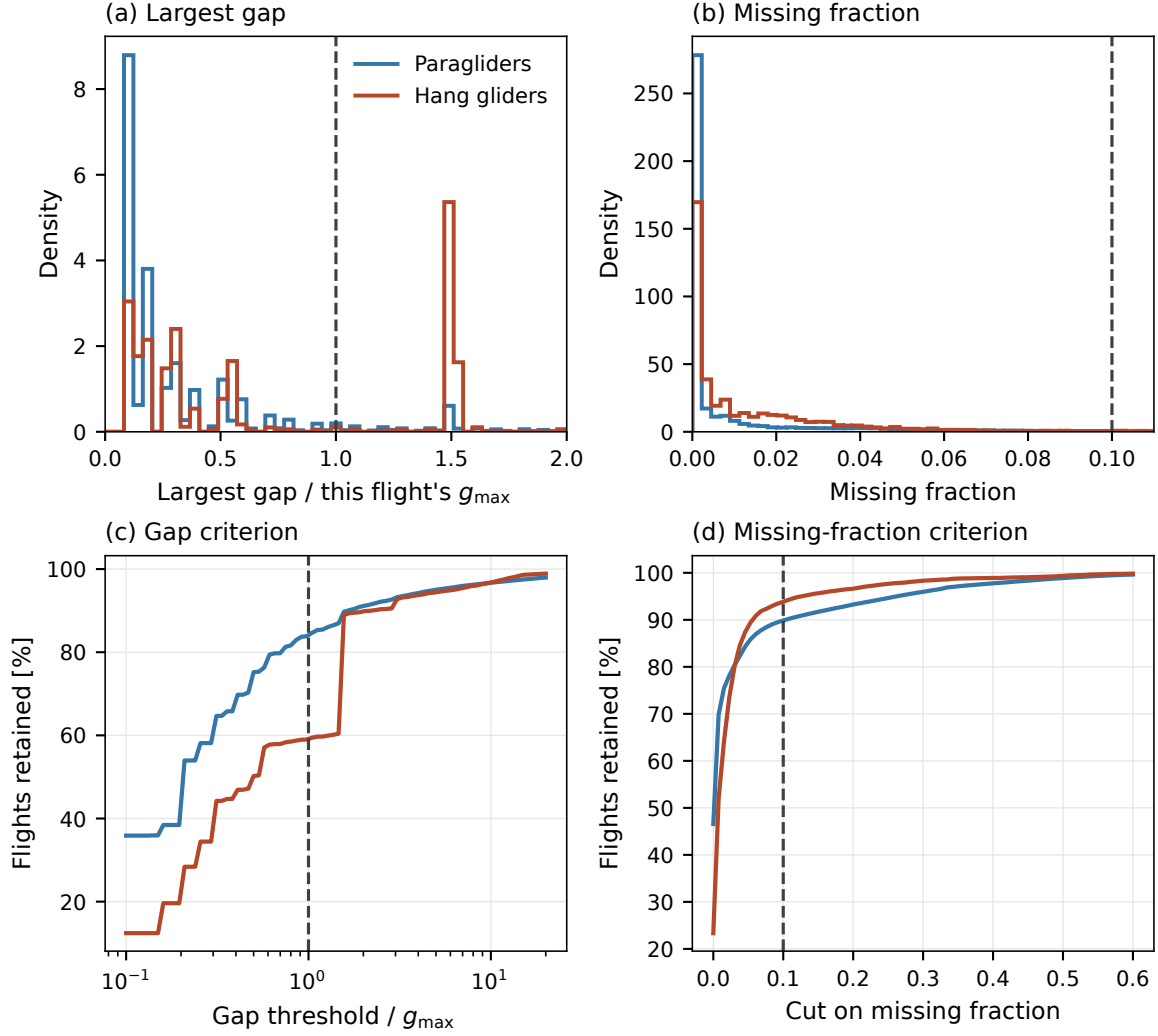


Figure 2.17. Sampling gaps in the raw archive, before cleaning and splitting. (a)/(b): distributions of the largest gap relative to $g_{\max}$ and of the missing fraction on a native-rate grid. (c)/(d): fractions of flights satisfying each criterion separately as its threshold varies. These are diagnostics of whole raw flights; the operational missing-fraction gate acts on segments after splitting, so these curves are not final flight-retention rates.

2.7.7 Smoothing and differentiation

Position errors are amplified by differentiation. We estimate position, velocity and acceleration from the same local polynomial, fitted independently to E , N and z within each retained segment. For $w = 2m + 1$ equally spaced samples around fix i , the Savitzky–Golay fit [18] is

$$P_i(u) = \sum_{k=0}^p a_k(i) u^k, \quad u = -m, \dots, m, \quad (2.24)$$

$$\mathbf{a}(i) = \arg \min_{\mathbf{a}} \sum_{u=-m}^m [x_{i+u} - P_i(u)]^2. \quad (2.25)$$

Here u is a sample offset. The physical quantities at the window centre are

$$\hat{x}_i = a_0(i), \quad \hat{v}_i = \frac{a_1(i)}{\Delta t}, \quad \hat{a}_i = \frac{2a_2(i)}{\Delta t^2}. \quad (2.26)$$

The weights. With $V_{u,k} = u^k$ and $\mathbf{x}_i = (x_{i-m}, \dots, x_{i+m})^\top$, the mathematical least-squares solution is

$$\mathbf{a}(i) = (V^\top V)^{-1} V^\top \mathbf{x}_i. \quad (2.27)$$

The inverse is a formula for the weights, not a recommendation to form a matrix inverse numerically. The implementation uses SciPy’s Savitzky–Golay coefficients. For fixed (w, p) , row k defines weights $c_k(u)$ with

$$a_k(i) = \sum_{u=-m}^m c_k(u) x_{i+u}. \quad (2.28)$$

This forward-offset convention is a correlation; convolution uses the reversed weights. For an odd derivative the reversal changes their signs. Table 2.6 gives the forward-offset weights, before conversion to seconds.

Table 2.6. Interior Savitzky–Golay weights for $w = 5$, $p = 3$. Velocity and acceleration weights must be divided by Δt and Δt^2 , respectively.

	sample offset u				
	−2	−1	0	1	2
c_0	−3/35	12/35	17/35	12/35	−3/35
c_1	1/12	−8/12	0	8/12	−1/12
$2c_2$	2/7	−1/7	−2/7	−1/7	2/7

The order $p = 3$ allows nonzero acceleration and locally varying acceleration. A quadratic already estimates velocity and acceleration; a cubic is a modelling choice, not a mathematical necessity. On a symmetric window, adding a cubic term changes the first-derivative weights but leaves the value and second-derivative weights of the quadratic fit unchanged.

Window length and cadence. A raw spectral diagnostic suggests a change in slope around 0.2 Hz (Fig. 2.19). It motivates the working timescale

$$\tau_c = 1/f_c = 5 \text{ s}, \quad w = \max\left(\left\lceil \frac{\tau_c}{\Delta t} \right\rceil_{\text{odd}}, 5\right). \quad (2.29)$$

Horizontal and vertical timescales are separate configuration entries, currently equal. The code converts this timescale to a *number of samples*: the distance in time between the first and last sample of a window is $(w - 1)\Delta t$. Thus $w = 5$ spans 4 s at a 1 s cadence, and 20 s at a 5 s cadence. Slower recording therefore changes the physical smoothing scale and can remove motion that faster loggers resolve. Comparisons of short-time observables must control for cadence.

The spectral knee is neither a fitted receiver-noise model nor the cutoff frequency of this filter. Rounding a field to steps of size q has variance $q^2/12$ only under a uniform-error approximation; a flat quantization spectrum additionally requires temporal independence. Similar high-frequency spectra do not establish those assumptions or identify the source of the observed knee.

Frequency response. In the segment interior, a Fourier mode $x_i = e^{2\pi i f_i \Delta t}$ is multiplied by

$$G(f) = \sum_{u=-m}^m c_0(u) e^{2\pi i f u \Delta t}, \quad S_x(f) = |G(f)|^2 S_x(f). \quad (2.30)$$

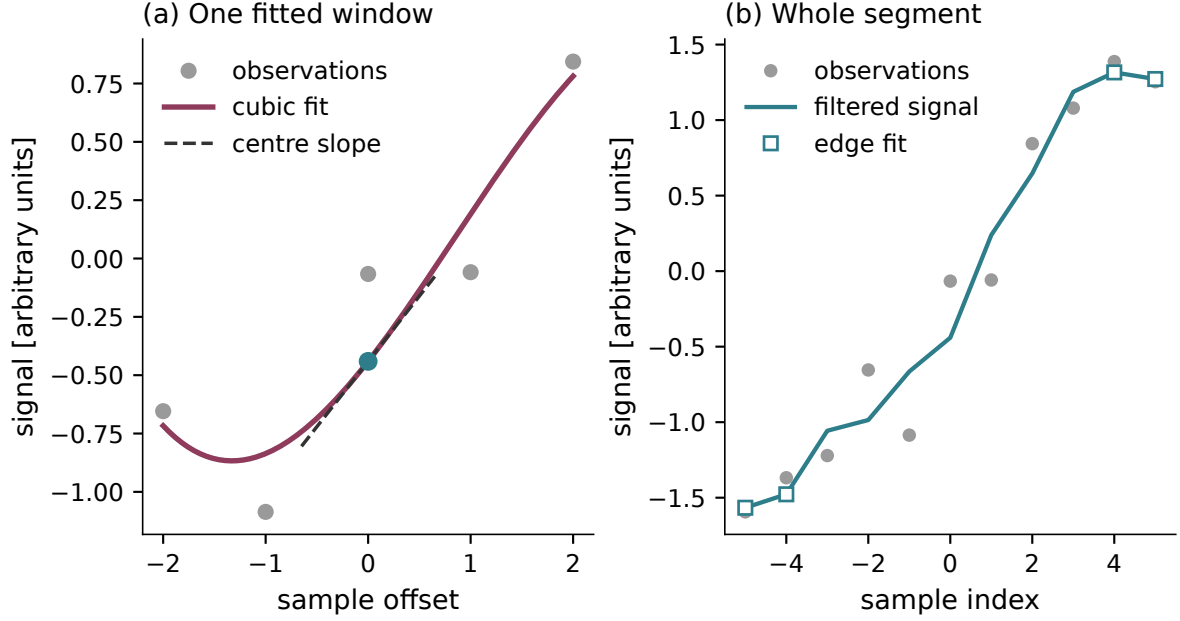


Figure 2.18. A cubic fit to a synthetic signal. Left: one five-sample window, its fit and the fitted value and slope at the centre. Right: the output of the implemented filter over the same series; squares mark the two edge samples at each end. Synthetic values are $\sin(0.28j) + \varepsilon_j$, with independent $\varepsilon_j \sim \mathcal{N}(0, 0.35^2)$.

The response depends on frequency and cadence through

$$\nu = f\Delta t \in [0, \tfrac{1}{2}], \quad (2.31)$$

in cycles per sample, with Nyquist at $\nu = 1/2$. Thus the same $G(\nu)$ applies to every cadence: $\nu = 0.2$ corresponds to 0.2 Hz at $\Delta t = 1$ s and 0.04 Hz at $\Delta t = 5$ s.

Symmetric weights give a real response, $G(\nu) = c_0(0) + 2 \sum_{u=1}^m c_0(u) \cos(2\pi\nu u)$. The phase is unchanged where $G > 0$ and reversed by π where $G < 0$. For the five-sample cubic fit (Table 2.6),

$$G(\nu) = \frac{17 + 24 \cos(2\pi\nu) - 6 \cos(4\pi\nu)}{35}. \quad (2.32)$$

Figure 2.20 shows the resulting power transmission. The filter preserves constants, $G(0) = 1$, and transmits half the power near $\nu = 0.24$. It vanishes at $\nu_0 = \arccos(1 - \sqrt{35/12})/(2\pi) \approx 0.375$, then rises to $|G(1/2)|^2 = (13/35)^2 \approx 0.14$. High frequencies are therefore attenuated unevenly by this short kernel.

Effect on the MSD. Write one recorded coordinate as $\tilde{x}_i = x_i + \varepsilon_i$. For zero-mean errors independent of the true motion and of one another, with $\mathbb{E}[\varepsilon_i \varepsilon_j] = \sigma^2 \delta_{ij}$, the one-coordinate MSD at lag $\ell \geq 1$ is

$$\mathbb{E}[(\tilde{x}_{i+\ell} - \tilde{x}_i)^2] = \mathbb{E}[(x_{i+\ell} - x_i)^2] + \underbrace{\mathbb{E}[(\varepsilon_{i+\ell} - \varepsilon_i)^2]}_{2\sigma^2}. \quad (2.33)$$

The constant noise contribution matters most at short lags, where the true displacement is small.

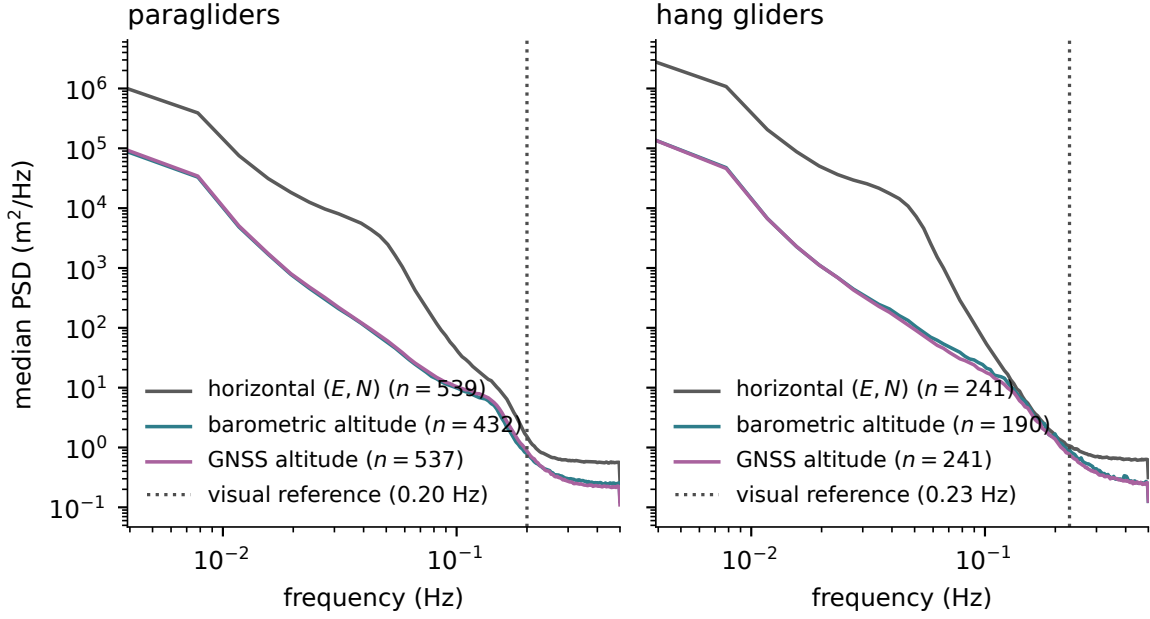


Figure 2.19. Median raw Welch PSDs, by discipline and coordinate channel, on a seeded sample of up to 900 flights per discipline. Spectra use recording intervals near 1 s; legend counts give the usable flights. Horizontal coordinates use a local equirectangular approximation before cleaning, and the curve pools both component spectra from each admitted flight. Channel eligibility is assessed separately, so the three medians need not represent identical flights or intervals. The vertical dotted lines are visual reference frequencies, 0.20 Hz and 0.23 Hz, not estimates with an uncertainty interval. Recording precision, sensor error, interpolation and real motion can all affect these spectra.

Smoothing replaces ε_i by the weighted average $\eta_i = \sum_u c_0(u) \varepsilon_{i+u}$, with c_0 extended by zero outside $|u| \leq m$. Independence of the raw errors leaves only the terms that read the same fix, so with the weight autocorrelation $R(\ell) = \sum_u c_0(u) c_0(u - \ell)$,

$$\mathbb{E}[\eta_i \eta_{i+\ell}] = \sigma^2 R(\ell), \quad (2.34)$$

$$B(\ell) = \mathbb{E}[(\eta_{i+\ell} - \eta_i)^2] = 2\sigma^2 [R(0) - R(\ell)], \quad (2.35)$$

For $|\ell| \geq w$, the two windows share no raw errors and $R(\ell) = 0$.

For $w = 5$, $p = 3$ the weights of Table 2.6 give

$$35^2 R(\ell) = 595, 336, 42, -72, 9, 0 \quad (\ell = 0, \dots, 5), \quad (2.36)$$

Thus $B(\ell)/\sigma^2 \approx 0.42, 0.90, 1.09, 0.96$ at $\ell = 1, \dots, 4$, settling at $34/35 \approx 0.97$ for $\ell \geq 5$. The filter roughly halves the noise contribution beyond its window but makes it lag dependent inside, with an overshoot at $\ell = 3$ (Fig. 2.20). A horizontal MSD adds both coordinate contributions.

This calculation assumes independent errors; it does not calibrate GNSS noise. The filter also changes the true motion unless each window follows a polynomial of degree at most p . A Hurst exponent alone cannot bound that change.

Edges and validation. At either end of a segment, the filter fits the nearest full window and evaluates that polynomial inside its fitted interval. These are off-centre evaluations, not extrapolation beyond the observations. The first and last m samples carry an **edge** flag because their weights and error variance differ from the interior. The filter can overshoot despite earlier cleaning, so speed bounds are not guaranteed by smoothing.

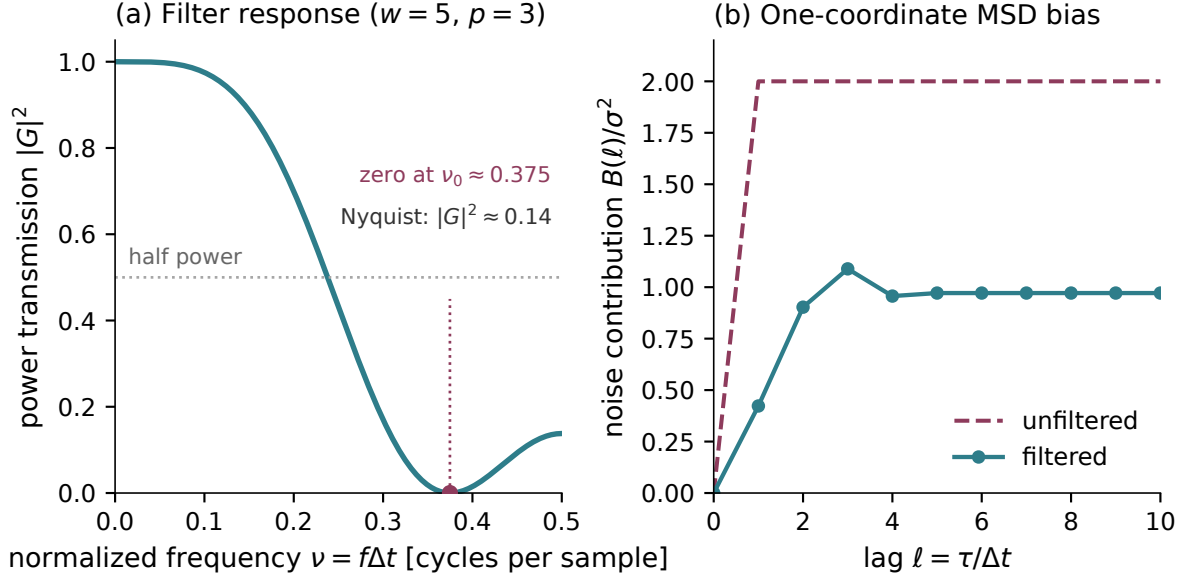


Figure 2.20. Exact interior response of the five-sample cubic filter. Left: power transmission $|G(\nu)|^2$, with $\nu = f\Delta t$ in cycles per sample and Nyquist at $1/2$. Right: additive one-coordinate MSD bias $B(\ell)/\sigma^2$ for independent measurement errors, compared with the unfiltered value 2. Overlapping windows produce the short-lag variation.

The kernel identities and derivative units are checked in automated tests. Archive-level validation remains separate: after rebuilding the cleaned archive, compare the MSD and phase features at neighbouring odd window lengths, within each cadence class, and with edge and reconstructed samples excluded. A flat residual spectrum is not a sufficient criterion: even white input noise has residual PSD $|1 - G(f)|^2 S_\varepsilon(f)$. Until these comparisons are reported, the adopted window is a working choice rather than a certified optimum.

2.7.8 Notation

The coordinates and their derivatives are

$$\begin{aligned} \mathbf{r}(t) &= (E(t), N(t)), & z(t) &= h_{\text{GNSS}}(t), \\ \mathbf{v}(t) &= \dot{\mathbf{r}}(t), & v_z(t) &= \dot{z}(t), \\ v_{xy}(t) &= |\mathbf{v}(t)|, & \Delta \mathbf{r}(t_0, \tau) &= \mathbf{r}(t_0 + \tau) - \mathbf{r}(t_0). \end{aligned}$$

The first trimmed fix sets $t = 0$ and the horizontal origin; smoothing can shift its position slightly. Altitude keeps its logged value. Every segment retains the parent flight's origin and clock. A bare *position* denotes the horizontal vector $\mathbf{r}$; three-dimensional position is stated explicitly.

2.8 Pipeline summary

Figure 2.21 gathers the processing steps of Sec. 2.7 in the order they are applied, from raw IGC tracks to the retained trajectories. Each stage identifies the records it acts on, its effect and the section that explains it. Section 2.9 then describes the resulting dataset.

Each flight is processed independently using its own cadence and geometry, with the shared thresholds in Table 2.8. Adding flights to the archive therefore does not change how an existing record is cleaned.

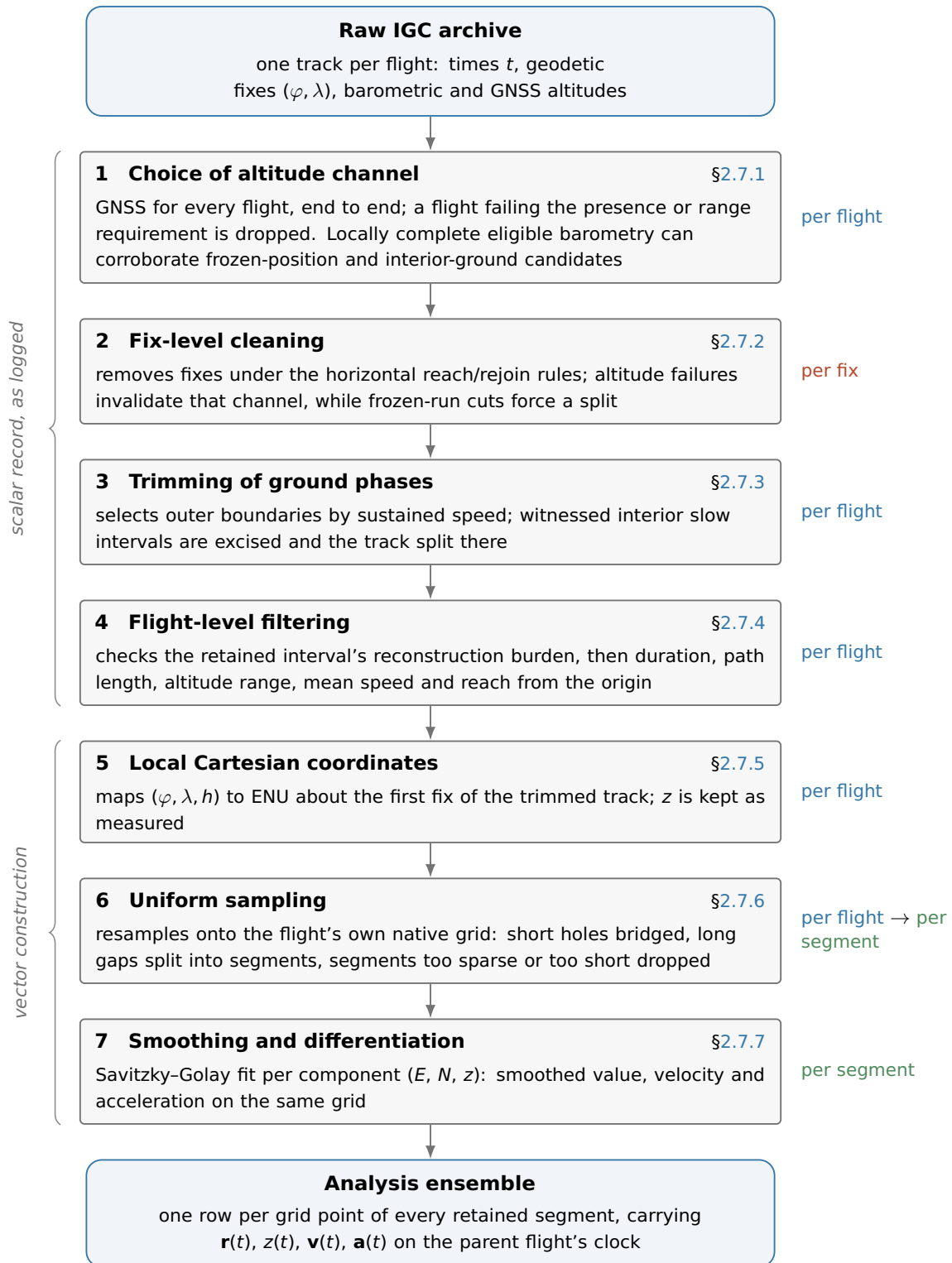


Figure 2.21. The processing chain, from the raw archive to the analysis ensemble. Each stage carries its effect and the section that argues the choice; the badge to the right is the level it acts at (*per fix*, *per flight*, *per segment*). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record about to be repaired or dropped. The thresholds each stage applies are listed, with their symbols, in Table 2.8.

What the chain retains. Run over the whole archive, the cascade retains 156,406 of 186,052 paraglider flights (84.1 %) and 6,094 of 6,716 hang-glider flights (90.7 %). Table 2.7 says where the rest went. Counts form a cascade: each rejected flight reached that stage after passing earlier criteria. They differ from the single-criterion raw diagnostics in Secs. 2.7.4 and 2.7.6.

Within the retained flights, cleaning deletes 1.83 % of paraglider fixes and 0.72 % of hang-glider ones, and invalidates an altitude on a further 0.056 % and 0.036 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.96 % of fixes are flagged and *kept*. The flag records local disagreement, while the impossibility gate controls horizontal deletion (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.30 % of grid points. A split is common but not the rule: 9.9 % of retained paraglider flights and 20.5 % of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.15 as Sec. 2.7.6 anticipated.

Table 2.7. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* counts flights that survived earlier criteria; *share* is the fraction removed from that remaining population. Resampling leaves no retained segment in 20,798 paraglider flights (11.7 % of those reaching that step) and 193 hang-glider flights (3.1 %). These conditional removal fractions differ from the separate raw-population diagnostics in Secs. 2.7.4 and 2.7.6.

Why the flight was dropped	paragliders			
	standing	removed	share (%)	remaining
unreadable track	186,052	27	0.01	186,025
no usable GNSS altitude	186,025	697	0.37	185,328
never airborne	185,328	197	0.11	185,131
integrity gate	185,131	550	0.30	184,581
shorter than the duration cut	184,581	2,084	1.13	182,497
longer than the duration cut	182,497	3	0.00	182,494
path below the floor	182,494	173	0.09	182,321
altitude activity below the floor	182,321	4,899	2.69	177,422
out of reach of its own first fix	177,422	217	0.12	177,205
no segment survived resampling	177,205	20,798	11.74	156,407
no segment carries the smoothing window	156,407	1	0.00	156,406
retained				156,406

Why the flight was dropped	hang gliders			
	standing	removed	share (%)	remaining
no usable GNSS altitude	6,716	135	2.01	6,581
never airborne	6,581	1	0.02	6,580
integrity gate	6,580	53	0.81	6,527
shorter than the duration cut	6,527	79	1.21	6,448
path below the floor	6,448	1	0.02	6,447
altitude activity below the floor	6,447	155	2.40	6,292
out of reach of its own first fix	6,292	5	0.08	6,287
no segment survived resampling	6,287	193	3.07	6,094
retained				6,094

Table 2.8 lists the working thresholds and their symbols by stage.

Table 2.8. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; configured values are read from `configs/preprocessing.yaml`. The minimum smoothing window is fixed by the polynomial-order rule in Sec. 2.7.7. The listed sections explain each criterion.

Symbol	Value	What it controls
Altitude admission (Sec. 2.7.1)		
—	95 %	minimum fraction of finite, nonzero GNSS readings; a flight below this completeness threshold is excluded
—	30 m	minimum GNSS range among finite, nonzero readings; a flight below it is excluded, without establishing a sensor failure
—	95 %	minimum barometric presence for witness eligibility; each candidate also requires complete local support
—	30 m	minimum barometric range for the same witness eligibility
Fix-level cleaning (Sec. 2.7.2)		
k	5	robust- σ multiplier of the Hampel test, Eq. (2.5)
w	20 s	half-width of the Hampel window; the implementation reuses it as the longest a pending block may stay open before the track is cut (Sec. 2.7.2)
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	45 m s ⁻¹ (para), 55 m s ⁻¹ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$ v_z ^{\max}$	10 m s ⁻¹	cutoff for the two adjacent window statistics in Eq. (2.9); also used by the out-and-back spike rule
w_v	5 s	half-width of that window
—	5 finite steps	smallest window population for which the median is estimable; below it the per-step value stands in
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration before a run can be cut by the frozen-position rule
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3)		
v_0	5 m s ⁻¹	ground-speed threshold used to identify take-off and landing candidates
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		

continued on the next page

Table 2.8 (continued)

Symbol	Value	What it controls
—	40 min	minimum recorded airborne duration, accumulated within blocks
—	20 km	minimum flown path length
—	16 h	maximum elapsed span, including gaps; records above it are excluded
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)
—	$v_{xy}^{\max}$ (above)	reach bound applied at every fix: $d_i \leq v_{xy}^{\max}(t_i - t_0) + 50$ m, where d_i is its distance from the first trimmed fix
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		
$g_{\max}$	$\min(10 \Delta t, \max(20 \text{ s}, 2\Delta t))$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained <i>segment</i> may have had to interpolate
—	90 s	minimum retained segment duration; smoothing additionally requires enough grid samples for its window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s (horizontal), 5 s (vertical)	working timescale setting the window through Eq. (2.29); equal values are adopted for the two channels, without identifying a physical correlation time
—	5 samples	minimum odd window for the adopted cubic fit. The temporal span is $(w - 1)\Delta t$, with w given by Eq. (2.29)

The analysis ensemble. The analysis ensemble comprises the retained segments written to `fixes.parquet`, with one row per grid time carrying (E, N, z) and their first two derivatives. A flight must pass stages 1–4 and retain at least one segment after resampling and smoothing. Segment origins and clocks remain those of the first trimmed fix of the parent flight.

Transport and phase estimators impose their own cadence, lag-support and quality-mask requirements, so their contributing populations can differ. The file schema and estimator inputs are documented in `docs/guide/data-on-disk.md`.

2.9 Preliminary characterization

This section describes the retained flights: their duration, path length, recording interval and geographical coverage. These measurements require no phase segmentation.

Trajectory statistics. Figure 2.22 compares the retained trajectories. Hang-glider flights are typically longer and logged less frequently:

	Paragliders	Hang gliders
Retained flights	156,406	6,094
Median retained span [h]	2.63	3.07
Median path [km]	87	151
Median cadence [s]	1	3
Recorded at 1 Hz [%]	69.9	28.0

Paraglider interquartile ranges are 1.77 h to 3.93 h for duration and 57 km to 134 km for path length. The median record contains 7,132 fixes. The native cadence sets each flight’s smoothing window (Sec. 2.7.7).

The dataset audit counts 0 non-finite coordinates and 0 non-monotone flight clocks in the paraglider audit. Its largest filtered interior step speed is 49 m s^{-1} , with 26 interior steps above the working bound but below the verifier’s gross-anomaly threshold. Across segment boundaries, 30 paraglider and 7 hang-glider flights exceed its separate speed-envelope diagnostic, which compares displacement with elapsed time and the declared coordinate tolerance. These boundary anomalies do not establish a unique sensor-error mechanism. Splitting prevents increment windows from bridging them; it does not independently validate absolute positions after reacquisition.

These checks identify structural failures and large kinematic anomalies. Finite coordinates and bounded speeds do not exclude plausible-looking recorder errors.

Panels (a) and (b) describe retained flights. Here duration is the elapsed span from the first to the last retained fix, including intervening gaps; path length sums only within-segment steps. These quantities can be smaller than those of the trimmed parent flight tested in Sec. 2.7.4, because short, sparse or unreconstructable segments can be discarded. The resulting distributions include 609 paraglider flights (0.39 %) and 6 hang-glider flights below the original 40 min or 20 km threshold. The parent-flight quantities passed those cuts; the cuts are not reapplied to the shorter retained trajectory.

Retention across recording intervals and seasons. The cascade in Table 2.7 retains most attempted flights. Loss of every segment at resampling is the largest conditional cut in both archives:

	Paragliders	Hang gliders
End-to-end retention [%]	84.1	90.7
No segment survives [%]	11.7	3.1
Altitude-range cut [%]	2.7	2.4
Ratio of the two cut rates	4.4	1.3

Each cut rate uses the flights reaching that stage. The 20.798 paraglider losses at resampling combine segments that are too short, too sparse or unreconstructable; they cannot all be attributed to the missing-fraction bound.

Across the six most common known cadences, the paraglider share with no surviving segment ranges from 0.0 % to 22.4 %. This variation calls for sensitivity checks of the missing-fraction and minimum-duration thresholds.

Across the 20 seasons with at least a thousand flights, overall retention ranges from 69.8 % to 91.9 % (Fig. 2.23b). Similar rates can conceal changes in cadence, region and pilot composition.

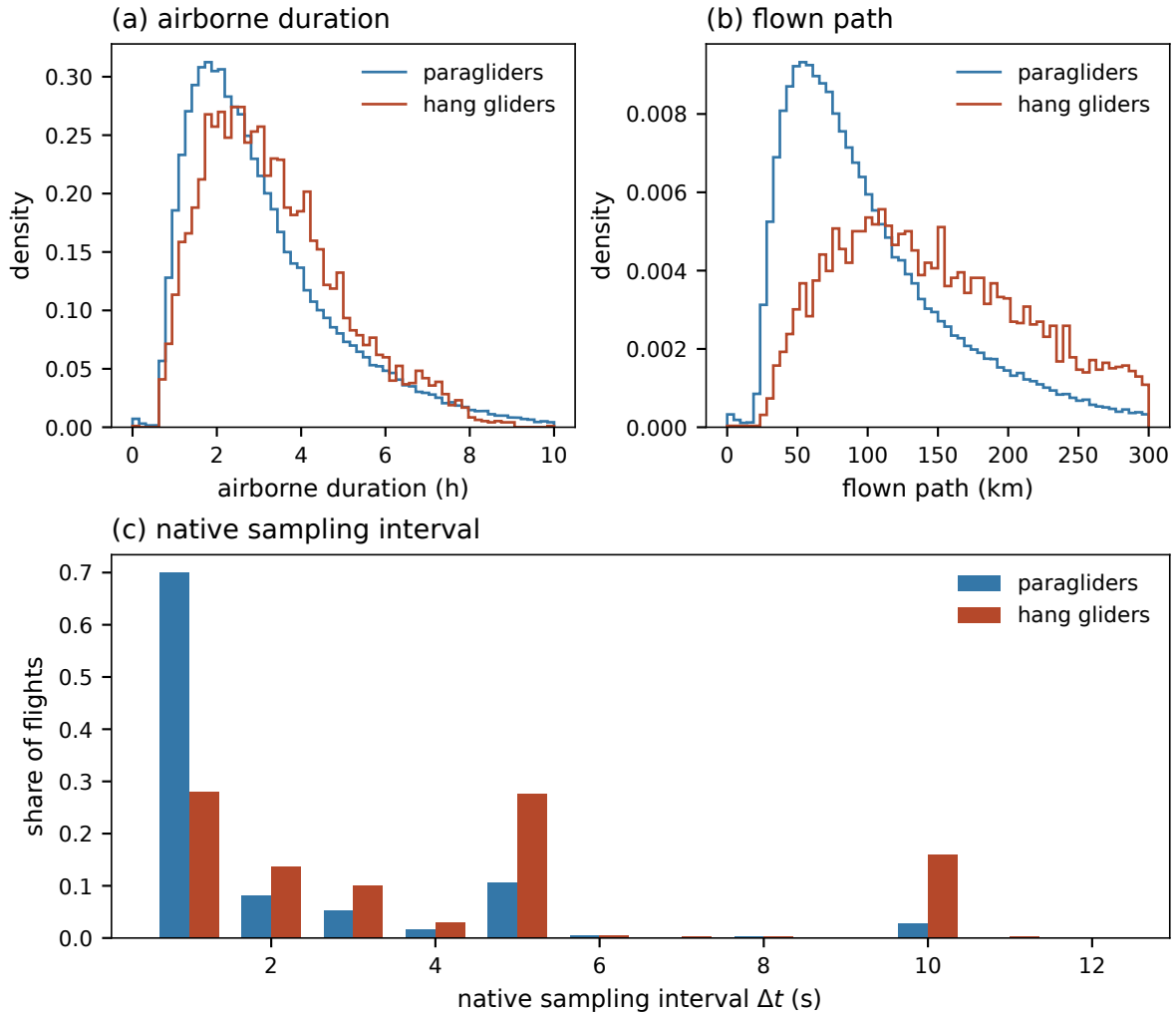


Figure 2.22. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt , per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.17 and 2.15. The duration labelled in panel (a) is the first-to-last retained-fix span, including gaps, rather than observed airborne time. Path length in (b) sums only within-segment steps. These retained-record quantities differ from the trimmed-parent quantities tested by Sec. 2.7.4; no parent-flight threshold is reapplied after resampling.

Where the flights are. The distribution of take-off sites is presented across three separate panels to account for the geographic coverage of the CFD: Figure 2.24 displays these sites against coastline and national borders.

- **Metropolitan France.** 94.7% of retained paraglider flights launch here, against 96.7% of hang-glider ones — the hang gliders are slightly more domestic.
- **La Réunion.** 3.2% (4,946 flights), the one overseas department with a substantial share.
- **Elsewhere.** 2.2% (3,379 flights): the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,341 distinct take-off sites, where distinct sites are launches rounded to a hundredth of a degree (~ 1 km); by share of paraglider launches:

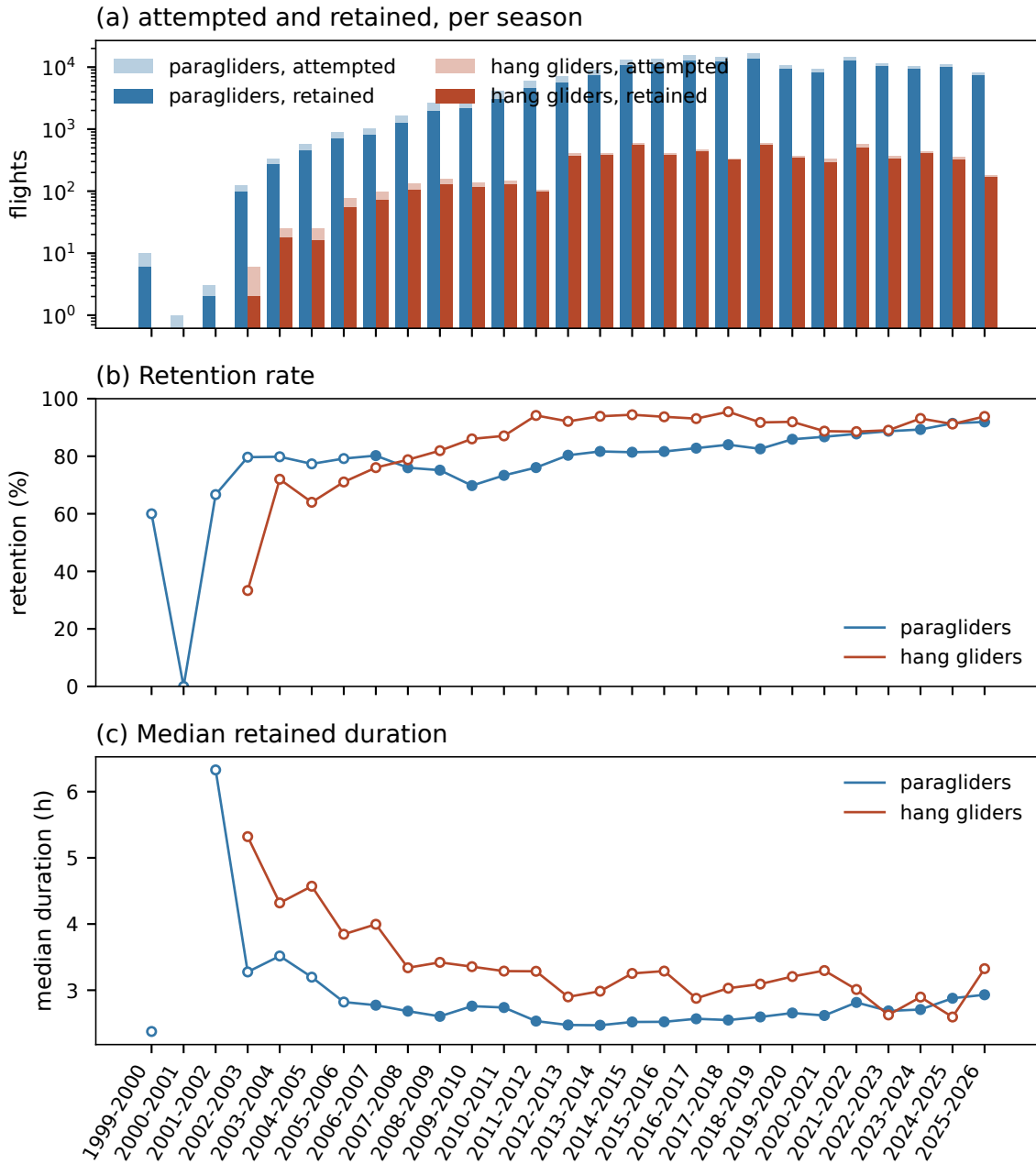


Figure 2.23. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights. The aggregate retention rate is approximately flat across the larger seasons; this does not establish equal selection within cadence or region groups.

- **Alps.** 58.8 %.
- **Pyrenees.** 7.9 %.
- **Massif Central.** 8.2 %.
- **Channel Coast.** 3.2 %: a box over low-relief terrain (Normandy, Picardy, the coastal Nord), not one carved out by excluding the other three.
- **Outside the massifs.** 16.7 %: everywhere else in metropolitan France, including the Champagne-Lorraine window used below; this is a residual defined by exclusion, not a terrain claim.

The geographic selections are longitude–latitude boxes, drawn in Figure 2.24a and defined numerically in Table 2.9. For initial longitude λ_0 and latitude φ_0 , membership requires

$$(\lambda_0, \varphi_0) \in [\lambda_{\min}, \lambda_{\max}] \times [\varphi_{\min}, \varphi_{\max}],$$

including the four boundaries. The Champagne-Lorraine box provides the second lowland region for the transport comparison; it extends beyond Champagne itself and is included in the residual share quoted above. These windows define reproducible geographic selections, not administrative boundaries or homogeneous terrain polygons.

Region	$\lambda_{\min}$	$\lambda_{\max}$	$\varphi_{\min}$	$\varphi_{\max}$
Alps	5.4	10.0	43.8	46.6
Pyrenees	−1.9	3.3	42.0	43.5
Massif Central	1.8	4.6	43.6	46.2
Channel Coast	−1.8	2.0	48.3	51.2
Champagne-Lorraine	2.1	6.2	48.0	50.6

Table 2.9. Geographic boxes shown in Figure 2.24. Coordinates are decimal degrees: λ is longitude, positive east and negative west; φ is latitude, positive north. The lower-left corner is $(\lambda_{\min}, \varphi_{\min})$ and the upper-right corner is $(\lambda_{\max}, \varphi_{\max})$.

A flight can lie inside a mountain-region box while starting in a lowland setting. For the regional comparisons of Section 3.2.2, geographic membership is therefore crossed with the initial cleaned altitude used in that chapter: Alps and Pyrenees retain only Low/High mountains ($z_0 \geq 800$ m), while Channel Coast and Champagne-Lorraine retain only Plains/Hills ($z_0 < 800$ m). The map itself shows the full retained ensemble, before this additional selection.

Altitude at the beginning of the record. Figure 2.25 groups flights by their first raw altitude, before trimming: GNSS where available, pressure altitude otherwise. This classifies 99.9 % of retained paraglider flights. Their shares are

Legend label	Logged altitude [m]	Share [%]
Plains	< 300	7.6
Hills	300–800	11.2
Low mountains	800–1500	50.7
High mountains	≥ 1500	30.5

The bands follow Hernández-Aguayo, Cristelli and Benzaquen [19]. They simplify the full classification of Kapos et al. [20]; their historical names do not measure local relief or slope.

The first fix can be away from launch, already airborne, or affected by sensor settling. Datum and atmospheric offsets can move it between bands. This exploratory map therefore describes recorded starting altitude, with pressure fallback confined to this map.

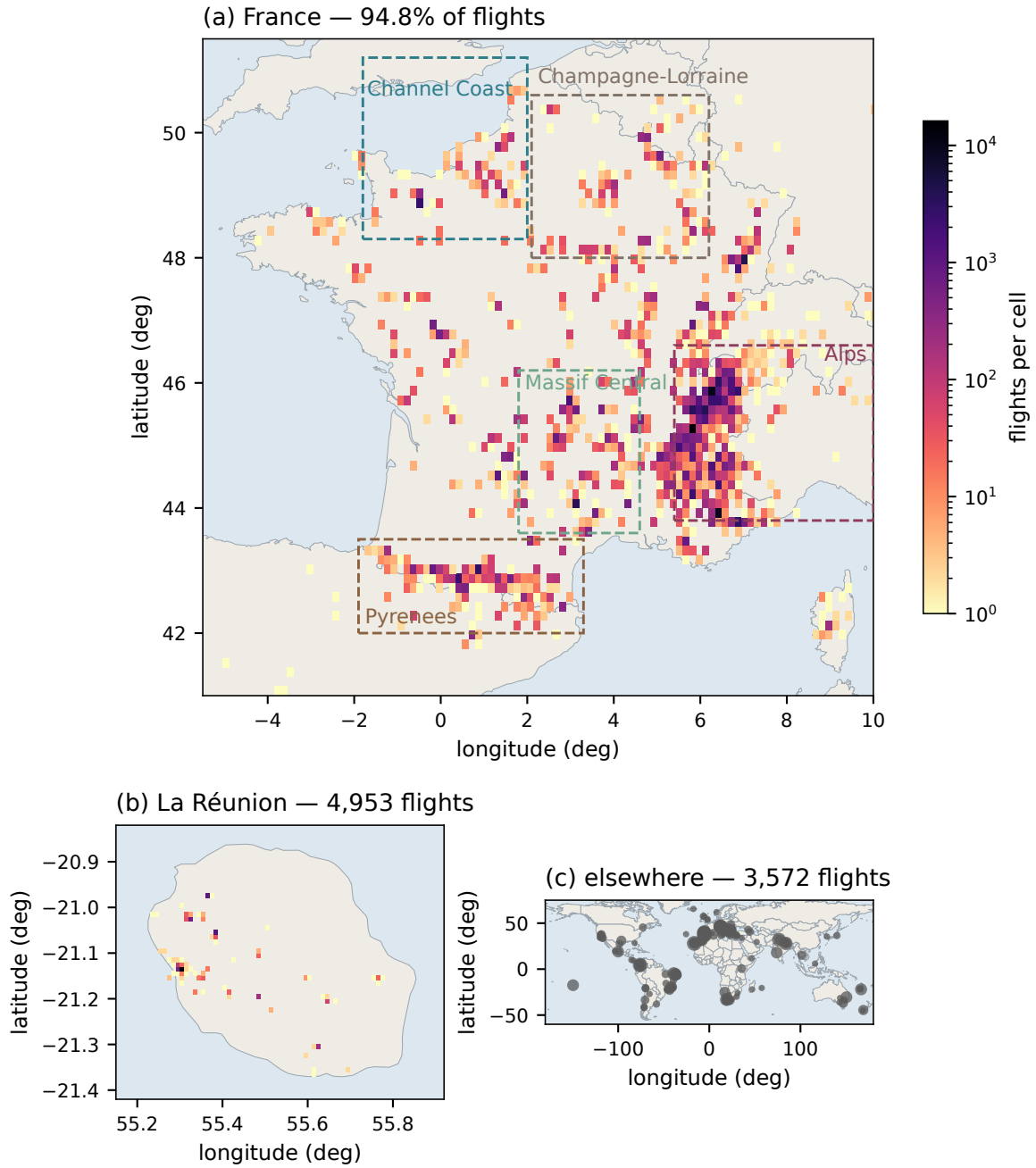


Figure 2.24. Take-off sites of the retained ensemble, on three frames, pooling both disciplines throughout. (a) Metropolitan France, density on a 0.15° cell; the dashed boxes mark the geographic windows in Table 2.9, including Champagne-Lorraine; no altitude preselection is applied to this map. (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count.

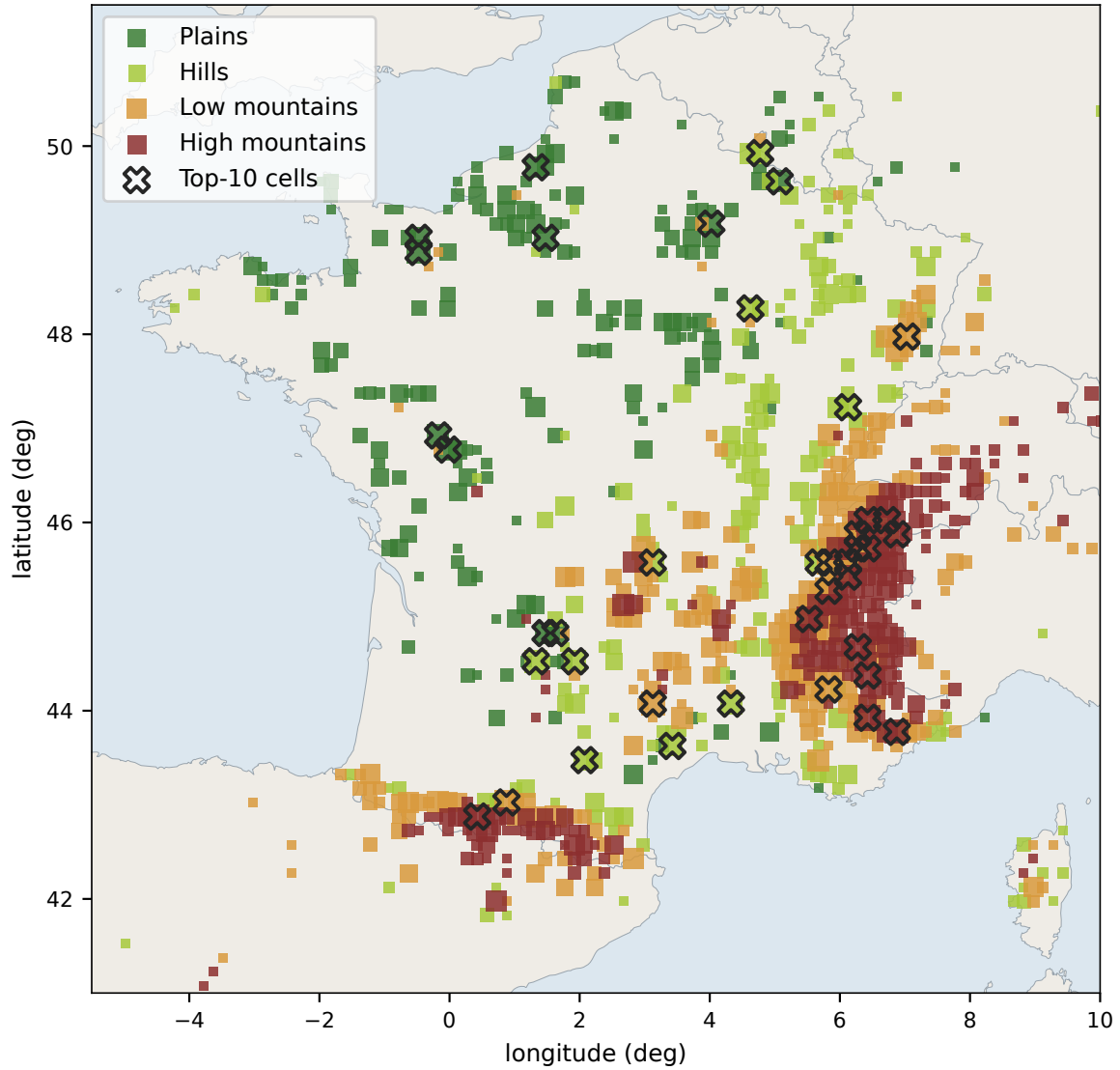


Figure 2.25. First recorded positions of the retained ensemble (both disciplines pooled), grouped by logged altitude using the elevation bands of [19]. These are approximate launch-altitude groups, not measured terrain classes. One marker per occupied 0.15° cell, coloured by its majority class and sized by the logarithm of its flight count; crosses mark the 10 busiest cells of each class.

2.10 Trajectories for the transport analysis

The retained trajectories have a uniform grid within each segment, local horizontal coordinates and explicit reconstruction flags. Phase features can exclude flagged vertical support, although plausible unflagged defects remain possible. Geography, cadence and flight duration also vary across the archive.

Any transport observable computed from these trajectories must specify its weights, lag support and contributing population.

Chapter 3

Horizontal transport and displacement scaling

We measure the mean-square displacement (MSD) and its dependence on circuit type, launch environment and equipment. The scaling of the displacement distribution is treated as an experimental extension in Chapter E. The general analysis separates paragliders from hang gliders; the conditional comparisons use the larger paraglider population.

3.1 Mean-square displacement

3.1.1 Estimator and temporal resolution

Let $\mathbf{x}_f(t) = (E_f(t), N_f(t))$ be the cleaned horizontal trajectory of flight f , and let s index its retained continuous segments. Within each segment, define $\Delta \mathbf{x}_{fi}(\tau) = \mathbf{x}_f(t_i + \tau) - \mathbf{x}_f(t_i)$ and $R_{fi}(\tau) = |\Delta \mathbf{x}_{fi}(\tau)|$, defined whenever t_i and $t_i + \tau$ lie in the same segment s ; no increment crosses a segment boundary. Let i index grid origins across the flight, and define the admissible index set

$$\mathcal{I}_f(\tau) = \{i : t_i \text{ and } t_i + \tau \text{ lie in the same retained segment of flight } f\}.$$

Let $n_{f,s}(\tau)$ count the admissible indices in segment s , so that $n_f(\tau) = |\mathcal{I}_f(\tau)| = \sum_s n_{f,s}(\tau)$ is their total across the flight. Set

$$\mathcal{C}(\tau) = \{f : n_f(\tau) > 0\}, \quad N(\tau) = |\mathcal{C}(\tau)|.$$

Let $m_f(\tau)$ be the time-averaged MSD of flight f , and $M_2(\tau)$ the mean of $m_f(\tau)$ over $\mathcal{C}(\tau)$. They are defined by

$$m_f(\tau) = \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} R_{fi}(\tau)^2, \quad M_2(\tau) = \frac{1}{N(\tau)} \sum_{f \in \mathcal{C}(\tau)} m_f(\tau). \quad (3.1)$$

We first average the squared displacements within each flight, pooling the valid increments from all its retained segments. We then average these flight-level MSDs, giving each contributing flight the same weight $1/N(\tau)$, regardless of its number of increments or segments. This is the *equal-flight* average. Two alternatives were rejected.

- *Equal-increment average*: pooling all increments directly, without first averaging within flights,

$$M_2^{\text{incr}}(\tau) = \frac{1}{\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)} \sum_{f \in \mathcal{C}(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} R_{fi}(\tau)^2 = \sum_{f \in \mathcal{C}(\tau)} \frac{n_f(\tau)}{\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)} m_f(\tau),$$

would give flight f weight $n_f(\tau)/\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)$ in the population TA-MSD: longer flights, which accumulate more valid increments, would dominate the pooled statistic, which would then track long flights rather than represent the population evenly.

- *Equal-segment average*: first computing a TA-MSD for each segment, then averaging over every segment in $\mathcal{C}(\tau)$,

$$m_{f,s}(\tau) = \frac{1}{n_{f,s}(\tau)} \sum_{i \in s} R_{fi}(\tau)^2, \quad M_2^{\text{seg}}(\tau) = \frac{1}{S(\tau)} \sum_{f \in \mathcal{C}(\tau)} \sum_s m_{f,s}(\tau),$$

with $S(\tau)$ the total number of segments with $n_{f,s}(\tau) > 0$ across $\mathcal{C}(\tau)$, would give equal weight to every segment regardless of its length: flights broken into many short segments, typically those with lower signal quality, would then be overrepresented.

For $10 \leq \tau \leq 10000$ s we subsample each segment to a 10 s grid, starting at its first fix. This spacing sets the step between the origins t_i from which the MSD at lag τ is computed within a segment, and it gives every flight the same physical lags to compare. Records with native cadence above 10 s are excluded. Missing positions are linearly interpolated within segments: with an 8 s cadence, for example, the 10 s position lies between the fixes at 8 and 16 s. Table 3.1 counts how often the grid requires this. The 33 evaluated lags are multiples of 10 s, approximately logarithmically spaced and including 100 and 1000 s. Figure markers show the evaluations; lines connect them.

The spacing itself is a trade-off. Finer sampling would raise the computational cost sharply without benefit once it resolves timescales below any actual change in the dynamics; coarser sampling would fail to resolve the fastest manoeuvres, which occur during the search and thermalling phases, where speed, acceleration and heading vary faster than during gliding. Vilpellet et al. [2] report that a single revolution while thermalling takes just under 30 s, so 10 s samples each turn about three times, resolving the fastest dynamics in the archive without oversampling the slower gliding phase.

Discipline	Grid fixes	Coincident	Interpolated	Fraction
Paragliders	85,915,677	81,888,079	4,027,598	4.69%
Hang gliders	4,099,322	3,639,355	459,967	11.22%

Table 3.1. Common-grid positions in the main fixed cohort. Coincidence is checked against actual cleaned timestamps. The last two columns count interpolation added by the 10 s grid, separately from reconstruction during cleaning.

Among the coincident grid positions, 355,170 paraglider fixes and 9,648 hang-glider fixes already carry the cleaning interpolation flag. The short-lag detail at 1–9 s in the general TA-MSD uses only native segments that resolve the requested lag exactly.

3.1.2 General transport and uncertainty

Figure 3.1 compares the TA-MSD, averaged over within-segment origins, with squared distance from the flight’s single post-trim origin (Section 2.7.3). For the latter, t is elapsed time from that origin. Gaps leave missing support without resetting either the clock or the reference position.

A flight contributes at t if a retained segment covers that time and its native sampling interval satisfies $dt \leq t$. Linear interpolation misses any curvature between consecutive fixes and its position error depends on motion within the local interval dt , so it doesn’t improve

as elapsed time t increases. What can decrease is the error *relative to the displacement from the origin*. At a characteristic speed v , unresolved motion has a spatial scale $v dt$, whereas approximately ballistic displacement grows as vt . Their ratio dt/t explains the cutoff: for $t < dt$, interpolation error can be comparable to the displacement; as t grows beyond dt , its relative contribution can decrease.

At each t we report the mean, median and 5–95% range across available flights. The contributing population changes as t grows, since flights whose retained span is shorter than t drop out, and the TA-MSD loses flights the same way as τ grows; Table 3.2 counts the flights that remain. These curves therefore describe a changing mixture of sites, days and flight conditions, which is why the controlled comparisons of Section 3.1.3 instead fix the population once for all lags.

We summarise the growth of the TA-MSD by an ordinary-least-squares linear fit of $\log_{10} M_2$ against $\log_{10} \tau$. A power law $M_2 \propto \tau^{2H}$ has slope $2H$ in this plane, so H is half the fitted slope. Here H denotes an effective growth exponent: its identification with a process’s self-similarity exponent is tested in the experimental Section E.2. The interval for H comes from repeating the fit on the TA-MSD recomputed for each bootstrap replicate, a resampling of site–day groups described in the paragraph “Uncertainty” below. The fits to the TA-MSD of the available population (Figure 3.1, right) are requested over 10–30000 s but the last lag of the grid inside that range is 28440 s, so they span 10–28440 s. In the right panels of Figure 3.1, the TA-MSD over the fitted lags lies close to a straight line in logarithmic scale. Figure 3.3 shows the fit residuals: they fall on both sides of zero over the whole fitted range, and their RMS say that a typical measured value lies within a factor 1.09 and 1.19 of the fitted line, respectively for paragliders and hang gliders. Such ratios, close to unity over a curve that grows by about six decades, combined with the fact that the residuals are evenly distributed above and below the fitted curve, make the line an adequate description of the measured growth. The residuals are nevertheless not scattered at random. In fact, they change sign three times as the lag grows, so the departure from a single power law is a smooth curvature, largest at the longest lags where the contributing population has shrunk most (Table 3.2). The fit therefore summarises the growth as one effective exponent, and the local exponent below (Figure 3.2) shows how the slope varies with lag.

The available-population fits give, with 90% bootstrap intervals in brackets:

- paragliders: $H = 0.869[0.864, 0.874]$ over 10–28440 s;
- hang gliders: $H = 0.841[0.816, 0.855]$ over the same lags.

Since the flights contributing at each lag change as τ grows (Table 3.2), these values summarise the growth of a curve whose population shrinks with lag.

To describe how growth varies with lag, we also estimate the local exponent

$$H_{\text{loc}}(\tau) = \frac{1}{2} \frac{d \log M_2(\tau)}{d \log \tau}. \quad (3.2)$$

At each lag, we fit a straight line by ordinary least squares to the log–log curve within ± 0.25 decades and take half its slope. At least three lags are required. For sparse interior windows in the general curves, we use the three nearest lags in logarithmic distance; this fallback requires at least two available grid lags on each side of the target. Otherwise, windows with fewer than three lags are left unestimated.

The TA-MSD band in Figure 3.2 is the pointwise 5–95% range obtained by repeating the local fits on the same bootstrap curves. The origin-distance mean and median use the same method with elapsed time t in place of lag τ .

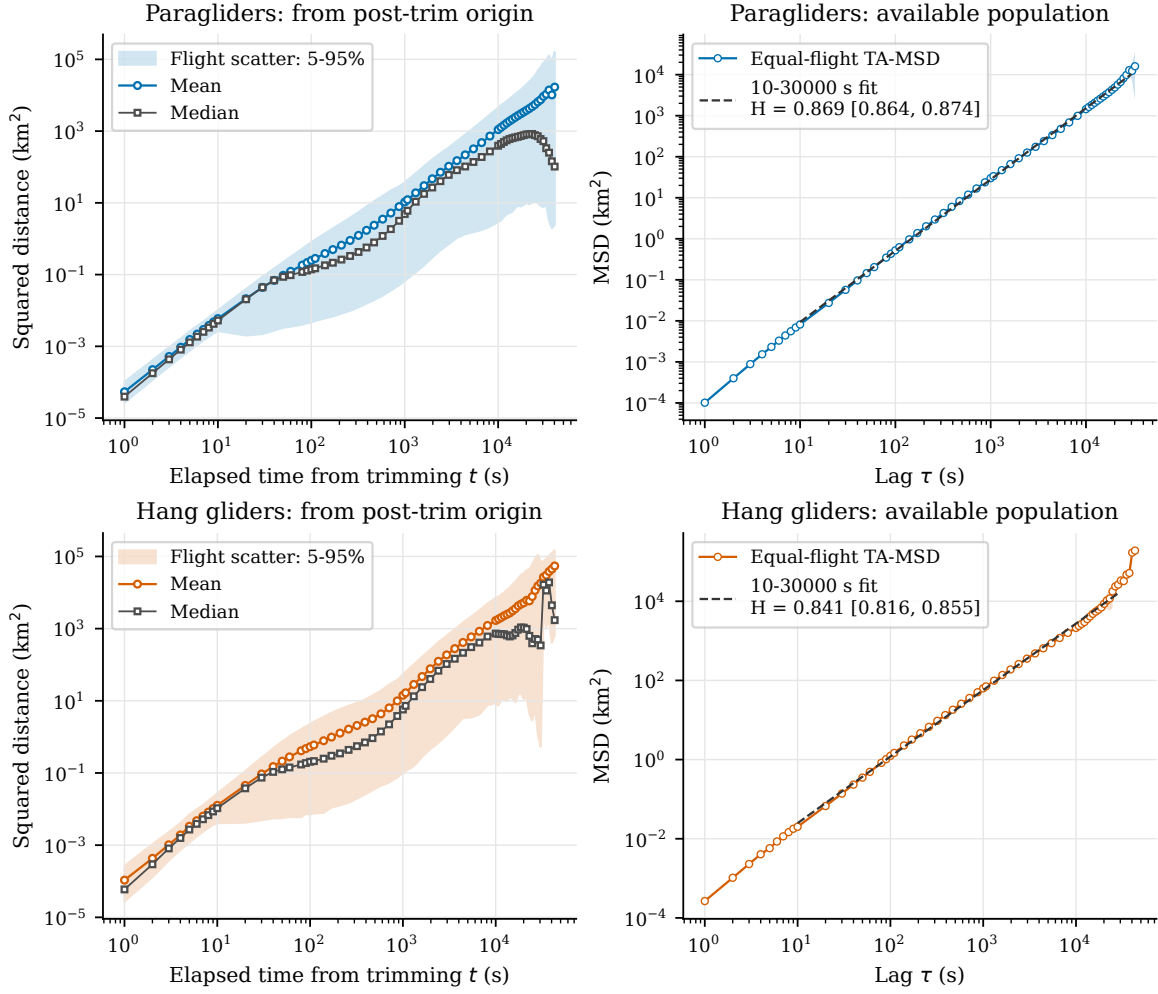


Figure 3.1. General horizontal transport. Left: squared distance from the post-trim origin, with equal-flight mean, median and the empirical 5–95% flight range. Right: equal-flight TA-MSD over the population available at each lag, with pointwise 90% cluster-bootstrap intervals and a log–log fit over 10–30000 s (evaluated lags: 10–28440 s). Native short-lag support differs from common-grid support.

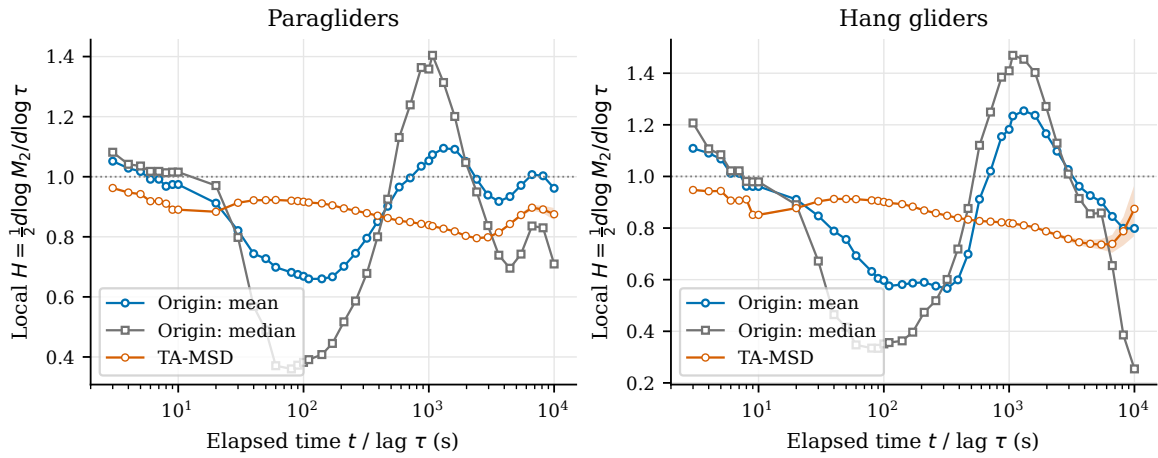


Figure 3.2. Local slopes of the general curves. The origin-distance mean and median have no confidence band here; the TA-MSD band is obtained by refitting bootstrap curves. The horizontal coordinate is elapsed time t from trimming for the origin-distance curves and lag τ for the TA-MSD. The dotted reference is ballistic growth, $H = 1$.

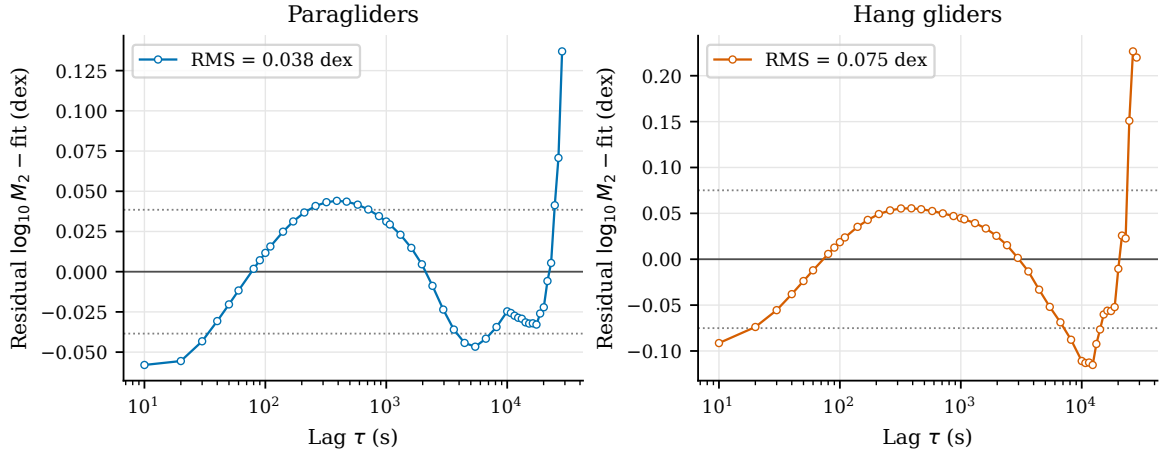


Figure 3.3. Residuals of the log-log fit of the general TA-MSD at the fitted lags, in dex: $\log_{10} M_2$ minus the fitted line. A residual of r dex means the measured curve sits a factor 10^r from the line, so 0 dex is a point lying on it. Dotted lines mark plus and minus the RMS residual.

Discipline	Lag (s)	TA-MSD flights	Origin-distance flights
Paragliders	10	155,085	152,771
Paragliders	100	155,044	155,117
Paragliders	1000	154,482	155,250
Paragliders	10000	46,273	72,160
Paragliders	32690	27	1,237
Hang gliders	10	6,060	5,330
Hang gliders	100	6,060	6,059
Hang gliders	1000	6,044	6,070
Hang gliders	10000	2,326	3,480
Hang gliders	43200	1	3

Table 3.2. Number of contributing flights in the general curves. The last row for each discipline is the largest evaluated lag with TA-MSD support. The origin-distance curve uses elapsed time; the TA-MSD requires a within-segment lag no larger than $0.8T_s$. Their support sets therefore differ. Early origin-distance support can increase when elapsed time exceeds an initial excluded interval or resolves a coarse cadence.

Figures 3.1 and 3.2 show two linked features of the origin-distance curves. After the approximately ballistic regime at $t \lesssim 10$ s, the mean rises visibly above the median, with a marked separation at $t \sim 10^2$ – 10^3 s. The mean’s local slopes reach minima at 110 s ($H_{\text{loc}} = 0.66$) for paragliders and 320 s ($H_{\text{loc}} = 0.57$) for hang gliders within 20–2000 s, then recover towards $H_{\text{loc}} \simeq 1$ around 600–700 s.

A possible physical explanation is the asynchronous alternation of gliding, search and climb. At the same elapsed time, some pilots glide while others search for lift or circle in a thermal, making less net horizontal progress. At short elapsed times, an episode of searching or lifting lasting a few hundred seconds can greatly reduce the displacement relative to a flight that keeps gliding. The latter pulls the mean above the median, especially because displacement is squared. Meanwhile, the slow horizontal progress of flights in search and climb lowers the growth rates, with a stronger dip in the median’s H_{loc} .

The duration distributions in Vilpellet et al. [2, Fig. 4f,i] support this timescale: search times and climb times have survival probabilities that remain of order 10^{-3} at durations of several hundred seconds. An archive with many thousands of flights can therefore contain several such long events alongside much shorter ones.

Search and climb continue throughout the flight, at different times for different pilots. At $t \sim 10^3\text{--}10^4$ s, however, many flights have already reached much larger distances from the origin. The extra displacement gained by gliding rather than searching or climbing during one interval of $10^2\text{--}10^3$ s can then be smaller relative to that distance, and so have a smaller relative effect on squared distance. The same alternation of phases can thus produce a weaker contrast between flights at long elapsed times. This is consistent with the recovery of the local exponents and the narrowing of the early mean–median gap, even though the phases remain asynchronous.

Uncertainty. The shading around the origin–distance curves of Figure 3.1 is the empirical 5–95% range across flights, a measure of scatter among flights. Every other shaded band in this chapter is a pointwise 90% bootstrap interval: the interval is constructed separately at each lag, with nominal 90% coverage at that lag, rather than simultaneous 90% coverage of the entire curve. Flights are grouped by catalogue take-off site (the one declared in the FFVL catalogue metadata (Section 2.3, Table 2.1)) and calendar date, retaining dependence within each site–day. Missing keys give singleton groups. A *replicate* is one resampled version of the archive.

Writing G for the number of site–day groups in the eligible archive, we construct $B = 1000$ replicates once, before evaluating the MSD at the different lags. Each replicate draws G groups with replacement and retains all flights in each selected group. Thus G counts group selections, including repetitions; the number of distinct groups and the total number of flights can vary between replicates.

At each lag τ , we evaluate the MSD using only the flights in that replicate that have admissible increments at τ , namely those in $\mathcal{C}(\tau)$. As τ increases, flights with insufficient segment duration cease to contribute, but the original group draws remain fixed: no replacement flights or groups are drawn. A group selected twice still counts each of its contributing flights twice; if none of its flights supports that lag, it contributes nothing there. The MSD is divided by the number of contributing flights in that replicate at that lag, counting repetitions. This count can vary both between replicates and with τ . An analysis of a sub-population, such as a duration cohort, reuses the same draws and keeps only the flights that belong to it. Let $g(f)$ be the group of flight f , and $c_g^{(b)}$ the number of times group g is drawn in replicate b . The replicate replaces the equal weight $1/N(\tau)$ of Eq. (3.1) by a weight proportional to the multiplicity of the flight’s group:

$$M_2^{(b)}(\tau) = \frac{1}{N^{(b)}(\tau)} \sum_{f \in \mathcal{C}(\tau)} c_{g(f)}^{(b)} m_f(\tau), \quad N^{(b)}(\tau) = \sum_{f \in \mathcal{C}(\tau)} c_{g(f)}^{(b)}. \quad (3.3)$$

Setting every $c_g^{(b)} = 1$ recovers Eq. (3.1). The multiplicities $c_g^{(b)}$ have no lag dependence; the contributing set $\mathcal{C}(\tau)$ and normalisation $N^{(b)}(\tau)$ do. Each of the 1000 replicates uses its own group draw consistently across lags, observables and cohorts. For a comparison between two cohorts, we subtract their estimates within each replicate, producing 1000 differences, one for each $b = 1, \dots, 1000$. Their 5th and 95th percentiles give the interval for the difference, retaining the sampling fluctuations shared by the two cohorts. The point estimate itself is the difference between the estimates from the original archive. We use this procedure for the duration-cohort contrasts in Section 3.1.3 and the open–closed circuit contrasts in Section 3.2.1. Quantiles and ratios are likewise recalculated within each replicate, and each fit uses points from that replicate’s curve, preserving the dependence between lags. For an estimate $\hat{\theta}$, the 5th and 95th percentiles of the bootstrap replicates $\{\hat{\theta}^{(b)}\}_{b=1}^{1000}$ give a pointwise 90% confidence interval

$$I_{90}(\theta) = \left[\text{perc}_5\{\hat{\theta}^{(b)}\}_{b=1}^{1000}, \text{perc}_{95}\{\hat{\theta}^{(b)}\}_{b=1}^{1000} \right]. \quad (3.4)$$

Percentiles are linearly interpolated between ordered estimates, excluding any replicate with no contributing flights at the lag considered.

Cluster bootstrap requires many approximately independent groups and sufficiently stable statistics [21]; flights within a group need not be independent. Table 3.3 describes their spatial and temporal separation. Nearby sites and successive days can share weather, so these distances do not verify independence; the paragraph below checks MSD and fitted-exponent errors using geographic, altitude and seasonal groups. Residual dependence may make the bands too narrow; upload selection also limits inference to the population represented by the archive. General TA-MSD bands are omitted below 20 contributing groups.

The statistical validity of these intervals is therefore not established at this stage. Independence between site–day groups is assumed and has never been verified, and the size of the groups makes the assumption doubtful: their median size is one flight, the 95th percentile is five flights for paragliders and three for hang gliders, and even the largest group holds 47 and 13 flights (Table 3.3). A partition this fine absorbs little of the dependence it was introduced to capture. Flights that share one weather episode across neighbouring sites, or across consecutive days at the same site, fall into different groups and are resampled as if they carried independent information. The groups are then probably not independent, and Eq. (3.4) probably understates the true uncertainty, so every band reported in this chapter should be read as a lower bound on the spread. The open question is whether bootstrap groups coarse enough to be approximately independent can be built at all. The paragraph below is a first attempt at it and has not answered it yet, which is why it is typeset in red.

Quantity	Paragliders	Hang gliders
Catalogue sites	3,424	284
Calendar dates	3,642	1,017
Flights per group: median / P95 / max	1 / 5 / 47	1 / 3 / 13
Largest group (% of flights)	0.10	0.56
Nearest same-day site (km)	11.2 [0.04, 173]	24.7 [0.04, 338]
Spatial comparisons	24,509	1,147
Prior observed date at same site: gap (days)	19 [1, 429]	18 [1, 382]
Temporal comparisons	22,258	1,484
Flights with missing site/date keys	0	0

Table 3.3. Site–day diagnostics in the main cohort. Distances refer to the nearest other catalogue site represented on the same date; time gaps refer to the previous observed date at the same site. Distances and time gaps are reported as median [10th, 90th percentile]. Cohort group counts are in Table 3.5.

Reliability of bootstrap uncertainty estimates (work in progress, not working). This paragraph is a working draft and no conclusion in it is settled. The grouping tried here does not yet produce approximately independent bootstrap groups, so its intervals validate nothing and its numbers are reported only to record where the attempt stands.

We repeat the uncertainty check with groups defined by a fixed geographic cell, launch-altitude class and month of the year, pooling all years. Thus July flights from different years share a group when their initial positions lie in the same cell and altitude class. The grid is independent of the observed sites: each initial position belongs to one half-open 50×50 km square in its WGS84 UTM zone, with sides measured in projected metres. Cells are clipped at zone and hemisphere boundaries. The four altitude classes (plains, hills, low and high mountains) retain the existing thresholds of 300, 800 and 1500 m; they are a proxy for launch

orography. This construction groups geographical and seasonal context, not necessarily a common weather episode.

All main-cohort flights and their equal-flight weights are retained. We draw 1000 bootstrap replicates, paired across lags, and repeat with two seeds. The archived site-day bootstrap provides the reference. Table 3.4 compares group composition. Singleton groups remain, but contain only 1.4% of paraglider flights and 5.0% of hang-glider flights. The standard error of H is 3.97–4.05 and 2.00–2.05 times the site-day value, respectively; point estimates are unchanged.

	Site-day	Grid	Median size	Singletons (%)	Max. share (%)
Paragliders	25682	1901	3	33.5	4.4
Hang gliders	1768	267	2	43.4	13.4

Table 3.4. Bootstrap groups of the main cohort before and after cell–altitude–month grouping. “Before” are the site-day groups that resample every other interval in this chapter; “after” are the alternative groups introduced in this check, used only here. Median size is the number of flights per new group. Singletons are expressed as a percentage of new groups; maximum share is the largest group’s percentage of cohort flights. All years are pooled within each month.

We test sensitivity using 10, 25 and 50 km cells, shifting each grid by half a cell along either or both axes. Separately, we merge two or three adjacent months and shift the seasonal boundaries, retaining 50 km cells and the altitude classes (Fig. 3.4). These are alternative resampling partitions, not exclusions of flights.

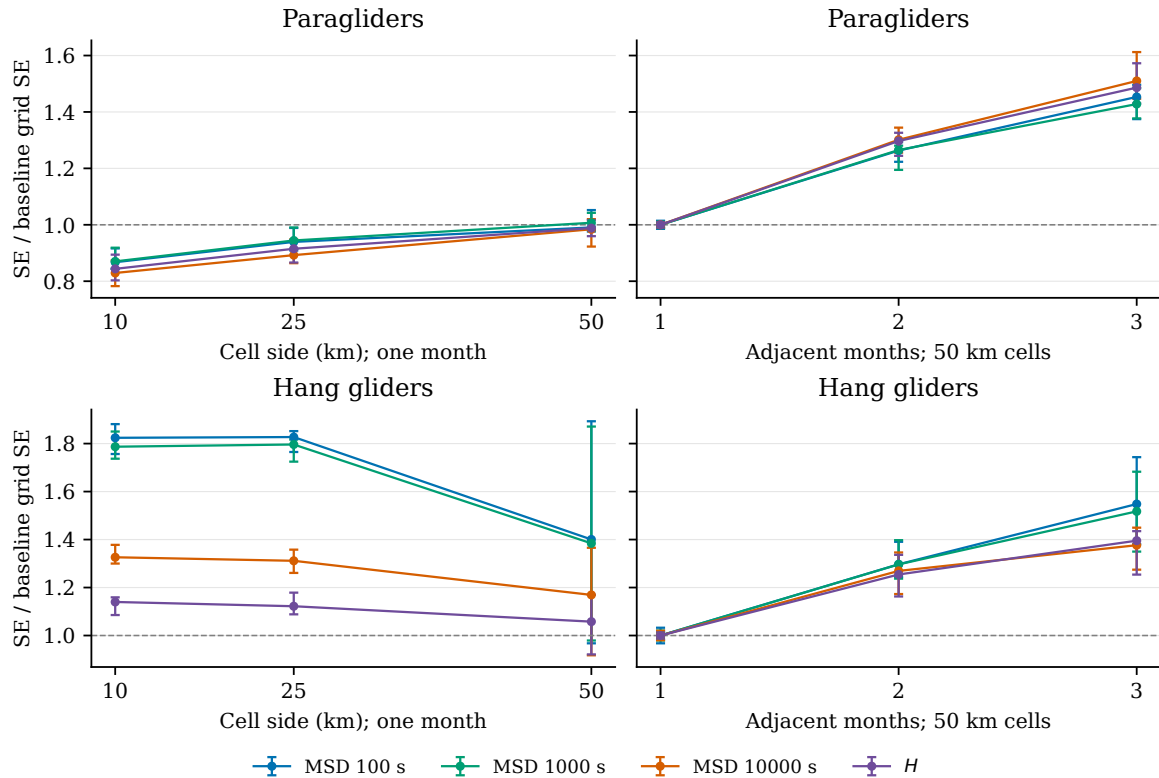


Figure 3.4. Standard errors relative to the unshifted 50 km, one-month grid baseline. Left: cell size and boundary shifts. Right: adjacent-month unions with seasonal boundary shifts. Markers are medians and bars span minima and maxima across partitions and two seeds; they are not confidence intervals. Altitude classes remain separate, and all years are pooled.

The uncertainty has not stabilised across these choices. Merging three months multiplies the median standard error of H by 1.49 for paragliders and 1.40 for hang gliders. For the latter, shifting the 50 km grid gives MSD errors at 1000 s ranging from 0.98–1.87 times the unshifted value. Larger groups therefore do not by themselves validate independence or interval coverage. Neighbouring cells, months and altitude classes can still share conditions, while persistent geographical differences can also affect the comparison [21]. The new intervals are conditional on this grouping; the archived site-day bands elsewhere remain the reference and are not replaced by this MSD/ H check.

3.1.3 Fixed population and the effect of duration selection

The general TA-MSD of Sec. 3.1.2 averages over whichever flights support each lag, so its population shrinks as τ grows: short flights and short segments drop out one after another. The mix of circuit types and launch sites that remains then changes with τ , and a change in the slope of $M_2(\tau)$ cannot be told apart from a change in the dynamics. Every statistic of the motion in the rest of this chapter is therefore measured on one population that does not depend on τ . Such a population is limited by its shortest member: the longest lag it supports is fixed by how long its segments are. Selecting only long segments gives more lags, and loses flights. This subsection asks whether that selection changes the measured H enough to matter, and chooses the cohort accordingly.

Cohorts. We select the segments once, using

$$\tau_{\max} \leq 0.8T_s, \quad \mathcal{C}_{\tau_{\max}} = \{f : \exists s \in f, T_s \geq \tau_{\max}/0.8\}, \quad (3.5)$$

where T_s is the segment's elapsed duration on the 10 s grid. The condition $\tau_{\max}/T_s \leq 0.8$ leaves $T_s - \tau_{\max} \geq 0.2T_s$, one fifth of the segment, for the origins at the largest lag. The threshold 0.8 fixes how many origins remain there. For the main cohort, $\mathcal{C}_{10000}$, segments last at least 12500 s, so each supplies at least 251 origins on the 10 s grid at $\tau = 10000$ s. If a flight has segments of 14000 and 3000 s, only the first contributes, even at $\tau = 10$ s. Flights and selected segments therefore remain the same at every lag.

Weights. For the fixed cohort, $\mathcal{I}_f(\tau)$ and $n_f(\tau)$ refer only to its selected segments, and Eq. (3.1) still applies once $\mathcal{C}(\tau)$ and $N(\tau)$ are replaced by the τ -independent cohort and its size:

$$m_f(\tau) = \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} R_{fi}(\tau)^2, \quad M_2(\tau) = \frac{1}{N} \sum_{f \in \mathcal{C}_{10000}} m_f(\tau), \quad N = |\mathcal{C}_{10000}|. \quad (3.6)$$

Expanding the flight average $m_f(\tau)$ assigns every admissible origin of flight f the same explicit weight

$$w_f(\tau) = \frac{1}{Nn_f(\tau)}, \quad \sum_{i \in \mathcal{I}_f(\tau)} w_f(\tau) = \frac{1}{N}, \quad (3.7)$$

so that $M_2(\tau) = \sum_f \sum_i w_f(\tau) R_{fi}(\tau)^2$.

The same weights define the mean $\langle A \rangle_\tau = \sum_f \sum_i w_f(\tau) A_{fi}$ of any per-origin quantity A_{fi} (e.g. the signed displacement $\Delta x_{j,fi}(\tau)$ below), used from Eq. (E.12) onwards.

Effect of the selection on H . We assess duration selection with the nested cohorts $\mathcal{C}_{100}$ and $\mathcal{C}_{1000}$. They contain the main cohort's segments, additional shorter segments of its flights, and additional flights. Selecting long segments also changes who is in the sample. For example,

Discipline	$\tau_{\max}$ (s)	Flights	Segments	Groups
Paragliders	100	155,044	188,312	82,650
Paragliders	1000	154,482	167,152	82,390
Paragliders	10000	46,273	46,367	25,682
Hang gliders	100	6,060	7,515	4,264
Hang gliders	1000	6,044	6,610	4,253
Hang gliders	10000	2,326	2,327	1,768

Table 3.5. Nested duration cohorts and their site–day groups, the resampling units of Sec. 3.1.2. The nominal minimum durations $\tau_{\max}/0.8$ are 125, 1250 and 12500 s; since T_s only takes multiples of 10 s, the effective cutoffs are 130, 1250 and 12500 s.

the declared circuit class, open-distance flights against closed routes such as triangles and out-and-return flights (Sec. 3.2), shifts with the cohort: the share of closed routes rises from 56.6% in $\mathcal{C}_{100}$ to 61.0% in $\mathcal{C}_{10000}$ for paragliders, and from 64.0% to 76.5% for hang gliders.

Duration selection can also change the mix of launch environments and seasons. The MSD and its fitted H in $\mathcal{C}_{10000}$ could therefore differ from those in $\mathcal{C}_{100}$, which retains almost the entire eligible archive. We compare the cohorts to assess how much extending the accessible lag range changes the MSD curve and its effective growth exponent.

We compare two cohorts A and B over the same lag interval. In each bootstrap replicate, both use the same site–day group draw: we compare $H_A^{(1)}$ with $H_B^{(1)}$, $H_A^{(2)}$ with $H_B^{(2)}$, and so on. This gives 1000 paired differences,

$$\Delta H^{(b)} = H_A^{(b)} - H_B^{(b)}, \quad b = 1, \dots, 1000.$$

The 5th and 95th percentiles of these differences form the interval in Table 3.6.

The cohorts share flights and site–day groups. Using the same draws gives each shared group the same multiplicity in both cohorts, preserving the dependence between their estimated exponents. When a draw shifts both exponents by similar amounts in the same direction, those shifts largely cancel in ΔH . Independent draws would lose this cancellation and overestimate the uncertainty of the difference.

Discipline	Fit range (s)	Contrast	ΔH [5–95%]
Paragliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{10000}}$	-0.00006 [-0.00009, -0.00004]
Paragliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{1000}}$	-0.01687 [-0.01712, -0.01660]
Paragliders	10–100	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01680 [-0.01705, -0.01654]
Paragliders	10–1000	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01703 [-0.01733, -0.01674]
Hang gliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{1000}}$	-0.00002 [-0.00013, 0.00011]
Hang gliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{10000}}$	-0.01646 [-0.01748, -0.01552]
Hang gliders	10–100	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01644 [-0.01746, -0.01550]
Hang gliders	10–1000	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01771 [-0.01879, -0.01668]

Table 3.6. Paired differences of the cohort exponents, with 5–95% bootstrap intervals. Each comparison uses identical lags and draws. The $\mathcal{C}_{100}$ – $\mathcal{C}_{1000}$ shift is negligible in magnitude ($|\Delta H| < 10^{-4}$), although statistically resolved for paragliders. Contrasts with $\mathcal{C}_{10000}$ exclude zero and represent a larger, but still small, shift of about 2% in H .

Result. On 10–100 s, $\mathcal{C}_{100}$ and $\mathcal{C}_{1000}$ differ in H by less than 10^{-4} in both disciplines. The paired interval resolves this tiny difference for paragliders; for hang gliders it includes zero. The two cohorts share 99.6% (paragliders) and 99.7% (hang gliders) of their flights, so in each replicate their exponents move together and their difference has a very narrow interval.

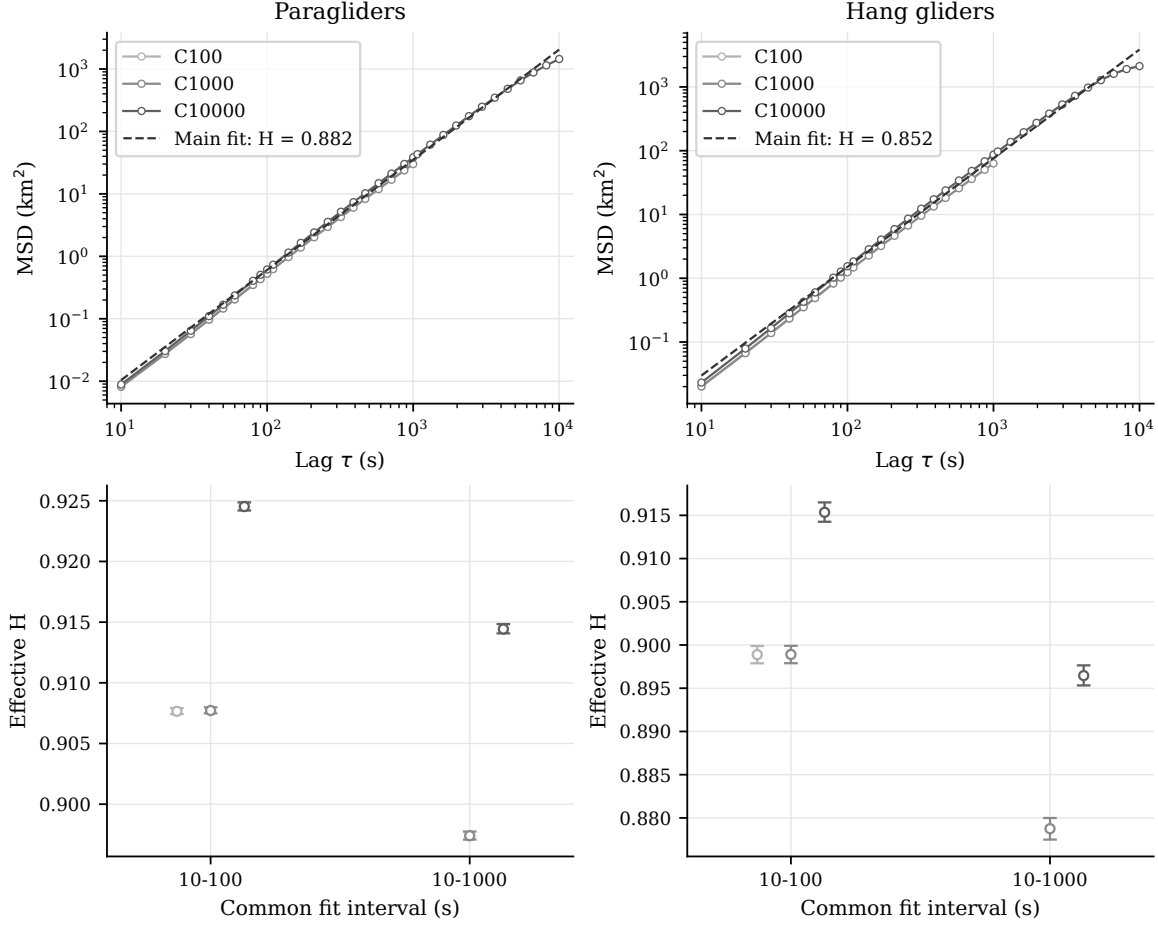


Figure 3.5. Effect of duration selection. Top: MSD in the three nested cohorts, drawn lightest to darkest as C_{100} , C_{1000} , C_{10000} ; dashed lines fit the main cohort over 10–10000 s. The darkest curve is drawn on top and hides the lighter cohorts wherever they coincide, which is almost the whole visible range: the result below is that duration selection barely changes H . Bottom: effective H with 90% intervals, fitted over 10–100 s for all three cohorts and 10–1000 s for the two longer cohorts. Fits within each comparison use identical lag values.

For the present analysis, we regard the two cohorts as having practically the same H on this lag range.

The main cohort behaves differently. Its effective exponent is larger than that of every broader cohort, in both fit intervals and both disciplines, and every one of these paired differences excludes zero. The difference between the main cohort and a broader one lies between 0.0164 and 0.0177 in H , which is 1.8–2.0% of the exponent itself.

Two candidate explanations concern who, and under what conditions, flies long enough to enter the main cohort. One is that sustaining a segment of 12500 s, nearly four hours, selects more experienced pilots. Vilpellet et al. [2] find that, taking the class of the glider as a proxy for experience, the MSD exponent of the search phase grows with experience, and interpret this as skill enhancing the ability to probe, interpret and exploit atmospheric cues. Experienced pilots may therefore locate and exploit thermals more efficiently, cutting the time spent searching rather than gliding, which raises H . The other is that it selects better flying days, where stronger or more reliable lift shortens that same search time for any pilot.

We accept this small bias and adopt C_{10000} for every analysis below that requires a fixed

population. A shift of about 2% in H is small beside what the cohort provides: a population that stays the same over three decades of lag, 10–10000 s, against two for $\mathcal{C}_{1000}$ and one for $\mathcal{C}_{100}$.

3.2 Conditional mean-square displacement

We examine how MSD amplitude and its effective growth exponent H vary with three recorded characteristics:

- *Circuit type*: open distance flights versus closed routes.
- *Launch environment*: initial altitude as a proxy for plains, hills and mountains, followed by regional comparisons within these settings.
- *Equipment*: wing class as a proxy for pilot experience.

Marginal comparisons pool the other characteristics; *stratified comparisons* keep a selected category fixed, for example comparing circuits within each altitude class. This tests whether a contrast survives stratification and whether its magnitude depends on the other factor. These observational comparisons differ from a *one-factor-at-a-time* (OFAT) experiment, which varies one input while holding the others at fixed settings and does not estimate interactions [22]. Here the remaining flight conditions can still differ between groups.

We use the larger paraglider population to retain support after stratification. Every group $\mathcal{C}_g$ belongs to the fixed cohort $\mathcal{C}_{10000}$, with the same retained segments throughout $10 \leq \tau \leq 10000$ s:

$$M_{2,g}(\tau) = \frac{1}{N_g} \sum_{f \in \mathcal{C}_g} m_f(\tau), \quad N_g = |\mathcal{C}_g|, \quad M_{2,g}(\tau) \propto \tau^{2H_g}, \quad (3.8)$$

Differences in H use the same 1000 site-day bootstrap draws across groups and lags, paired as in Section 3.1.3. The intervals retain the limitations discussed in Section 3.1.2.

3.2.1 Declared circuit: the marginal comparison

The catalogue separates open distance flights from closed routes, including triangles and out-and-return flights. The declaration describes the objective; the recorded, trimmed trajectory need not end at its starting point.

We first pool all initial altitudes within each circuit class. The fixed cohort contains 17,961 open flights, 28,234 closed flights and 78 unclassified flights; the last group enters neither curve in Figure 3.6.

The open-route MSD grows faster and reaches a larger displacement amplitude at long lags. Its effective exponent is 0.910 [0.909, 0.911], compared with 0.856 [0.855, 0.856] for closed routes. The paired difference $H_{\text{open}} - H_{\text{closed}}$ is 0.054 [0.053, 0.056], so the separation is resolved under the adopted bootstrap.

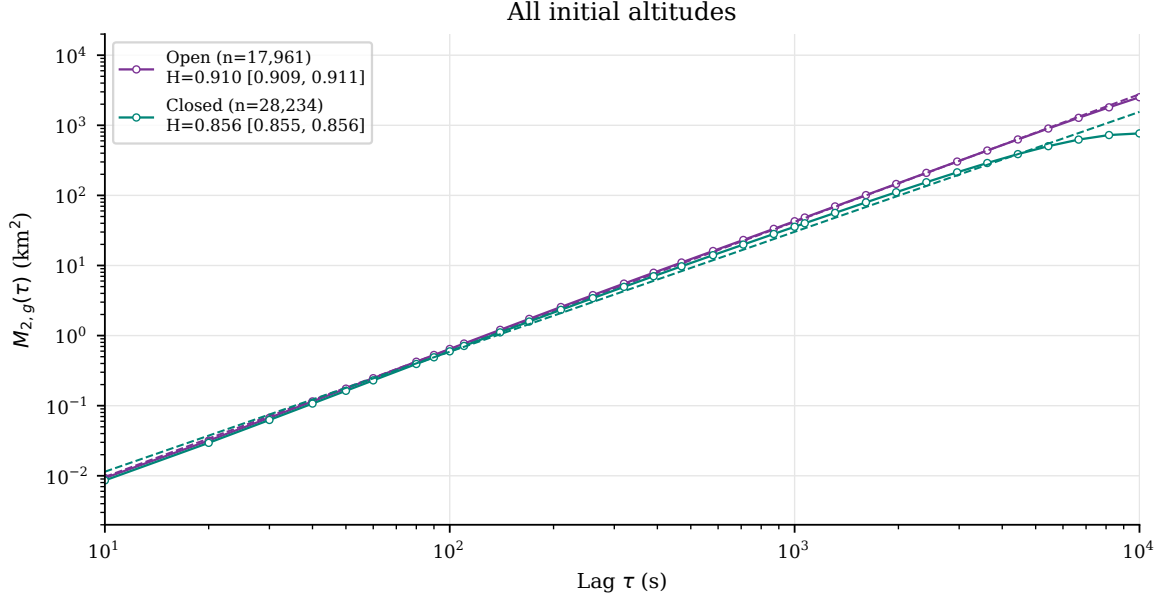


Figure 3.6. Paraglider MSD by declared circuit, pooling initial altitudes. Solid curves join the equal-flight estimates; dashed lines are fits over 10–10000 s.

Circuit geometry and the separation of the curves. Closed routes must reverse part of their outward progress. Intervals spanning turns or returns can therefore have smaller net displacements, consistent with the slower long-lag MSD growth. The relevant timescale depends on the route geometry and leg durations. For a flight of duration T with equal leg times, the first major turn occurs at $T/2$ for an out-and-return flight and $T/3$ for a triangle. These are times along the route: even a short lag can straddle a turn, because the MSD averages over starting points along each segment. Testing the role of triangular geometry requires separating the two closed route types and comparing their actual leg durations.

3.2.2 Initial altitude, circuit composition and interaction

The launch altitude z_0 defines four classes. It is the GNSS altitude of the first fix of the trimmed track. The classes are: Plains ($z_0 < 300$ m), Hills ($300 \leq z_0 < 800$ m), Low mountains ($800 \leq z_0 < 1500$ m), and High mountains ($z_0 \geq 1500$ m). All circuit declarations, including unclassified ones, contribute to Figure 3.7. These are descriptive comparisons of the flight populations launched in each setting, pooling their different circuit compositions.

Plains have the largest exponent, 0.950 [0.948, 0.951], followed by Hills, 0.929 [0.926, 0.931]. Both exceed Low mountains, 0.866 [0.865, 0.867], and High mountains, 0.875 [0.874, 0.876]. The broad hierarchy is **Plains** > **Hills** > **Mountains**, with High mountains slightly above Low mountains.

The Plains–Low mountains difference is 0.084 [0.082, 0.085], about 10% of the smaller exponent. Interpreting this gap requires accounting for the different circuit mixtures.

Open flights account for 91.1% of Plains and 71.1% of Hills, against 33.5% and 35.0% in the two mountain classes. The lowland averages therefore give much greater flight weight to open routes.

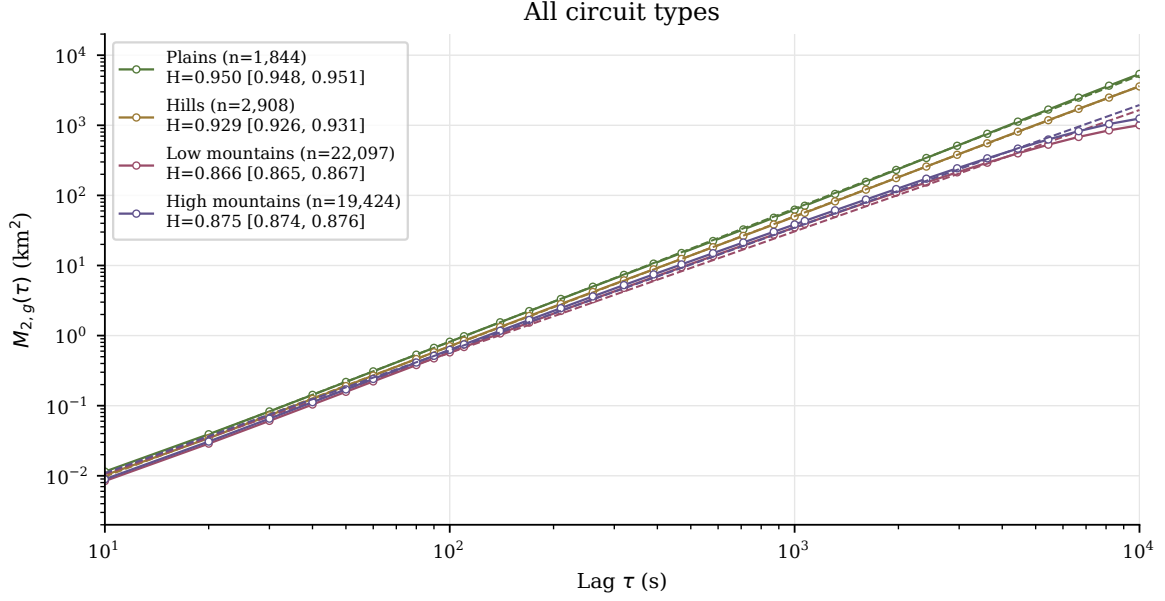


Figure 3.7. Paraglider MSD by initial altitude, pooling circuit types and equipment. The four groups partition the fixed cohort. These marginal curves combine launch environment with unequal circuit proportions; they do not isolate an orographic effect.

To explain the altitude curves, the relevant groups are the circuit types *within each altitude class*. Let a denote that class and $c \in \{\text{open, closed, unclassified}\}$ its circuit categories, omitting empty categories. With $N_{c,a}$ flights in each cell and $N_a = \sum_c N_{c,a}$, the equal-flight estimator gives

$$M_{2,a}(\tau) = \sum_c p_{c|a} M_{2,c,a}(\tau), \quad p_{c|a} = \frac{N_{c,a}}{N_a}, \quad \sum_c p_{c|a} = 1. \quad (3.9)$$

The flight fractions $p_{c|a}$ stay fixed over the lag range. The fraction of the class MSD supplied by circuit c is

$$w_{c|a}(\tau) = \frac{p_{c|a} M_{2,c,a}(\tau)}{M_{2,a}(\tau)}, \quad \sum_c w_{c|a}(\tau) = 1. \quad (3.10)$$

In Figure 3.8, shares sum to one *within each altitude class*, including its small unclassified component.

- **A circuit's contribution depends on its MSD as well as its number of flights.** The larger open MSD amplifies the circuit imbalance in Plains and Hills. At $\tau = 10000$ s, open routes supply 99.4% and 97.6% of their respective class MSDs. In Low mountains, open flights are only 33.5% of the class, yet supply 60.6% of its MSD at that lag: a minority of flights can contribute most of the MSD.
- **Circuit contributions change with lag and shape the pooled growth as well as its amplitude.** When open and closed MSDs grow at different rates, $w_{c|a}(\tau)$ changes even though $p_{c|a}$ stays fixed. The open contribution rises towards long lags, particularly in the mountain classes. In High mountains, closed routes still supply 51.9% at 10000 s, whereas their contribution in Plains and Hills is already small. The exponent H must therefore be fitted to the combined curve $M_{2,a}$; it is not generally a flight-count-weighted average of circuit exponents.

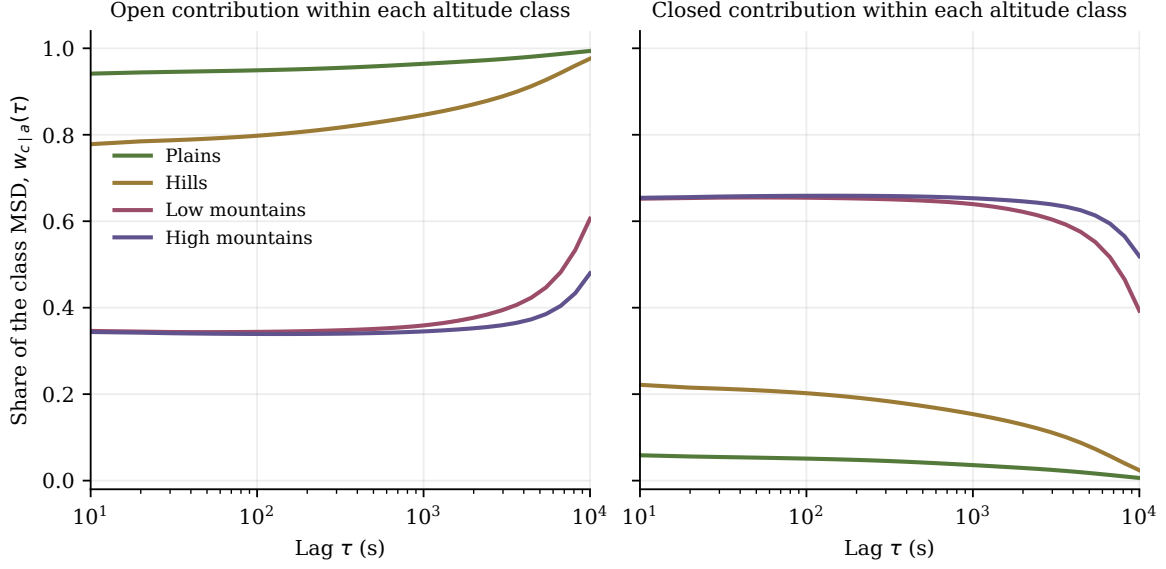


Figure 3.8. Circuit contributions to the MSD within each initial-altitude class. Left: $w_{\text{open}|a}(\tau)$. Right: $w_{\text{closed}|a}(\tau)$. The denominator is the full MSD of the indicated altitude class, including unclassified flights. Their unplotted contribution is at most 0.21% over these curves. These are observed shares, without uncertainty bands.

Circuit contrasts within altitude classes. Does the lowland advantage persist when circuit type is held fixed? Figure 3.9 crosses circuit type with altitude class; Table 3.7 reports counts, exponents and amplitudes.

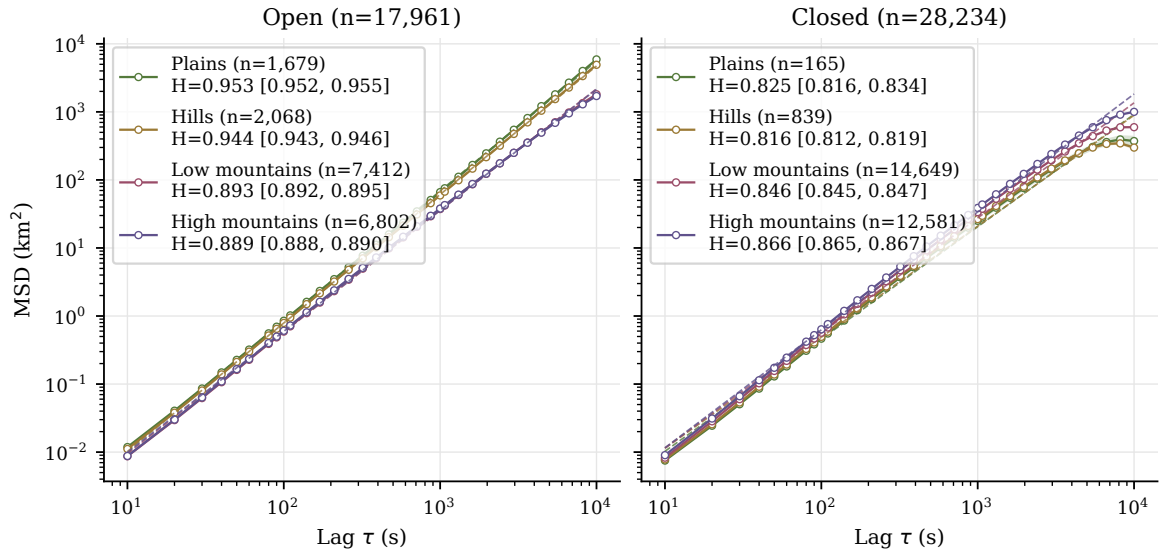


Figure 3.9. Paraglider transport by declared circuit and initial altitude, with fixed flights and segments throughout 10–10000 s. Markers give MSD estimates, shaded bands their pointwise 90% intervals, and dashed lines the global log–log fits. Legends report the flight count and the fitted H with its 90% interval. The identical axes permit comparison of displacement amplitude and growth rate.

- **Open routes have larger H than closed routes in every altitude class.** The paired gaps (nominal 90% intervals) are 0.128 [0.120, 0.138] in Plains, 0.129 [0.125, 0.132] in Hills, 0.048 [0.046, 0.049] in Low mountains and 0.023 [0.021, 0.024] in High

Circuit	Initial altitude	N	G	H [5–95%]	$M_2(10^4)$ (km ²)
Open	Plains	1,679	1,140	0.953 [0.952, 0.955]	5900
Open	Hills	2,068	1,634	0.944 [0.943, 0.946]	4920
Open	Low mountains	7,412	5,255	0.893 [0.892, 0.895]	1814
Open	High mountains	6,802	4,794	0.889 [0.888, 0.890]	1710
Closed	Plains	165	157	0.825 [0.816, 0.834]	372
Closed	Hills	839	756	0.816 [0.812, 0.819]	299
Closed	Low mountains	14,649	7,954	0.846 [0.845, 0.847]	596
Closed	High mountains	12,581	7,406	0.866 [0.865, 0.867]	1001

Table 3.7. Main-cohort flights N , site-day groups G , global effective exponents and displacement amplitude at 10000 s. Intervals come from the coupled cluster bootstrap. Even the smallest cell retains 165 flights from 157 groups.

mountains. All exclude zero, and the gap is smaller in mountains. The pooled circuit contrast therefore cannot be explained solely by the open population’s greater lowland share.

- **The altitude ordering reverses with circuit type.** Both lowland classes have larger H than either mountain class for open routes, but smaller H for closed routes. The four lowland-minus-mountain differences lie between 0.051 and 0.064 for open routes; for closed routes they are negative, with magnitudes between 0.021 and 0.051. All eight paired intervals exclude zero.

This reversal quantifies an interaction: the association between launch environment and H depends on circuit type. The predominance of open flights contributes to the pooled lowland advantage, but does not explain away the opposite within-circuit orderings. Mountain flights do not have smaller H in general.

Interpreting circuit choice and terrain constraints. The return objective explains the expected sign of the circuit contrast; terrain may explain why its magnitude changes. Sun-heated mountain slopes can feed convection that rises along the slope and detaches near a crest; thermal sources need not lie on the crest itself. Access to lift and terrain clearance can channel both circuit types along slope-and-ridge corridors [23, Chapters 9–10]. In plains, smaller physical barriers and separated heat sources, such as dry dark fields and paved areas [24], may leave more route choices.

- **Open routes.** Following ridges and detouring between usable lift sources could interrupt outward progress, making mountain open flights less persistent than lowland open flights. This is consistent with their smaller H .
- **Closed routes.** The same terrain constraints can alter the size and timing of a return circuit. Longer legs or later returns could suppress MSD growth less over 10–10000 s, consistently with the larger mountain closed H . Terrain constraints alone do not guarantee that timing change; it must be tested against measured leg and return times.

Mountain open routes may thus appear “less open” and closed routes “less closed” in their finite-lag statistics: their fitted exponents are closer, while their declared classes remain unchanged. This interpretation is a hypothesis; the conditional orderings are measured, but the proposed route changes have not yet been measured.

Observed climb locations in two viewer cells. Figure 3.10 compares two selected 5×5 km cells from the viewer: **High mountains #2 (H2)**, in the Aulp du Seuil area of the Chartreuse (Alps), and **Plains #1 (P1)**, in Suisse Normande within the broad Channel Coast box. Their ranks refer to all-time distinct crossing flights within each altitude band: 15,840 for H2 and 2,246 for P1. P1 is inland and contains local slopes despite its low starting-altitude classification.

The maps pool **all available dates and years, without a seasonal restriction**, using the viewer’s Vilpellet climb classification for both disciplines, without the MSD cohort’s duration restriction. Each point marks a climb crossing at 100 or 600 m above a fixed cell reference: 1756 m ASL for H2 and 185 m for P1, estimated from five and 1165 screened starts, respectively. The viewer labels this offset AGL; it is not clearance above the terrain beneath each point. Red lines trace approximate crests derived independently from the IGN RGE ALTI terrain model [25]: local transverse height maxima on a 25 m grid, smoothed at 50 m. They include secondary ridges but are scale-dependent, not an exhaustive inventory; their ground positions stay fixed at both heights.

Climb locations spread out more visibly with height in P1 than in H2. At 100 m, P1 crossings cluster near local slopes; at 600 m, gliders cross the plane while climbing over a much broader part of the cell. In H2, both heights retain concentrations near the same terrain corridors, despite some broadening at 600 m. The two main bands at 100 m follow mainly the outer slopes of separate ridges—Lances de Malissard to the west and Aulp du Seuil–Bellefont to the east of the Vallon de Marcieu—rather than opposite slopes of a single central ridge. Thus gliders exploit lift across a wider area aloft in P1, supporting greater spatial freedom for route choices, while repeatedly using narrower terrain-associated corridors in H2.

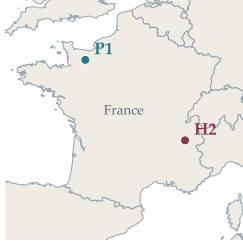
Distributed sources and wind advection could jointly explain the broader use of space aloft. In an idealised steady plume, horizontal advection $\mathbf{u}_h$ and upward air speed $w > 0$ give

$$\Delta \mathbf{r}_h \simeq \frac{\mathbf{u}_h}{w} \Delta z.$$

Drift therefore accumulates during ascent. Variation in wind direction, speed or updraft strength across flights and days can broaden the pooled crossing pattern with height. A common, constant wind and rise speed would instead translate the source pattern; inclination alone does not imply lift everywhere at a given height.

One possible mountain profile combines sheltered, heated slopes with stronger winds above the crests: convection initially rises along or away from the slope, then tilts more strongly aloft; strong shear can disrupt it [23, Chapter 9]. Exposed lowland sources may experience advection from lower heights. These are possible conditions, not a general wind hierarchy: valleys can also channel and accelerate flow [26]. The maps do not establish either profile, and the inland P1 cell cannot be assigned a coastal wind regime from its regional label.

The maps show sampled climbs, including possible slope lift, rather than tracked thermal centres. Different flights and weather support each height, and low horizontal planes can intersect terrain. Their spatial contrast is observed; attributing it to wind requires following the same climbs across heights and matching wind profiles. Its contribution to H and generality across sites remain to be quantified.

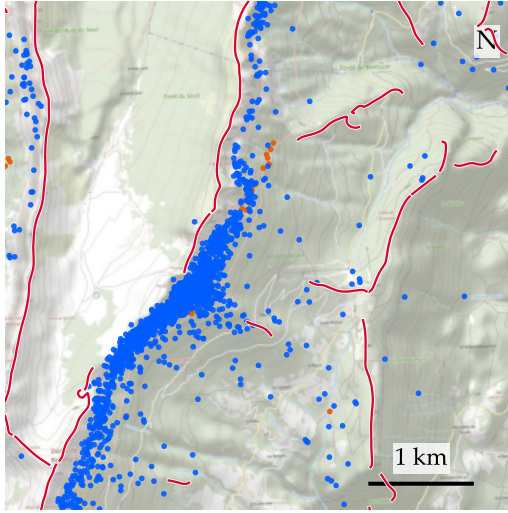


H2: High mountains #2, Chartreuse (Alps)
P1: Plains #1, Suisse Normande (Channel Coast)

All available dates and years; no seasonal filter.
 Locator markers show cell centres, enlarged for visibility.

Alps: Chartreuse (H2)

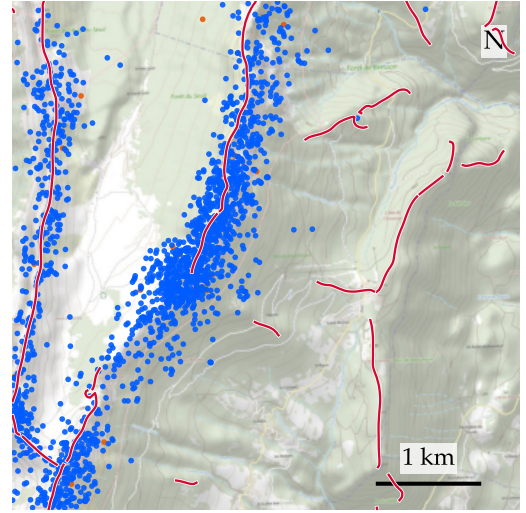
$z_{AGL} = 100$ m



1,856 crossings / 1,261 flights

Alps: Chartreuse (H2)

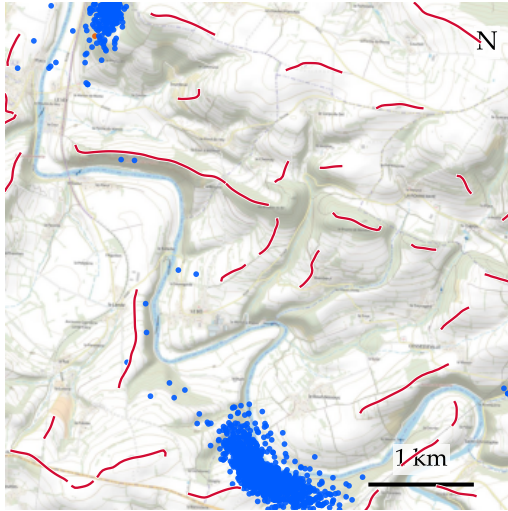
$z_{AGL} = 600$ m



1,868 crossings / 1,452 flights

Channel Coast: Suisse Normande (P1)

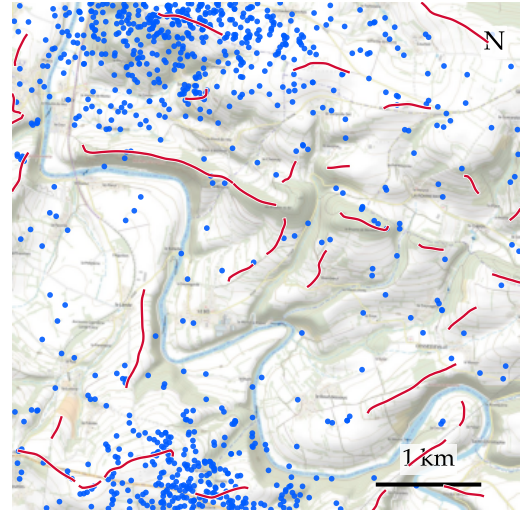
$z_{AGL} = 100$ m



1,441 crossings / 734 flights

Channel Coast: Suisse Normande (P1)

$z_{AGL} = 600$ m



901 crossings / 574 flights

Figure 3.10. Climb crossings in H2 and P1, at 100 and 600 m above each fixed cell reference, pooling all dates. Each pair covers the same 5×5 km footprint. Blue: paragliders; orange: hang gliders; red: approximate DEM-derived crests. Counts distinguish crossings from contributing flights, which differ between heights. Background: © IGN/Géoplateforme, Plan IGN; terrain: RGE ALTI, Licence Ouverte 2.0.

3.2.3 Regional differences within a broad launch environment

We compare Alps with Pyrenees, retaining only Low/High mountains, and Channel Coast with Champagne-Lorraine, retaining only Plains/Hills. This restricts the broad launch environment within each geographic comparison. For a box $\mathcal{B}$ defined in Table 2.9 and Figure 2.24a, the selections are

$$\begin{aligned}\mathcal{C}_{\mathcal{B},\text{mountains}} &= \{f \in \mathcal{C}_{10000} : (\lambda_{0f}, \varphi_{0f}) \in \mathcal{B}, z_{0f} \geq 800 \text{ m}\}, \\ \mathcal{C}_{\mathcal{B},\text{lowlands}} &= \{f \in \mathcal{C}_{10000} : (\lambda_{0f}, \varphi_{0f}) \in \mathcal{B}, z_{0f} < 800 \text{ m}\}.\end{aligned}\quad (3.11)$$

Within the fixed cohort, this removes 385 flights from the Alpine box and 103 from the Pyrenean box. The lowland restriction also removes 17 flights from Channel Coast and 24 from Champagne-Lorraine. Figure legends report the remaining counts.

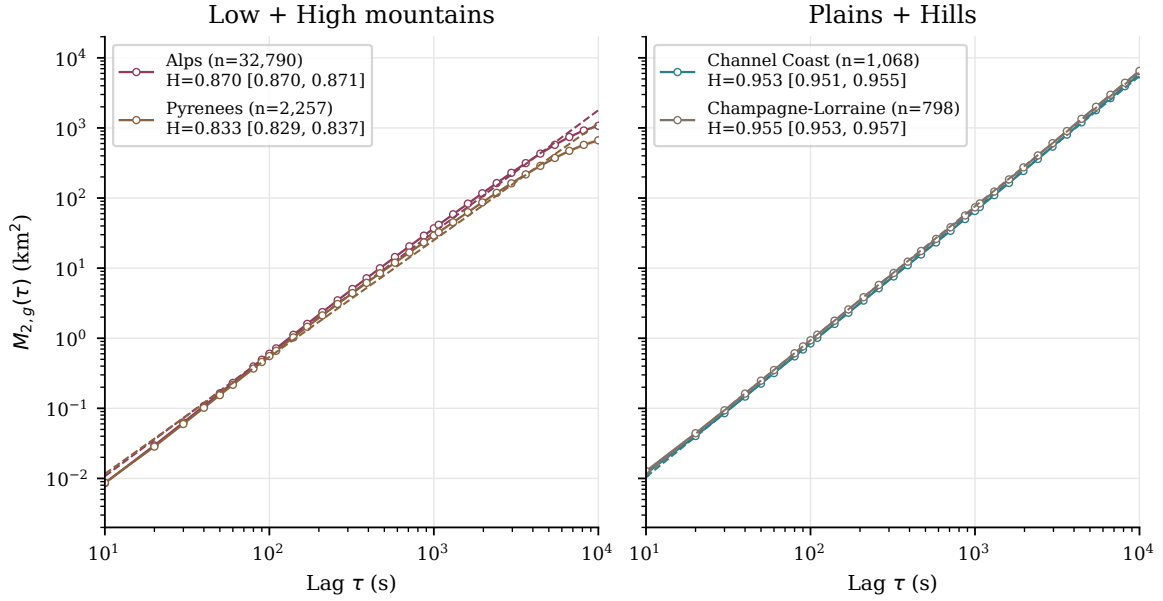


Figure 3.11. Regional paraglider MSDs. Left: Alps and Pyrenees. Right: Channel Coast and Champagne-Lorraine. The left panel retains only Low/High mountains within each box; the right retains only Plains/Hills. Circuit types and equipment classes are pooled. Counts and fits refer to the preselected populations; both panels share the same axes.

The Alpine exponent is 0.870 [0.870, 0.871], against 0.833 [0.829, 0.837] in the Pyrenees. Their paired difference is positive (Table 3.8), and Alpine flights also reach a larger MSD at long lags. Channel Coast and Champagne-Lorraine instead give very similar exponents, 0.953 [0.951, 0.955] and 0.955 [0.953, 0.957]. Their paired interval includes zero, although their MSD amplitudes differ.

Setting	Contrast	ΔH [5–95%]
Mountains	Alps – Pyrenees	0.037 [0.033, 0.041]
Lowlands	Coast – Champagne-Lorraine	−0.002 [−0.005, 0.001]

Table 3.8. Paired regional contrasts in H , after the terrain preselection: Low/High mountains for Alps–Pyrenees and Plains/Hills for the two lowland regions. Intervals are nominal 90% site-day bootstrap intervals.

The two lowland regions have similar growth exponents, whereas the identity of the mountain range matters in this comparison.

One possible explanation is a broadly distributed supply of thermal triggers over lowlands. Sun-heated asphalt and dark, ploughed fields can heat more rapidly than their surroundings and initiate convection [24]. Roads, fields and other heated patches recur across different plains. If their effective spacing and availability are statistically similar in Channel Coast and Champagne-Lorraine, changing the region could leave the growth exponent largely unchanged. This proposes comparable trigger statistics across the two lowlands, rather than establishing a spatially uniform thermal field. The source locations have not been mapped here, and the observed similarity concerns H , with different MSD amplitudes.

In mountains, heated slopes and terrain-associated lift tie the places where height can be recovered more closely to the arrangement of ridges and valleys. Thermals can flow upslope, while wind-driven slope lift also follows the terrain [23, Chapter 9]. These distinct mechanisms both make the geometry of the mountain range relevant to available routes. **We interpret the lower Pyrenean H and smaller long-lag MSD as consistent with stronger constraints on net progress than in the Alps.** Under this interpretation, the Alpine arrangement of usable slopes and connections between them offers more freedom in choosing where to travel, whereas the Pyrenean arrangement restricts those choices more strongly. This is a proposed explanation of the regional contrast, not a measured ranking of route accessibility: H does not count available directions or identify the geometry responsible for a trajectory.

Broad altitude selection still leaves different altitude distributions, circuit proportions, equipment and weather within each regional pair. The comparison cannot isolate terrain geometry or establish that all plains share the same transport statistics. Testing the proposed explanation would require independent maps of thermal triggers and usable terrain connections, together with route comparisons under matched flight and weather conditions.

3.2.4 Beginners and experts across launch environments

We use the equipment groups defined in Section 2.4:

beginners : EN A/B/C, experts : EN D/CCC.

Membership follows the catalogue wing class for each flight: an experienced pilot can fly an A/B/C wing, so these labels combine an experience proxy with equipment differences. We exclude tandem, non-certified and unclassified entries from this comparison: 2,167 fixed-cohort flights enter neither group. The retained groups contain 27,609 beginners flights and 16,497 experts flights.

Over all initial altitudes (Figure 3.12), the experts group reaches a larger MSD and has the larger effective exponent: 0.895 [0.893, 0.896], against 0.870 [0.869, 0.871] for beginners. The paired difference is 0.025 [0.024, 0.026]. A speed difference alone would multiply an otherwise identical MSD by a constant and leave its log-log slope unchanged. The observed separation in H therefore also concerns the lag dependence of transport, although it cannot distinguish pilot decisions from equipment, route or environmental differences.

Figure 3.13 separates altitude classes within the beginners group (left) and the experts group (right).

For both groups the altitude ordering is that of Figure 3.7: Plains and Hills have the larger H , followed by the two mountain classes. Comparing the two panels, the experts curve has the larger displacement amplitude at long lags and the larger H in every altitude class. All four paired differences in Table 3.9 exclude zero.

The size of the contrast depends on the environment: it is largest in Low mountains, smaller in High mountains, and smaller again in Plains and Hills. The pooled difference is larger than any within-class difference, showing that the all-altitude comparison also incorporates the groups' different altitude composition and MSD amplitudes.

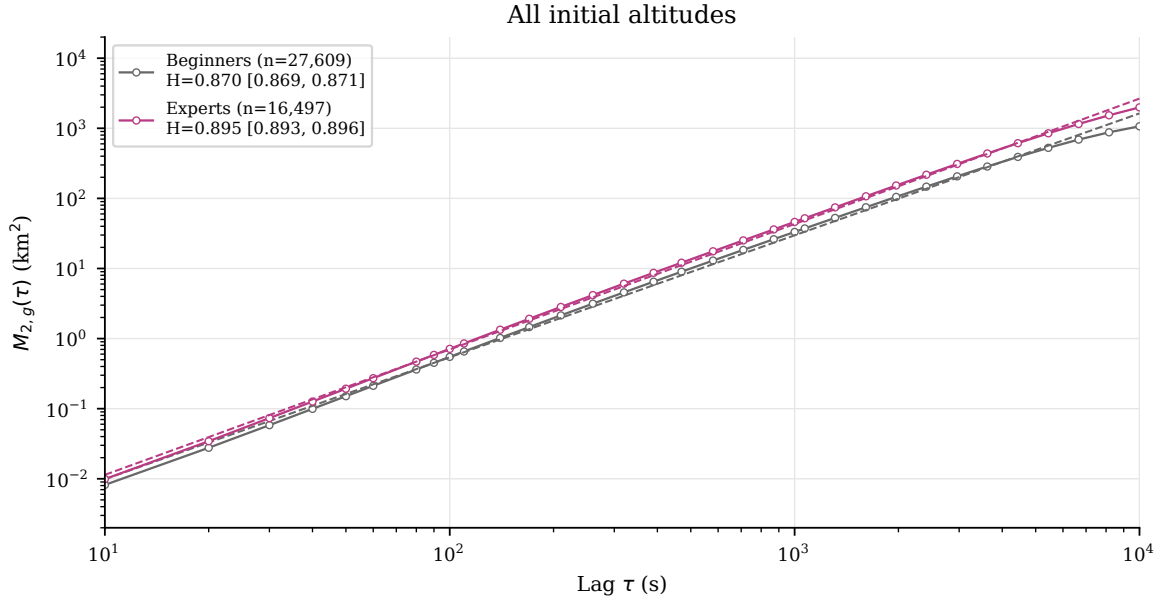


Figure 3.12. Paraglider MSD for beginners (EN A/B/C) and experts (EN D/CCC), pooling regions, initial altitudes and circuit types. These names refer to equipment proxies; individual pilot experience is not recorded by the grouping.

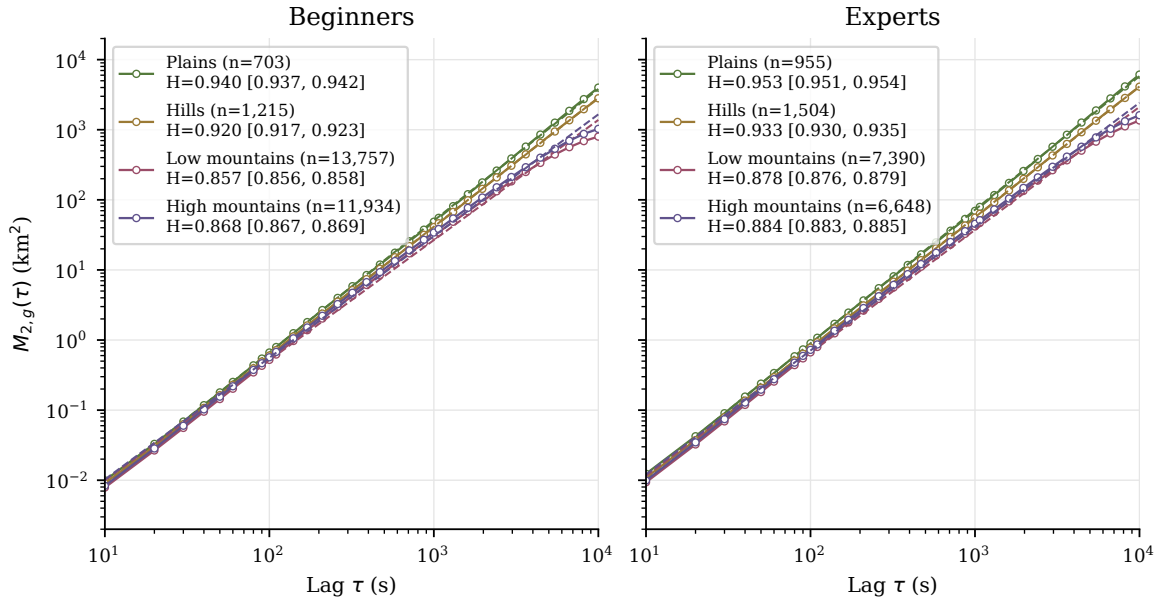


Figure 3.13. Paraglider MSD by initial altitude for beginners (left) and experts (right), pooling regions and circuit types. Estimator, fit range, axes and uncertainty convention are the same as in Figure 3.9; the number of contributing flights is given for every curve.

Setting	Contrast	ΔH [5–95%]
All altitudes	Experts – beginners	0.025 [0.024, 0.026]
Plains	Experts – beginners	0.013 [0.011, 0.015]
Hills	Experts – beginners	0.013 [0.009, 0.016]
Low mountains	Experts – beginners	0.021 [0.019, 0.022]
High mountains	Experts – beginners	0.016 [0.014, 0.017]

Table 3.9. Experts-minus-beginners contrasts in the effective exponent, first pooling all altitudes and then within each class. The same bootstrap draws provide every contrast.

To assess whether those contrasts themselves differ, define

$$\Delta H_a = H_{\text{experts},a} - H_{\text{beginners},a}, \quad D_{ab} = \Delta H_a - \Delta H_b, \quad (3.12)$$

where a and b denote altitude classes. We compute D_{ab} within each paired bootstrap replicate, rather than comparing the overlap of separate intervals. The Low mountains contrast exceeds those in Plains, Hills and High mountains with all three intervals excluding zero. The Plains–Hills difference is not resolved, and neither is the difference between High mountains and either lowland class at the adopted 90% level. There is therefore evidence that the equipment-group separation changes with launch environment, but no monotonic increase of that separation with altitude.

These contrasts concern long flights made on different wing classes. In particular, the beginners group does not represent the full population of novice pilots, and circuit composition remains mixed within each altitude cell.

3.2.5 From stratified comparisons to a joint analysis

The comparisons reveal two features that a joint analysis must account for:

- **Unequal composition:** the factor proportions differ between groups, and each subgroup contributes according to both its abundance and its MSD (Eq. (3.10)).
- **Interactions:** a contrast in H changes with another factor’s level, as observed for both circuit and equipment across altitude classes. This departure from additivity is distinct from unequal proportions.

The two-factor comparisons quantify these conditional patterns while pooling the remaining characteristics. They cannot yet separate circuit, launch environment and equipment jointly.

Factorial analysis as the next stage. A full *factorial experiment* combines every selected level of each factor with every level of the others [27]. This permits estimation of main effects, which summarise contrasts over specified settings of the other factors, and interactions, which describe how those contrasts change with the settings [28, Sections 9.3 and 9.7].

For this archive, the corresponding *factorial analysis* would cross four altitude classes, two circuits and two equipment groups, giving 16 cells, with H fitted over the same lag range in each. It would test whether the circuit–altitude patterns persist at common equipment settings and quantify interactions among all three factors. The observational data require explicit weighting of unequal cells and uncertainty estimates that respect shared flight conditions. Unmeasured weather and route selection would still limit causal interpretation. The provisional implementation in Section E.1, in *Experimentals*, remains under review; none of its numerical results is used here.

Why this matters for group versus solo flight. Group flights could have a higher pooled H because they contain more open routes or different terrain and equipment mixtures. Adding group/solo status as a fourth factor gives 32 cells and allows contrasts at common settings:

$$\Delta H_{a,c,e}^{\text{social}} = H_{\text{group},a,c,e} - H_{\text{solo},a,c,e},$$

where e denotes equipment class. Interactions with social status would test whether this contrast changes with circuit, altitude or equipment. Such joint comparisons are needed to distinguish the group/solo association from these composition effects; they require reliable social-flight labels and sufficient support in each compared cell.

3.3 Anisotropy and principal directions

Reserved for the analysis of anisotropy and principal directions. The existing PCA results remain in the experiments volume; they are not updated in this chapter.

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Appendices

Appendices 2.A and 2.B supply the spectral and geodetic definitions used in Chapter 2.

Appendix 2.A

Power spectral density and Welch's method

The spectra in Chapter 2 compare fluctuations in recorded altitude. They contain both flight dynamics and measurement error. A spectral estimate does not separate those contributions without further assumptions.

2.A.1 From covariance to a spectrum

Let $x_n = z_n - \mathbb{E}[z_n]$ be a real, uniformly sampled, second-order stationary process, with sampling interval $\Delta t = 1/f_s$ and covariance $C_k = \mathbb{E}[x_n x_{n+k}]$. When the covariance is summable, its two-sided spectral density is

$$S_2(f) = \Delta t \sum_{k=-\infty}^{\infty} C_k e^{-2\pi i f k \Delta t}, \quad |f| \leq f_s/2. \quad (2.A.1)$$

For real signals S_2 is even. The one-sided density doubles the positive-frequency contribution, giving

$$\int_0^{f_s/2} S_1(f) df = C_0 = \text{Var}(x_n). \quad (2.A.2)$$

Its units are $\text{m}^2 \text{Hz}^{-1}$ for an altitude signal. The Nyquist frequency $f_s/2$ limits distinguishable sampled frequencies: physical motion above that frequency may be aliased into the observed band.

A full flight altitude series need not be stationary. Equation (2.A.2) therefore gives the stationary reference interpretation; it is not a claim that the variance of an entire climbing and descending flight equals the integral of its detrended Welch estimate. Here the spectrum is a finite-record diagnostic, computed after detrending shorter windows.

2.A.2 The estimator used here

For window j of length L , subtract its least-squares linear trend and denote the residuals by $y_{j,n}$. With a periodic Hann taper w_n , define

$$I_j(f_k) = \frac{\left| \sum_{n=0}^{L-1} w_n y_{j,n} e^{-2\pi i k n / L} \right|^2}{f_s \sum_{n=0}^{L-1} w_n^2}, \quad f_k = k f_s / L. \quad (2.A.3)$$

This is the two-sided normalisation. For a one-sided estimate, positive bins are doubled except for the Nyquist bin when L is even; the zero-frequency bin is also left unchanged. Welch's

estimate averages the resulting periodograms over K overlapping windows [29]:

$$\hat{S}_1(f_k) = \frac{1}{K} \sum_{j=1}^K I_{j,1}(f_k). \quad (2.A.4)$$

The implementation uses $L = 256$, half-window overlap, linear detrending and the arithmetic mean within each flight. The frequency spacing is f_s/L ; tapering broadens the effective resolution. Details of the eligible intervals and the separate aggregation across flights are given in [Choice of altitude channel](#).

An individual periodogram has substantial sampling variability. Under suitable stationarity and weak-dependence conditions, averaging independent windows reduces its variance approximately by K . Overlapping windows are correlated, so their effective number is smaller. Increasing L improves frequency resolution but leaves fewer windows to average; keeping L fixed also retains smoothing bias as the record grows. The mean Welch estimator is not, by itself, resistant to arbitrarily large outliers.

Linear detrending suppresses slow variation within each window; it does not remove all physical motion or isolate sensor noise. Likewise, a high-frequency plateau can be compatible with measurement noise without uniquely identifying its cause. Comparisons between GNSS and barometric spectra must account for missing-channel selection, sampling, interpolation and the response of any subsequent filter.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.20), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a , flattening f , $b = a(1 - f)$ the semi-minor axis, $e^2 = f(2 - f) = 1 - b^2/a^2$ the first eccentricity squared [15].

In the exact transform, h denotes ellipsoidal height. The GNSS channel in an arbitrary HPG record need not follow this convention (Sec. 2.7.1). For orthometric height H , the relation is $h = H + \mathcal{N}(\varphi, \lambda)$, where $\mathcal{N}$ is geoid undulation. The pipeline currently inserts the recorded GNSS value as a height proxy; it does not infer a datum or apply a geoid correction. A height error δh perturbs a horizontal displacement from the origin by approximately $|\delta h|d/R_\oplus$ on a short, level route. This is a sensitivity estimate, not an established error bound for every recorder. Moreover, a spatially varying geoid correction does not cancel exactly from altitude increments.

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O : the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X - Z plane by λ ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p \cos \lambda, \quad Y = p \sin \lambda. \quad (2.B.1)$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h .

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \quad (2.B.2)$$

The geodetic latitude is defined through the surface *normal* (Figure 2.13). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2, Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan \varphi = \frac{Z_s/b^2}{p_s/a^2} \implies Z_s = \frac{b^2}{a^2} p_s \tan \varphi = (1 - e^2) p_s \tan \varphi. \quad (2.B.3)$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1 - e^2)$,

$$\frac{p_s^2}{a^2} \left(1 + (1 - e^2) \tan^2 \varphi \right) = 1 \implies p_s^2 = \frac{a^2 \cos^2 \varphi}{\cos^2 \varphi + (1 - e^2) \sin^2 \varphi} = \frac{a^2 \cos^2 \varphi}{1 - e^2 \sin^2 \varphi}, \quad (2.B.4)$$

that is,

$$p_s = N(\varphi) \cos \varphi, \quad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad (2.B.5)$$

and, back through (2.B.3),

$$Z_s = (1 - e^2) N(\varphi) \sin \varphi. \quad (2.B.6)$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a computation shows that, away from the poles, it is the distance along the normal from the surface point to the polar axis (the segment $Q-P$ of Figure 2.13): the normal direction in the meridian plane is $(\cos \varphi, \sin \varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N \cos \varphi = 0$ by (2.B.5). At the poles the normal lies on the polar axis, so this intersection construction is degenerate; the formula for N remains defined by continuity.

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi). \quad (2.B.7)$$

Adding $h \hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h) \cos \varphi \cos \lambda, \quad Y = (N + h) \cos \varphi \sin \lambda, \quad Z = (N(1 - e^2) + h) \sin \varphi, \quad (2.B.8)$$

which is Eq. (2.20). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $(N(1 - e^2) + h)$ is the ellipsoid's flattening at work: on a sphere ($e = 0$) both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan \psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2) \sin \varphi}{N \cos \varphi} = (1 - e^2) \tan \varphi, \quad (2.B.9)$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where $\tan$ is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2 . Using $\tan(\varphi - \Delta) \simeq \tan \varphi - \Delta (1 + \tan^2 \varphi)$ in (2.B.9),

$$\tan \varphi - \Delta (1 + \tan^2 \varphi) \simeq (1 - e^2) \tan \varphi \implies \Delta \simeq \frac{e^2 \tan \varphi}{1 + \tan^2 \varphi} = e^2 \sin \varphi \cos \varphi, \quad (2.B.10)$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad (2.B.11)$$

which is the expression quoted in Sec. 2.7.5. At this order it is largest at $\varphi = 45^\circ$, where it equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192^\circ \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus (e^2/2) \approx 21$ km at mid-latitudes—the ~ 20 km quoted in the body. The error vanishes at the equator and at the poles; the leading-order approximation is symmetric about 45° .

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing toward increasing latitude; Up, along the ellipsoid normal (2.B.7). Away from the poles, East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial \mathbf{X} / \partial \lambda}{\|\partial \mathbf{X} / \partial \lambda\|} = (-\sin \lambda_0, \cos \lambda_0, 0), \quad (2.B.12)$$

At a pole, longitude fixes a conventional orientation of the horizontal axes; the displayed vector still defines that convention although the derivative vanishes. Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0) , and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin \varphi_0 \cos \lambda_0, -\sin \varphi_0 \sin \lambda_0, \cos \varphi_0). \quad (2.B.13)$$

The ENU components of a displacement $\Delta \mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E \cdot \Delta \mathbf{X}$ and likewise for N and U ; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix}}_R \Delta \mathbf{X}, \quad (2.B.14)$$

whose rows are orthonormal by construction, so R is a rotation ($R^\top R = 1$) [16]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.20), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Two different distances must be distinguished. On a sphere, an arc of length d subtends a three-dimensional chord

$$c = 2R_\oplus \sin \frac{d}{2R_\oplus} = d - \frac{d^3}{24 R_\oplus^2} + O(d^5 / R_\oplus^4). \quad (2.B.15)$$

The horizontal EN projection from a frame anchored at one endpoint is instead

$$d_{EN} = R_\oplus \sin(d / R_\oplus), \quad d - d_{EN} = \frac{d^3}{6 R_\oplus^2} + O(d^5 / R_\oplus^4). \quad (2.B.16)$$

This deficit is 0.033 m at 20 km, 0.51 m at 50 km, and about 33 m at 200 km. These estimates assume surface points and an origin at one endpoint. For increments between two remote points in a fixed frame, projection distortion also depends on their distance and bearing from the frame origin. It cannot be bounded from increment length alone. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2 / (2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the adopted altitude channel itself (Sec. 2.7.5, Sec. 2.7.1), and the U component serves only the geometric construction.

2.B.4 Size of the spherical-approximation error

Sec. 2.7.2 computes inter-fix distances by the haversine formula, Eq. (2.2), which treats the Earth as a sphere of fixed radius $R_\oplus = 6,371$ km regardless of latitude or heading. This section bounds what that costs against the WGS84 ellipsoid.

Local metric. For a short step at geodetic latitude φ , the ellipsoid and the sphere give

$$ds_{\text{ell}}^2 = M(\varphi)^2 d\varphi^2 + N(\varphi)^2 \cos^2 \varphi d\lambda^2, \quad ds_{\text{sph}}^2 = R_\oplus^2 (d\varphi^2 + \cos^2 \varphi d\lambda^2). \quad (2.B.17)$$

Define $d\sigma = ds_{\text{sph}}/R_\oplus$ and the bearing α in this spherical metric by $d\varphi = d\sigma \cos \alpha$ and $\cos \varphi d\lambda = d\sigma \sin \alpha$. Then

$$R_{\text{eff}}(\varphi, \alpha) \equiv \frac{ds_{\text{ell}}}{d\sigma} = \sqrt{M(\varphi)^2 \cos^2 \alpha + N(\varphi)^2 \sin^2 \alpha}, \quad (2.B.18)$$

interpolating between the meridian value $M(\varphi)$ ($\alpha = 0$, north–south) and the prime-vertical value $N(\varphi)$ of Eq. (2.B.5) ($\alpha = 90^\circ$, east–west) [15]. $N(\varphi)$ is already in hand; $M(\varphi)$ follows the same route, differentiating the surface point $(p_s(\varphi), Z_s(\varphi))$ of Eqs. (2.B.5)–(2.B.6) with respect to φ and using $ds = M d\varphi$ along the meridian:

$$\frac{dp_s}{d\varphi} = -\frac{N(\varphi)(1-e^2)\sin \varphi}{1-e^2\sin^2 \varphi}, \quad \frac{dZ_s}{d\varphi} = \frac{N(\varphi)(1-e^2)\cos \varphi}{1-e^2\sin^2 \varphi}, \quad (2.B.19)$$

so that $M(\varphi)^2 = (dp_s/d\varphi)^2 + (dZ_s/d\varphi)^2$ collapses, via $\sin^2 \varphi + \cos^2 \varphi = 1$, to

$$M(\varphi) = \frac{N(\varphi)(1-e^2)}{1-e^2\sin^2 \varphi} = \frac{a(1-e^2)}{(1-e^2\sin^2 \varphi)^{3/2}}. \quad (2.B.20)$$

Comparison to the sphere. The reference radius used in Eq. (2.2), $R_\oplus = 6,371$ km, agrees with the ellipsoid’s mean radius $R_m = (2a + b)/3 = a(1 - f/3) = 6,371.009$ km to within 10 m. The fractional difference, expressed as ellipsoidal length relative to spherical length, is approximately $[R_{\text{eff}}(\varphi, \alpha) - R_m]/R_m$, bounded between its values at $\alpha = 0$ and $\alpha = 90^\circ$. Expanding Eqs. (2.B.5) and (2.B.20) to first order in the flattening f (using $e^2 \simeq 2f$),

$$\frac{N(\varphi)}{R_m} - 1 \simeq f \left(\sin^2 \varphi + \frac{1}{3} \right), \quad \frac{M(\varphi)}{R_m} - 1 \simeq f \left(3 \sin^2 \varphi - \frac{5}{3} \right), \quad (2.B.21)$$

both of order the flattening $f = 1/298.257223563 \approx 0.335\%$ at every latitude: from $-5f/3 \approx -0.56\%$ at the equator ($\varphi = 0$, meridian direction) to $+4f/3 \approx 0.45\%$ at the poles ($\varphi = 90^\circ$, where $M = N$ by symmetry). Thus 0.3% is a regional estimate, not a global bound.

At the latitudes flown. Most of the archive launches from the Alps box $\varphi \in [43.8^\circ, 46.6^\circ]$ N used to classify take-off sites (Sec. 2.9, Fig. 2.24). Evaluating Eq. (2.B.21) across that range moves the bound by under 0.02 percentage points, so its midpoint $\varphi = 45^\circ$ is representative:

$$\frac{N(45^\circ)}{R_m} - 1 \simeq \frac{5f}{6} \approx 0.28\%, \quad \frac{M(45^\circ)}{R_m} - 1 \simeq -\frac{f}{6} \approx -0.06\% \quad (2.B.22)$$

(numerically, $N(45^\circ) = 6,388.84$ km and $M(45^\circ) = 6,367.38$ km against $R_m = 6,371.01$ km). At 45° , the sphere underestimates an east–west ellipsoidal step by about 0.28% and overestimates a north–south step by about 0.06%. Across the latitude interval above the absolute discrepancy remains below 0.3%, or roughly 3 cm on a 10 m step. The bound only opens up toward the 0.45–0.56% extremes for the small remainder of flights nearer the equator or at higher latitude (Sec. 2.9, “elsewhere”), corresponding to at most roughly 6 cm on a 10 m step. The uncertainty of a GNSS increment must be assessed separately, since errors at successive fixes can be correlated.

Experimentals

E.1 Experimental: joint factorial analysis

Provisional implementation under methodological review. The factorial analysis below is retained for further study and validation. Its numerical results are not used to draw the conclusions of Section 3.2, which motivates the method from marginal and stratified comparisons. This exploratory extension brings the altitude, circuit and equipment comparisons of Section 3.2 into one table. Its purpose is to describe departures from additivity in the effective exponent; the coverage of its uncertainty intervals remains conditional on the resampling assumptions. We retain 44,030 flights in $\mathcal{C}_{10000}$ with an open or closed circuit and an admitted equipment class, excluding 2,243 other cohort members. The four initial-altitude classes, two circuits and two equipment groups define 16 disjoint cells. Each $\hat{H}_{ijk}$ is fitted to that cell's equal-flight MSD over 10–10000 s, with the estimator and retained segments of Eq. (3.8).

Factorial model and additive projection. We use the fixed-effects factorial decomposition into main effects, two-factor interactions and a three-factor interaction, with the sum-to-zero constraints of Mohr, Wilson and Freund [28, Sections 9.3.1 and 9.7]. Writing the empirical components without hats to simplify notation,

$$\begin{aligned}\hat{H}_{ijk} = & \mu + \alpha_i + \gamma_j + \eta_k \\ & + (\alpha\gamma)_{ij} + (\alpha\eta)_{ik} + (\gamma\eta)_{jk} + (\alpha\gamma\eta)_{ijk},\end{aligned}\tag{E.1}$$

where i indexes altitude, j circuit and k equipment. Every effect sums to zero over each of its own indices. These constraints make the decomposition unique. Bars below denote unweighted means of the fitted cell exponents: $\mu = \bar{H}_{...}$, $\alpha_i = \bar{H}_{i..} - \mu$, with analogous expressions for γ_j and η_k ; for example, $(\alpha\gamma)_{ij} = \bar{H}_{ij.} - \bar{H}_{i..} - \bar{H}_{.j.} + \mu$. The three-factor term is the remaining component.

The flights are unbalanced across cells. As discussed in Section 9.3.7 of the same reference, balanced-design sums of squares cannot be applied unchanged to such flight-level data. Here equal weights instead define a descriptive projection of the *16 estimated exponents*. They give the same importance to each combination, irrespective of flight count or precision. Consequently the marginal effects average cell exponents; they are generally different from fitting the MSD of a pooled or reweighted flight population. The logarithm of a mixture of MSDs does not equal a mixture of their logarithms.

The additive approximation is the ordinary least-squares projection onto the grand mean and main effects under this equal-cell metric:

$$\begin{aligned}\hat{H}_{ijk}^{\text{add}} &= \bar{H}_{i..} + \bar{H}_{.j.} + \bar{H}_{..k} - 2\bar{H}_{...}, \\ r_{ijk} &= \hat{H}_{ijk} - \hat{H}_{ijk}^{\text{add}}.\end{aligned}\tag{E.2}$$

It has six free coefficients. The residual occupies the ten-dimensional interaction space: $3+3+1$ dimensions for the three pairs and three for the three-factor interaction. The complete

decomposition reconstructs the observed table exactly, while sampling uncertainty remains in every estimated component. We do not use the residual as an estimate of a common error variance or apply the classical ANOVA F tests. In particular, bootstrap draws are not independent experimental replications of a cell; the missing-error issue for an unreplicated factorial table is discussed in [28, Section 9.6].

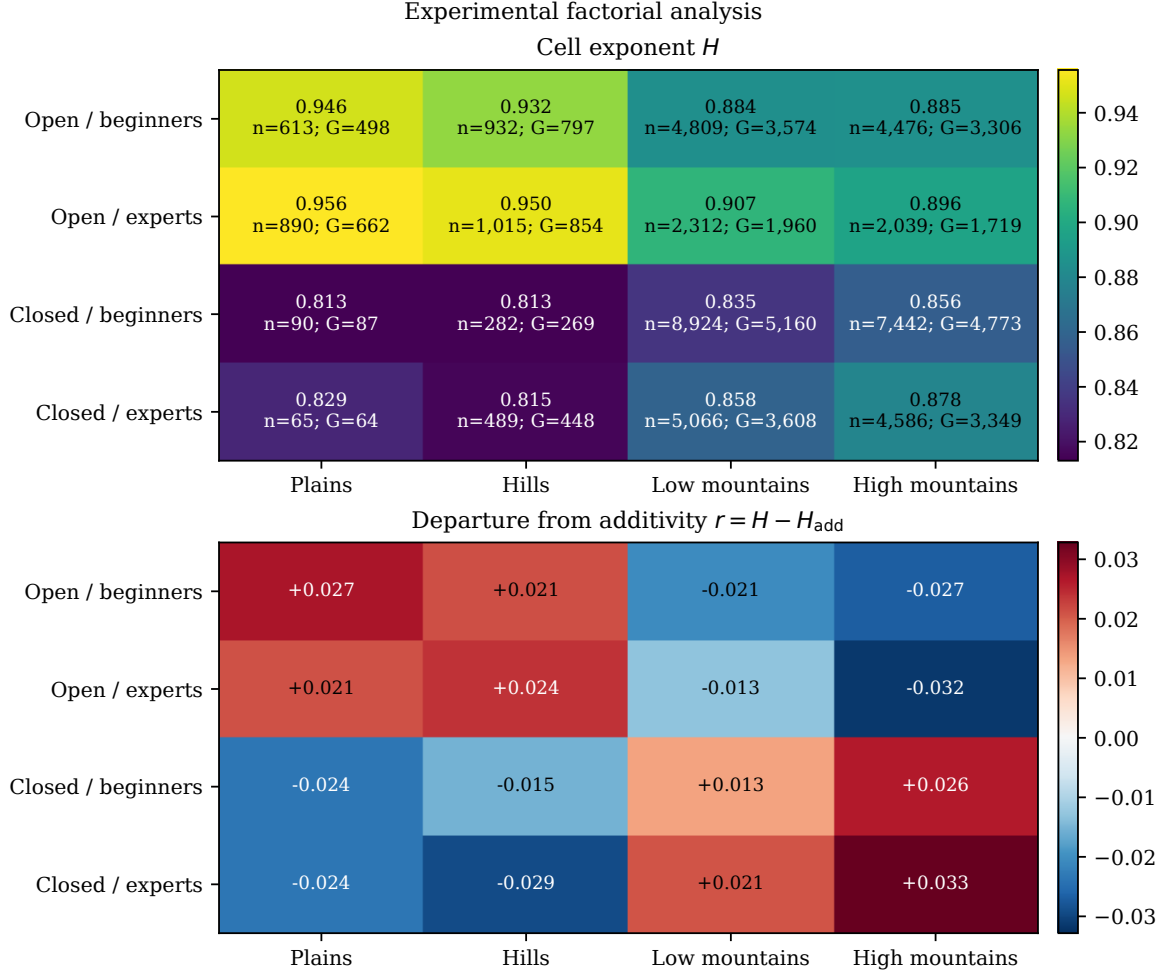


Figure E.1. Experimental factorial analysis. Top: the 16 effective exponents, with flight counts n and contributing site-day groups G . Bottom: residuals from the equal-cell additive projection, on a colour scale centred at zero. Residuals combine all interaction terms and sampling error; the colours carry no cellwise significance declaration. Shared groups can contribute to more than one cell, so group counts are not additive.

The orthogonal components partition the squared variation between the cell estimates. For each component u , Table E.1 reports its share $\sum_{ijk} u_{ijk}^2 / \sum_{ijk} (\hat{H}_{ijk} - \mu)^2$ and its root-mean-square magnitude. The additive share is

$$R_{\text{add}}^2 = 1 - \frac{\sum_{ijk} r_{ijk}^2}{\sum_{ijk} (\hat{H}_{ijk} - \mu)^2}, \quad R_{\text{rms}} = \sqrt{\frac{1}{16} \sum_{ijk} r_{ijk}^2}. \quad (\text{E.3})$$

These are descriptive summaries of estimated cell variation, which includes sampling noise. They do not measure a causal fraction of transport explained.

The additive terms account for 75.8% of this variation, with residual RMS 0.024. Altitude $\times$ circuit alone accounts for 23.6%, almost all of the nonadditive component. The altitude

Component	df	Share (%)	RMS in H
Altitude	3	1.17	0.0053
Circuit	1	72.10	0.0412
Equipment	1	2.54	0.0077
Altitude $\times$ circuit	3	23.64	0.0236
Altitude $\times$ equipment	3	0.26	0.0025
Circuit $\times$ equipment	1	< 0.01	0.0001
Three-way interaction	3	0.29	0.0026

Table E.1. Experimental partition of variation among the 16 fitted exponents, using equal cell weights. The degrees of freedom give subspace dimensions; no mean-square error or F test is constructed. RMS values have the units of H . Small shares do not establish the absence of an interaction.

main effect contributes only 1.2%: this small marginal share reflects compensation between circuit-specific patterns, rather than an absence of altitude dependence. The residual map has opposite signs for open and closed circuits, reversing between lowlands and mountains in both equipment groups (Figure E.1).

Simple effects and interaction contrasts. Following the distinction between marginal comparisons and comparisons within interacting factors [28, Section 9.4], we examine the simple effects

$$\begin{aligned}\Delta_C(i, k) &= \hat{H}_{i,\text{open},k} - \hat{H}_{i,\text{closed},k}, \\ \Delta_E(i, j) &= \hat{H}_{i,j,\text{experts}} - \hat{H}_{i,j,\text{beginners}}.\end{aligned}\tag{E.4}$$

For a compact interaction contrast, let $D_k = \Delta_C(\text{Plains}, k) - \Delta_C(\text{High mountains}, k)$. This difference of differences measures how much the circuit separation decreases between the endpoint altitude classes at fixed equipment. Its change $D_E - D_B$ is one contrast in the three-factor interaction space; it does not exhaust that three-dimensional space. These are exploratory comparisons informed by the earlier stratified results, rather than hypotheses specified before examining the archive.

For uncertainty we reaggregate the saved per-flight MSDs using the original 1000 paired site-day cluster-bootstrap draws, then refit each cell in every draw. Each linear contrast is evaluated on the same draw across all cells. We construct *single-step bootstrap simultaneous confidence intervals* using the maximum absolute standardized deviation, following the joint-region construction of Romano and Wolf [30, Sections 3–5]. For a declared family $\mathcal{F}$ of 19 contrasts—the eight Δ_C , eight Δ_E , two D_k and $D_E - D_B$ —our implementation is

$$\begin{aligned}q_{0.90} &= Q_{0.90} \left[\max_{\ell \in \mathcal{F}} \left| \frac{\hat{\theta}_\ell^{(b)} - \hat{\theta}_\ell}{\hat{s}_\ell} \right| \right], \\ I_\ell &= \hat{\theta}_\ell \pm q_{0.90} \hat{s}_\ell,\end{aligned}\tag{E.5}$$

where $\hat{s}_\ell$ is the standard deviation across bootstrap draws. We use fixed marginal bootstrap scales, a first-order plug-in standardization, without recomputing a standard error inside each draw or performing stepdown testing. The nominal 90% simultaneous level applies to this declared family within one fit window and resampling scheme; it does not cover all earlier or subsequently selected comparisons. Its population coverage still requires an adequate cluster-resampling model [21].

All eight circuit contrasts remain positive under these simultaneous intervals. Their decrease from lowlands to mountains survives equipment stratification: $D_B = 0.104$ [0.082, 0.125] and $D_E = 0.109$ [0.087, 0.132]. The equipment contrasts are smaller. The intervals include zero for

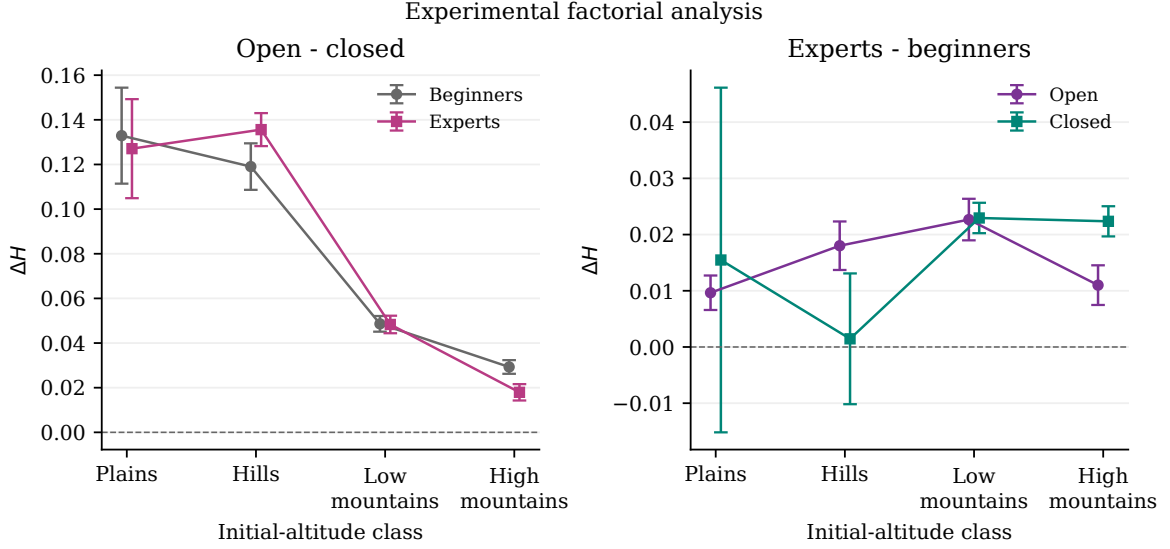


Figure E.2. Experimental simple-effect contrasts. Left: open minus closed within each altitude and equipment group. Right: experts minus beginners within each altitude and circuit. Bars are nominal 90% simultaneous site-day bootstrap intervals for the family of 19 contrasts in Eq. (E.5). Lines connect categorical levels for readability. The two panels use different vertical scales.

closed circuits in Plains and Hills, while the other six are positive under this resampling model. The endpoint three-factor contrast is $D_E - D_B = 0.006 [-0.026, 0.037]$; its interval includes zero, leaving the difference between the two endpoint attenuations unresolved.

Sensitivity and experimental status. We repeat the joint bootstrap using calendar days across all sites, three-day calendar blocks, and the 50 km cell $\times$ altitude $\times$ month-of-year groups of Section 3.1.2, pooling years in the last case. Each alternative uses 1000 draws and two seeds, with the whole eligible archive as sampling frame before selecting the fixed cohort and cells. The point estimates are unchanged. The endpoint contrasts stay positive and their equipment difference stays unresolved (Table E.2); the inclusion or exclusion of zero also remains unchanged for the 16 simple contrasts across these alternatives. These checks assess sensitivity, not independence of the clusters or nominal coverage. The smallest cell has 65 flights in 64 site-day groups, falling to 44 contributing groups under the spatial/seasonal partition; support cannot be judged from flight count alone.

Resampling groups	D_B	D_E	$D_E - D_B$
Site-day	[0.082, 0.125]	[0.087, 0.132]	[-0.026, 0.037]
Calendar day	[0.081, 0.126]	[0.085, 0.133]	[-0.028, 0.039]
Three days	[0.080, 0.128]	[0.085, 0.134]	[-0.029, 0.040]
50 km / altitude / month	[0.074, 0.134]	[0.082, 0.136]	[-0.020, 0.031]

Table E.2. Experimental resampling sensitivity over 10–10000 s. Intervals use the same family of 19 contrasts. For each alternative the displayed endpoints enclose the two seed-specific simultaneous intervals; this envelope is a sensitivity summary, not a separately calibrated confidence interval.

Finally, fitting each of the three lag decades separately, with exactly the same flights and segments, retains positive endpoint attenuations but changes their magnitude substantially (Table E.3). The larger long-lag contrasts show that the full-range H compresses a scale-dependent pattern. The decomposition describes these fitted effective exponents; it does not

establish an asymptotic scaling law. Geographic coverage, weather, wing performance and pilot decisions remain mixed within cells, so the factorial terms support conditional associations rather than causal attribution.

Fit range (s)	R_{add}^2	RMS	D_B	D_E
10–10000	0.758	0.024	0.104	0.109
10–100	0.651	0.008	0.027	0.040
100–1000	0.633	0.016	0.070	0.076
1000–10000	0.829	0.058	0.248	0.246

Table E.3. Experimental fit-window sensitivity, keeping C_{10000} and its retained segments fixed. All entries are point estimates; RMS denotes R_{rms} . The endpoint interaction contrasts D_B and D_E are defined after Eq. (E.4).

E.2 Experimental: scaling of displacement distributions

Experimental analysis. This section was moved from Chapter 3. Its results depend on the same cohort and resampling assumptions as the MSD analysis there, and they are not part of the main argument.

The MSD measures one moment of the displacement law. To test whether the whole law rescales with one exponent, we use quantiles, a spectrum of moments and excess kurtosis. All use the main cohort’s flights, segments and weights in Eq. (3.7).

E.2.1 Self-similarity and its observable consequences

A process $\mathbf{X}(t)$, with $\mathbf{X}(0) = 0$, is H -self-similar if, for some $H > 0$,

$$(\mathbf{X}(at_1), \dots, \mathbf{X}(at_m)) \stackrel{d}{=} a^H (\mathbf{X}(t_1), \dots, \mathbf{X}(t_m)) \quad (a > 0). \quad (\text{E.6})$$

This identity must hold for every $m \geq 1$ and $t_1, \dots, t_m \geq 0$. Equality in distribution concerns positions at any collection of times, including their dependence [31]. Gaussianity and independence of increments are not required.

Two separate hypotheses stand between this definition and a measurement, and each does different work. Stationary increments make a pooled lag statistic informative about $\mathbf{X}$ itself: with $\mathbf{X}(0) = 0$ they give $\mathbf{X}(\tau) \stackrel{d}{=} \mathbf{X}(t + \tau) - \mathbf{X}(t)$, so displacements collected from many origins estimate the law of the position reached after an elapsed time τ . The archive supplies no repeated releases from a common initial condition, which leaves this the only route from the increments we observe to the process we describe. Self-similarity then compresses the resulting family of laws: one exponent and one shape determine every lag, including elapsed times the cohort does not cover, and that exponent is the quantity stochastic transport models predict. Without it this chapter reports a separate propagator at each measured lag, with no relation between them. Testing Eq. (E.6) is therefore a test of whether one anomalous exponent describes these flights at all, and the sections below report necessary consequences of that identity rather than a proof of it.

To infer a lag-only scaling law, we additionally assume stationary increments: their joint laws must be unchanged by shifting the origin. Self-similarity alone does not impose this condition. With stationary increments, each amplitude $Y \in \{|\Delta E|, |\Delta N|, R\}$ obeys

$$Y(\tau) \stackrel{d}{=} (\tau/\tau_0)^H Y(\tau_0), \quad \begin{aligned} Q_{Y,p}(\tau) &= (\tau/\tau_0)^H Q_{Y,p}(\tau_0), \\ M_{q,Y}(\tau) &= (\tau/\tau_0)^{qH} M_{q,Y}(\tau_0). \end{aligned} \quad (\text{E.7})$$

Here $Q_{Y,p}$ is a population quantile and $M_{q,Y}$ a finite moment of order q ; below, the same symbols denote their equal-flight estimates. Thus every quantile has the same exponent, quantile ratios are constant, and the moment spectrum is $\zeta_Y(q) = qH$. Signed-component excess kurtosis is also constant when the fourth moment exists and variance is nonzero; it need not vanish. These are necessary consequences for the lag distributions. A finite selection of such statistics cannot establish the joint-law identity in Eq. (E.6).

E.2.2 A common empirical distribution

With the weights in Eq. (3.7), the common empirical distribution is

$$\hat{F}_{Y,\tau}(y) = \frac{1}{N} \sum_{f=1}^N \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} 1\{Y_{fi}(\tau) \leq y\}, \quad Q_{Y,p}(\tau) = \inf\{y : \hat{F}_{Y,\tau}(y) \geq p\}. \quad (\text{E.8})$$

We average the within-flight fractions below y , then invert the resulting CDF. The quantile is the first observed amplitude at which the cumulative weight reaches p , without interpolation between ranks. This order of operations defines the quantile of the equal-flight mixture.

Absolute components measure amplitudes along east and north. For example, $\Pr(|\Delta E| \leq a) = \Pr(-a \leq \Delta E \leq a)$, so small absolute quantiles measure proximity to zero displacement. The radius $R = (\Delta E^2 + \Delta N^2)^{1/2}$ is invariant under rotation and measures horizontal displacement without assuming isotropy.

E.2.3 Quantiles, growth exponents and relative spread

We evaluate $p = 0.25, 0.50, 0.75, 0.90$ and fit each quantile over the same 33 lags. Writing $\mathcal{B}$ for the least-squares slope against $\log_{10} \tau$ used in Section 3.1.2,

$$H_{Y,p} = \mathcal{B} \left[\log_{10} \frac{Q_{Y,p}}{1 \text{ m}} \right], \quad \mathcal{B} \left[\log_{10} \frac{Q_{Y,0.25}}{Q_{Y,0.90}} \right] = H_{Y,0.25} - H_{Y,0.90}. \quad (\text{E.9})$$

No factor of two enters a length exponent. The ratio compares small and large amplitudes at each lag, while the fitted exponents summarise their growth across 10–10000 s. The equality follows from linearity of least squares on identical lags; ratio drift and the exponent difference are therefore linked diagnostics. Figures E.3 and E.4 give the quantile curves and their ratio for each discipline, and Figure E.5 collects the fitted quantile exponents.

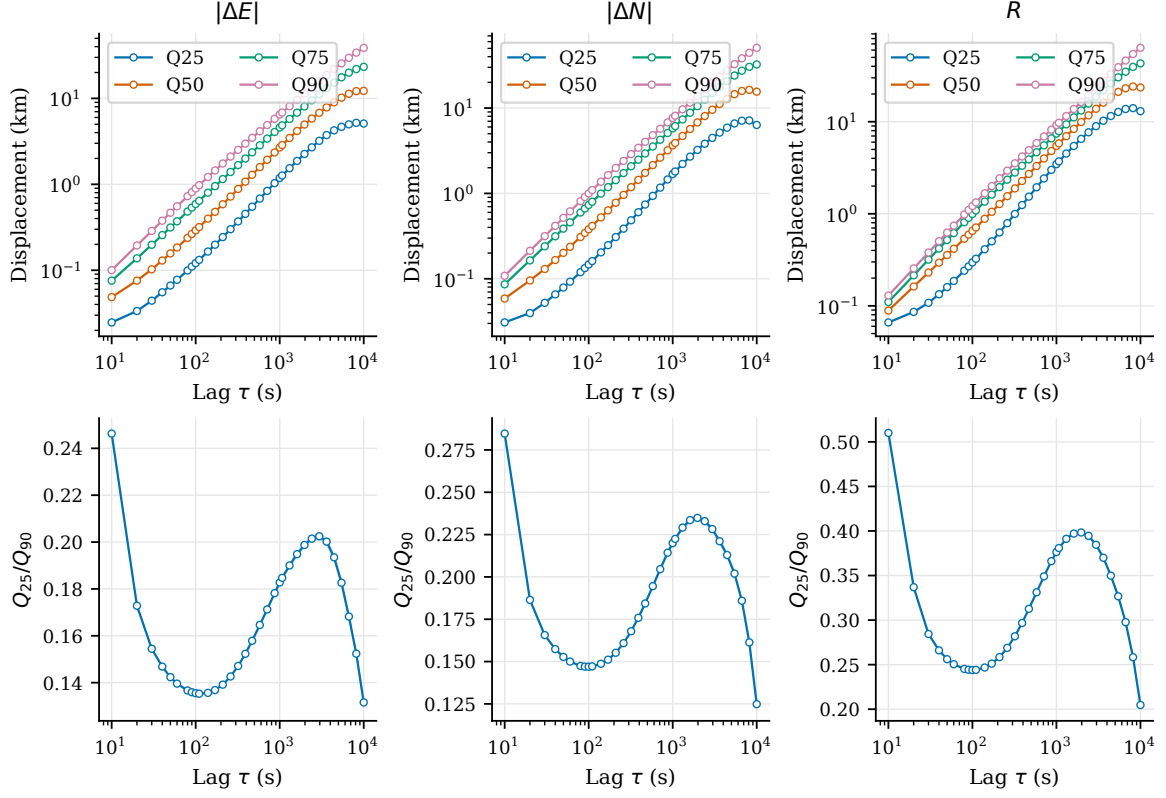


Figure E.3. Paraglider equal-flight displacement quantiles and their relative spread, on the fixed main cohort. Every panel uses the empirical measure in Eq. (E.8). Shading gives pointwise 90% cluster-bootstrap intervals.

For paragliders, the radial ratio falls from 0.510 at 10 s to 0.244 at 100 s, rises to 0.398 at 1970 s, then falls to 0.205 at 10000 s. Quantile-fit RMS residuals span 0.023–0.084 dex.

For hang gliders, the radial ratio falls from 0.472 at 10 s to 0.192 at 90 s, rises to 0.385 at 1970 s, then falls to 0.233 at 10000 s. Quantile-fit RMS residuals span 0.037–0.101 dex.

A positive global slope of these nonmonotonic ratios can coexist with a net endpoint decrease.

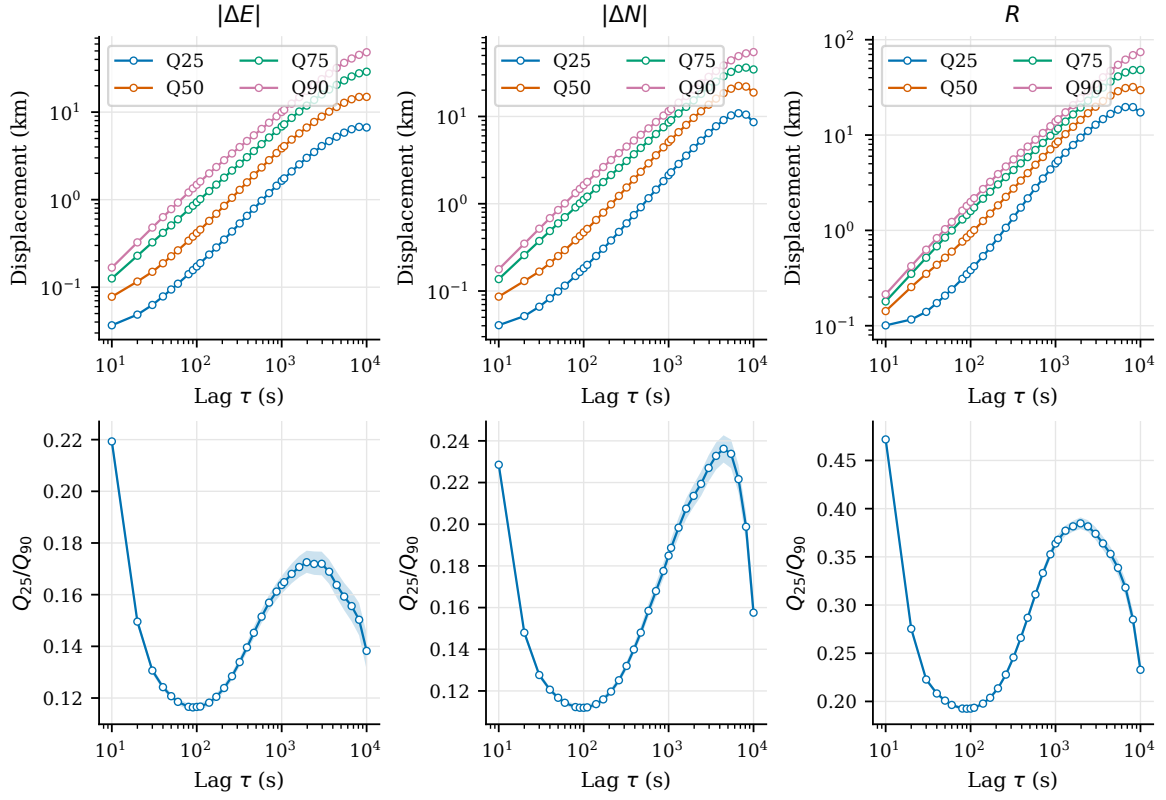


Figure E.4. The same amplitude quantiles and ratio for hang gliders, with an independent main cohort and the same sampling convention.

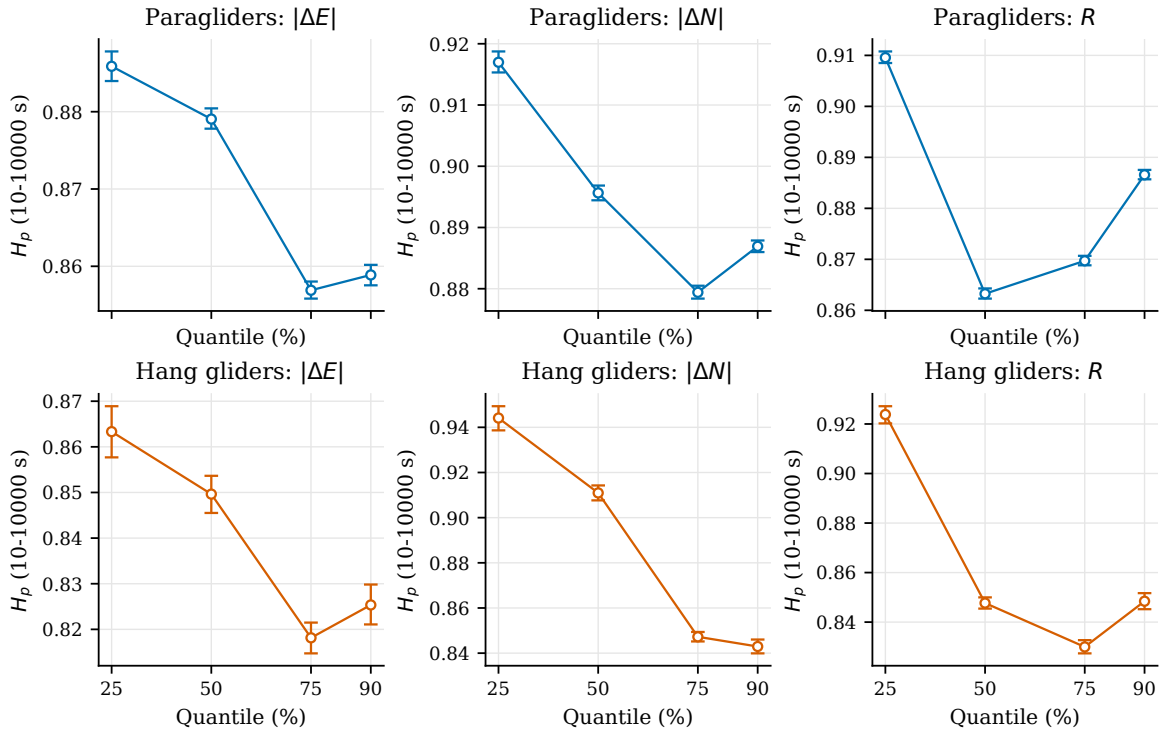


Figure E.5. Global quantile exponents from 10–10000 s fits. Error bars are the 5th–95th percentiles of the refitted bootstrap slopes. A common scale exponent would give a horizontal sequence within each panel.

E.2.4 Moment spectrum and finite-range scaling

For each amplitude and order we calculate

$$M_{q,Y}(\tau) = \frac{1}{N} \sum_f \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} Y_{fi}(\tau)^q, \quad \zeta_Y(q) = \mathcal{B} \left[\log_{10} \frac{M_{q,Y}}{(1\text{ m})^q} \right]. \quad (\text{E.10})$$

We use $q \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 3.5, 4\}$. Each point of the spectrum is obtained by first computing a moment at all lags, then fitting its log-log slope. The reference $\zeta = qH$ is fitted through the origin with equal weight for the ten orders. Higher orders emphasise large displacements. The radial spectrum contains the MSD, with the identity

$$M_{2,R} = M_{2,|\Delta E|} + M_{2,|\Delta N|} = M_2 \quad (\text{E.11})$$

verified at every lag and in every bootstrap replicate. Table E.4 reports the main-cohort MSD exponents, Figure E.6 compares their local and global values, and Figure E.7 gives the moment spectra. Table E.5 collects the paired quantile and moment contrasts that test $\zeta = qH$.

Discipline	Fit range (s)	H [5–95%]	RMS (dex)
Paragliders	10–10000	0.882 [0.881, 0.883]	0.046
Paragliders	10–100	0.925 [0.924, 0.925]	0.007
Paragliders	100–1000	0.898 [0.897, 0.898]	0.006
Paragliders	1000–10000	0.804 [0.802, 0.807]	0.026
Hang gliders	10–10000	0.852 [0.849, 0.855]	0.072
Hang gliders	10–100	0.915 [0.914, 0.916]	0.006
Hang gliders	100–1000	0.873 [0.871, 0.874]	0.007
Hang gliders	1000–10000	0.728 [0.720, 0.737]	0.051

Table E.4. Main-cohort MSD exponents: a single global fit and three common decade intervals. RMS residuals are measured in $\log_{10} M_2$. Each fit retains the same flight and segment population.

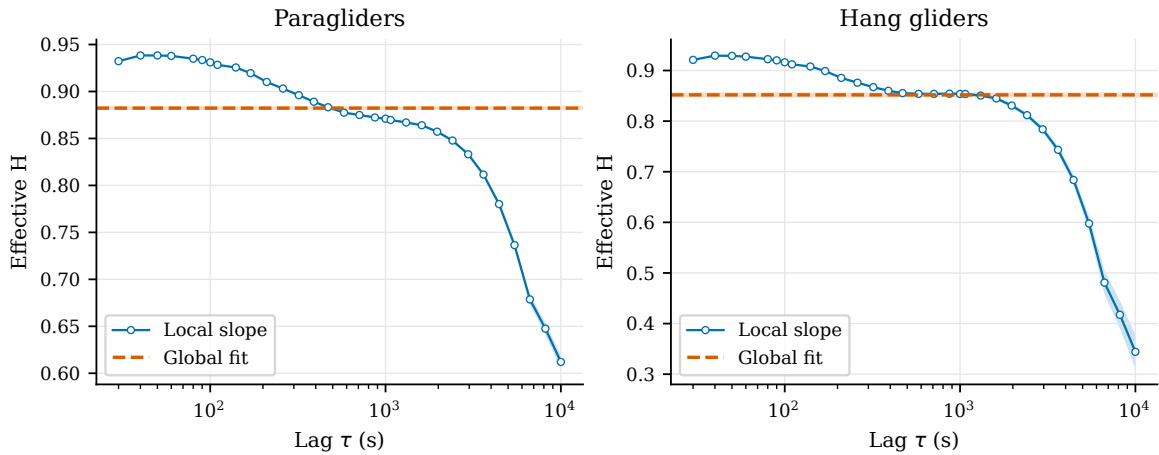


Figure E.6. Local and global effective MSD exponents on the fixed main cohort. Bands are pointwise 90% bootstrap intervals. The global fit compresses the full curve into one number; the local slopes expose its scale dependence.

For paragliders, the radial contrast $\zeta(4)/4 - \zeta(0.25)/0.25$ is 0.0095 [0.0086, 0.0105]. Moment-fit RMS residuals span 0.010–0.079 dex across all amplitudes and orders.

For hang gliders, the radial contrast $\zeta(4)/4 - \zeta(0.25)/0.25$ is -0.0085 [-0.0112, -0.0057]. Moment-fit RMS residuals span 0.011–0.126 dex across all amplitudes and orders.

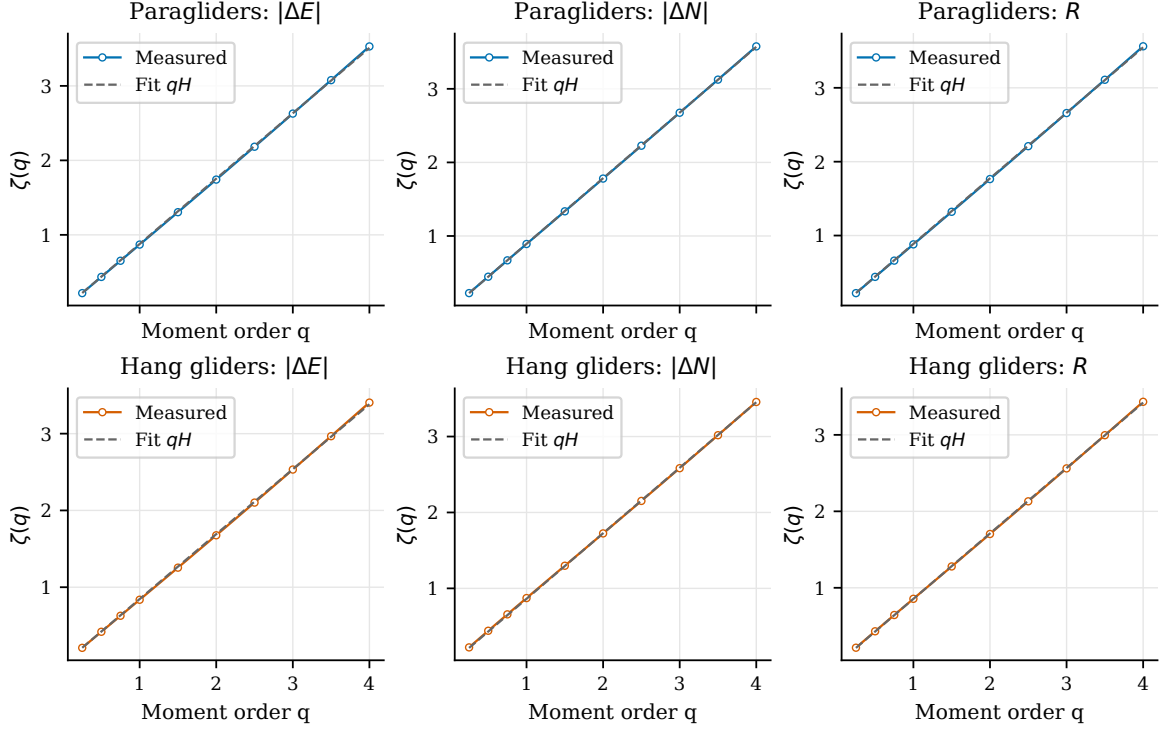


Figure E.7. Moment spectra for the three displacement amplitudes. Each point is the slope of a 10–10000 s log–log moment fit; shading gives its pointwise 90% interval. Dashed lines are constrained linear fits $\zeta = qH$.

Discipline	Amplitude	$H_{25} - H_{90}$	$\zeta(4)/4 - \zeta(0.25)/0.25$
Paragliders	$ \Delta E $	0.027 [0.025, 0.029]	0.0113 [0.0097, 0.0128]
Paragliders	$ \Delta N $	0.030 [0.028, 0.032]	-0.0004 [-0.0016, 0.0009]
Paragliders	R	0.023 [0.022, 0.024]	0.0095 [0.0086, 0.0105]
Hang gliders	$ \Delta E $	0.038 [0.032, 0.044]	0.0095 [0.0040, 0.0150]
Hang gliders	$ \Delta N $	0.101 [0.096, 0.106]	-0.0269 [-0.0305, -0.0232]
Hang gliders	R	0.075 [0.071, 0.079]	-0.0085 [-0.0112, -0.0057]

Table E.5. Paired global-slope contrasts, with 5–95% bootstrap intervals. A common scale exponent predicts zero for both the quantile and moment contrasts.

E.2.5 Excess kurtosis of the signed components

Departures from Gaussian shape are assessed on the signed components: taking an absolute value or a radius changes even a Gaussian law. With the equal-flight mean $\langle A \rangle_\tau = \sum_f \sum_{i \in \mathcal{I}_f(\tau)} w_f(\tau) A_{fi}$, define

$$\mu_j(\tau) = \langle \Delta x_j \rangle_\tau, \quad K_j(\tau) = \frac{\langle (\Delta x_j - \mu_j)^4 \rangle_\tau}{\langle (\Delta x_j - \mu_j)^2 \rangle_\tau^2} - 3, \quad j \in \{E, N\}. \quad (\text{E.12})$$

Centring uses the mean of the full mixture at each lag. A Gaussian marginal has $K_j = 0$ [32]; nonzero excess measures a departure in its standardised fourth moment. Zero excess alone does not establish Gaussianity. For self-similarity, the relevant prediction is constancy with lag, regardless of the value. Figure E.8 shows $K_j(\tau)$ for both components.

For paragliders, (K_E, K_N) changes from $(-0.68, -0.97)$ at 10 s to $(2.77, 0.94)$ at 10000 s. For hang gliders, (K_E, K_N) changes from $(-0.60, -0.72)$ at 10 s to $(4.03, 2.24)$ at 10000 s.

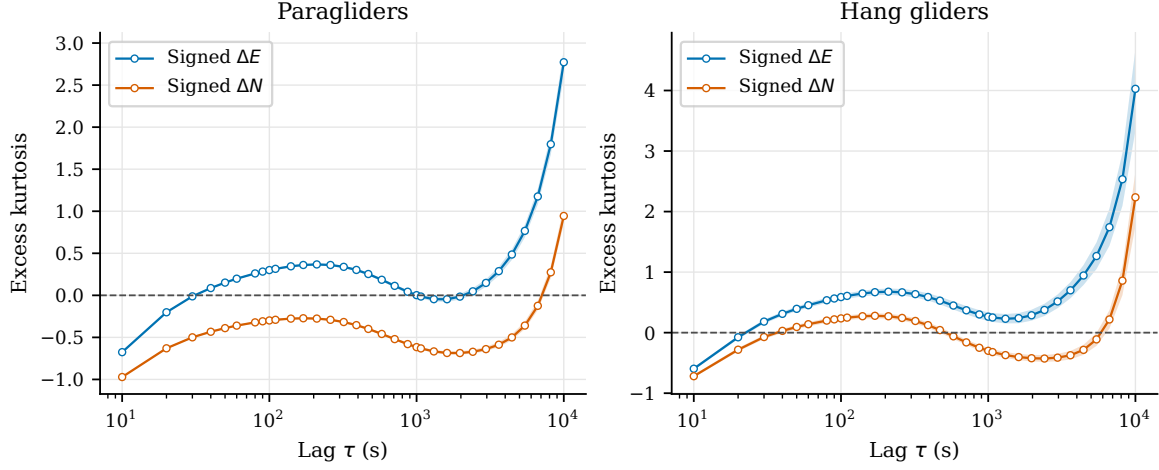


Figure E.8. Signed-component Fisher excess kurtosis at the same lags, with the same flights, segments and weights as the quantiles and moments. The dashed line is the Gaussian marginal reference. Bands use the coupled cluster bootstrap.

Different flight means and variances can themselves produce nonzero excess in the mixture. The measured kurtosis therefore characterises the pooled displacement law.

E.2.6 Assessment of self-similarity

The quantile curves and the nearly straight moment spectra can be reconciled through

$$M_{q,Y}(\tau) = \int_0^1 Q_{Y,p}(\tau)^q dp. \quad (\text{E.13})$$

Each moment averages over probability levels; the order q controls the emphasis on larger amplitudes. Fitting its slope across three decades further averages over crossovers. Changes in different parts of the distribution can consequently leave the fitted spectrum nearly linear. Moreover, Table E.5 resolves small departures from $\zeta = qH$; the spectrum is not exactly linear under the adopted bootstrap.

For these data, the lag-dependent quantile ratios provide the more direct evidence against a fixed distributional shape. Their large, nonmonotonic changes are visible without fitting a power law, and their estimation does not require finite high-order moments. Unequal fitted quantile exponents support this reading, but the curved quantile trajectories limit their interpretation as constant exponents. The change in kurtosis supplies complementary evidence. Visual linearity of ten fitted moment slopes is therefore insufficient to outweigh the observed shape changes.

Over 10–10000 s, a single self-similar model with stationary increments does not adequately describe the pooled displacement law. Approximate scaling over narrower intervals remains possible. Fixed flights and segments remove changing membership, and the contrast below measures how much of the remaining structure follows from where within a segment the origins fall. The present analysis does not determine multi-time laws. It therefore does not exclude every nonstationary self-similar process, nor establish an asymptotic multifractal law.

Does the increment law depend on the origin? Equation (E.7) reaches the lag distributions through stationary increments, and the estimators above pool every admissible origin inside a selected segment. That pooling estimates one population law only if the law of $\mathbf{X}(t+\tau) - \mathbf{X}(t)$ does not depend on t . We test this directly. Within each selected segment the admissible origins at lag τ are split into three contiguous blocks of equal size, and the same equal-flight amplitude statistics are measured separately in each. The early and late blocks receive identical origin counts in every segment, so each cohort flight contributes to both and the contrast is paired inside the site-day draws of Sec. 3.1.2. Under stationary increments the late-to-early ratio equals one at every lag. Pooling the three blocks returns the equal-flight moments of Sec. E.2 exactly, which fixes the diagnostic to the published measurement rather than to a separate estimator.

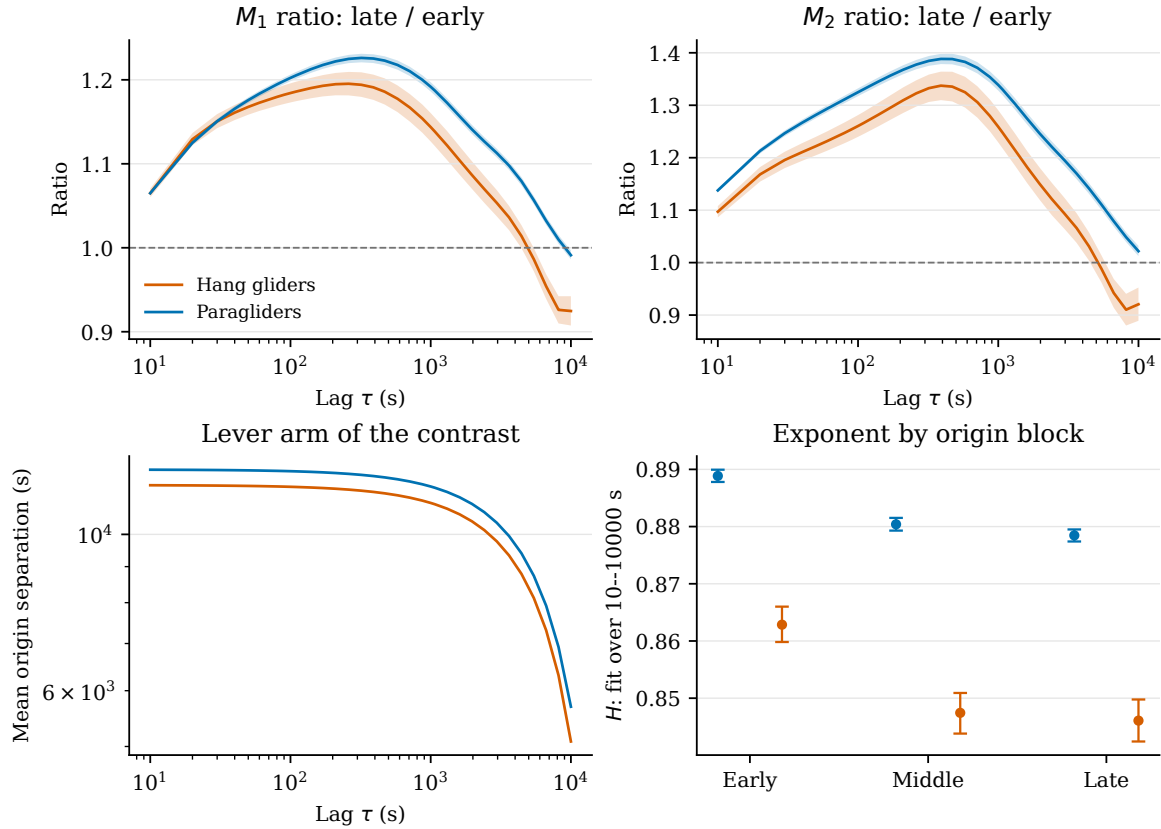


Figure E.9. Origin dependence of the increment law in the main cohort. Top: the ratio of the late-block to the early-block radial moments, with 95% paired bootstrap bands and the stationary-increment value as a dashed line. Bottom left: the mean separation between the two blocks of origins, which shrinks as the lag consumes the segment. Bottom right: H fitted in each block over 10–10000 s. Bands and bars are pointwise.

For paragliders, the late-origin second moment differs from the early-origin one at 33 of 33 lags. The largest departure is a ratio of 1.388 (1.378–1.398) at $\tau = 390$ s, where the two blocks are separated by 12,100 s on average. Fitting H over 10–10000 s in each block gives a late-minus-early difference of -0.0104 , with a 95% interval from -0.0110 to -0.0098 .

For hang gliders, the late-origin second moment differs from the early-origin one at 32 of 33 lags. The largest departure is a ratio of 1.337 (1.310–1.364) at $\tau = 390$ s, where the two blocks are separated by 11,489 s on average. Fitting H over 10–10000 s in each block gives a late-minus-early difference of -0.0168 , with a 95% interval from -0.0186 to -0.0150 .

The ratio departs from one over the whole lag range in both disciplines, so the increments

of these segments are not stationary. Displacements measured from late origins exceed those from early origins across most of the range, with the largest excess near $\tau \approx 400$ s, and the excess closes at the longest lags. The two blocks are separated by about 12000 s at small lags and by roughly half that at $\tau = 10000$ s, because the admissible origins shrink as the lag consumes the segment; the contrast at the longest lags therefore rests on a shorter lever arm and is not comparable in strength to the rest of the curve.

Two consequences follow for this chapter. First, the amplitude of the displacement law is a property of the flight phase from which origins are drawn, so the pooled one-lag laws of Sec. E.2 describe a mixture over flight phases rather than one stationary increment law. Second, the fitted exponent is far less sensitive to this mixture than the amplitude: the block-to-block change in H is resolved but small against the amplitude ratios in the same figure. Reading H as a lag exponent of the pooled mixture survives the test; reading it as the self-similarity exponent of a stationary-increment position process does not.

The sign of the effect does not match waiting-time ageing, which suppresses late displacements in a subordinated walk. Launch and initial circling confine the early part of a flight, while route-following occupies its later part, and this ordering alone produces the observed direction. The diagnostic separates origin position from lag; it does not identify the mechanism, and the flight-phase, wind and pilot explanations are not distinguished here.